Operator Interventions in Linear SPDEs: Galerkin Structural Semantics, Mixed-Rate Local Asymptotic Normality, and Uniform Inference for Post-Intervention Laws

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Abstract

We develop a causal-statistical theory for operator interventions in linear parabolic stochastic partial differential equations. The intervention replaces the generator coefficient while preserving the initial condition, noise spectrum, scale, observation mechanism, and primitive Wiener process. We first prove that the post-intervention SPDE is the infinite-dimensional limit of intervened Galerkin structural causal models driven by common exogenous Brownian segments. For Gaussian field functionals we give explicit post-intervention laws and necessary-and-sufficient target-identification criteria. Under exact discrete spectral observation, the likelihood is regular but anisotropic: the drift parameter is learned at rate $\sqrt{\sum_{k\le N}\lambda_k}$, whereas the noise scale is learned at rate $\sqrt{Nn}$. We establish local asymptotic normality, efficient estimation, uniform counterfactual CDF, quantile, tail, and distributional-effect processes, and feasible simultaneous confidence bands based on modal-score multipliers. We derive Fréchet derivatives of the operator-to-law map, combine statistical, Galerkin, temporal, and bounded operator-approximation errors, and state rate conditions for full-field inference. Finally, we construct observationally equivalent mechanisms with different post-intervention laws when forcing, initial, sensing, or cross-world modules are left unrestricted. Complete proofs are contained in the paper; numerical experiments verify the mixed rates, simultaneous coverage, approximation slopes, operator sensitivities, and impossibility constructions.

Keywords: stochastic partial differential equations; operator intervention; spectral inference; local asymptotic normality; empirical processes; multiplier bootstrap; Galerkin approximation.
MSC2020: 60H15; 62M05; 62F12; 62G15; 65M60.

1 Introduction

Many scientific interventions change a law of motion rather than a scalar treatment. Diffusivity can be altered by a material, a boundary protocol can change a transport operator, and a spatial policy can modify the generator of a stochastic field. In these settings the object of interest is not merely a parameter. It is a post-intervention distribution of a functional of the field.

We study the linear parabolic model

$$dU_t = \theta A_0 U_t\,dt + \sigma Q^{1/2} dW_t, \qquad U_0 = u_0, \tag{1}$$

where A_0 is self-adjoint, negative, and has compact resolvent. The primary intervention is

$$\mathrm{do}(\theta = \widetilde{\theta}),$$

which replaces θA_0 by $\widetilde{\theta} A_0$ while leaving the other structural modules invariant. For a space-time weight w, the target is

$$F_{\widetilde{\theta},w}(z;\vartheta) = \mathbb{P}_\vartheta\left\{\int_0^T \left\langle w_t, U_t^{\widetilde{\theta}}\right\rangle dt \le z\right\}, \qquad \vartheta = (\theta,\sigma). \tag{2}$$

The paper addresses four distinct questions. First, what makes the formal replacement of an operator a structural intervention rather than a relabelled stochastic equation? Second, when is the counterfactual law

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identified from a factual observation experiment? Third, how can it be estimated uniformly over interventions, weights, and thresholds? Fourth, how do statistical and numerical errors interact at the inferential scale?

Our construction extends the finite-dimensional intervention logic of Sokol and Hansen [33] to a diagonal parabolic SPDE. The post-intervention field is represented as the limit of intervened Galerkin structural causal models using the same Brownian path segments in every world. This common-noise convention is stronger than specifying marginal laws; it fixes the primitive disturbances that remain invariant under the intervention.

The statistical experiment uses exact discrete observations of the first N spectral coordinates. This setting is close to the spectral inference literature surveyed by Cialenco [13], but the target and asymptotic geometry are different. The exact transition likelihood produces two rates. With $\Lambda_N = \sum_{k \le N} \lambda_k$,

$$\sqrt{\Lambda_N}(\widehat{\theta} - \theta_0) \quad \text{and} \quad \sqrt{Nn}(\widehat{\sigma} - \sigma_0)$$

have non-degenerate limits. Fixed absolute-intervention laws overwrite the factual drift and are therefore estimated at the faster scale rate. Factual and factual-versus-counterfactual effects retain the slower drift uncertainty.

The main contributions are as follows.

(i) We prove exact finite-dimensional intervention semantics and uniform convergence of intervened Galerkin models to the post-intervention SPDE.

(ii) We give a fibre criterion for target identification and explicit finite-sample identification results under exact and noisy spectral observations.

(iii) We derive the exact likelihood, mixed Fisher-information limit, LAN expansion, efficient MLE, and moment convergence.

(iv) We establish uniform plug-in processes in $C(\mathcal{T})$, feasible multiplier bands, and contact-set tests for one-sided counterfactual hypotheses.

(v) We prove operator-to-law derivatives for bounded perturbations and a separate analytic-semigroup theorem for the unbounded rescaling direction A_0.

(vi) We combine statistical, Galerkin, temporal, and operator approximation errors in one family-specific inference theorem.

(vii) We construct sharp non-identification examples. In particular, unrestricted feedback or open-loop forcing can absorb the factual operator and separate only after intervention.

The contribution is deliberately narrower than causal discovery. The intervention is specified by the analyst. The theory identifies its target inside an admissible model class. It does not infer a causal graph from passive field observations, and it does not claim that every decomposition of an observed drift is structural.

Related literature

Three literatures meet in this paper, and the delimitation matters.

Statistics for SPDEs. The spectral approach to drift estimation under continuous observation goes back to Hübner and Rozovskii [22]; see Lototsky [25] and the survey of Cialenco [13]. An active recent strand studies discretely observed parabolic equations: power variations and volatility estimation under high-frequency sampling [8, 12], joint estimation from space–time grids [21, 23, 34, 35], drift estimation from discretely sampled Fourier modes [14], local measurements [6], and semilinear extensions [27]. Relative to this literature our observation scheme is deliberately the cleanest one — exact discrete observation of N spectral coordinates — but the output is of a different kind: a *joint* local asymptotic normality expansion with anisotropic diagonal normalisation $\mathrm{diag}(\sqrt{\Lambda_N}, \sqrt{Nn})$, efficiency and moment convergence of the exact conditional MLE, pushed through to uniform inference on post-intervention laws of field functionals. To our knowledge both the mixed-rate LAN statement and any inferential theory for operator interventions in SPDEs are new; the nearest discrete-observation results concern estimation rates rather than experiment-level asymptotics.

Causal models for dynamical systems. Interventions in ODE systems were formalised as limits of structural equation models by Mooij et al. [26], and extended to dynamic equilibria and random differential equations by Rubenstein et al. [31] and Bongers and Mooij [9]; Sokol and Hansen [33] treat SDEs via Euler schemes, and Peters et al. [29] survey the programme. Theorem 3.1 is the infinite-dimensional counterpart: the post-intervention SPDE is the limit of intervened Galerkin structural causal models driven by *common* Brownian segments, so the intervention is module surgery in the sense of Haavelmo [18] rather than a relabelling of stochastic equations. The non-identification constructions of Section 8 calibrate exactly which invariances carry identifying content.

Uniform inference on distributional targets. The plug-in processes and feasible bands use the functional delta method [36, 37], multiplier approximations for suprema [10], directional differentiability [17], and contact-set logic [7, 24]. The finite-rank limit geometry — every target process is a deterministic derivative surface loading a two-dimensional Gaussian — makes these tools exact rather than high-dimensional approximations, and the mixed rates propagate to target-specific band widths. In econometric terms the targets are counterfactual distributions and quantiles [11]; identification from discretely sampled continuous-time dynamics connects to the aliasing literature (Remark 4.1).

The paper is organised as follows. Section 2 states the model and Gaussian target law. Section 3 proves the intervention semantics. Section 4 treats identification. Section 5 develops the statistical experiment. Section 6 gives uniform inference and bootstrap procedures. Section 7 treats operator response and total error. Section 8 gives the impossibility boundary. Section 10 reports numerical validation. Complete proofs of every statement are collected in Section 9 (parts S0–S8), in dependency order and preceded by part S0, which restates every imported external theorem as an attributed local proposition whose hypotheses are verified at each point of use.

Notation

$H = L^2(D)$ with inner product $\langle\cdot,\cdot\rangle$ and norm $\|\cdot\|$; A_0 is self-adjoint, negative, with compact resolvent, eigenpairs $(-\lambda_k, e_k)$, $0 < \lambda_1 \le \lambda_2 \le \cdots$, and $\lambda_k \asymp k^{2/d}$ (Weyl growth). $\Lambda_N := \sum_{k\le N} \lambda_k$. The parameter is $\vartheta = (\theta, \sigma)$, its true value ϑ_0, and $A_{N,n} := \mathrm{diag}(\sqrt{\Lambda_N}, \sqrt{Nn})$ is the information normalisation. $\mathrm{do}(\theta = \tilde{a})$ denotes module surgery on the generator coefficient. Weak convergence is $\Rightarrow$; $o_{\mathbb{P}}(1), O_{\mathbb{P}}(1)$ are stochastic order symbols; C, c denote finite positive constants that may change between displays and whose dependencies are stated where they matter; $a \wedge b = \min(a,b)$, $a \vee b = \max(a,b)$, and $x_+ = \max(x, 0)$. For a compact index set $\mathcal{T}$, $C(\mathcal{T})$ carries the supremum norm $\|\cdot\|_\infty$.

2 Model and counterfactual targets

Let $D \subset \mathbb{R}^d$ be bounded and regular, and let $H = L^2(D)$. Let $A_0 : D(A_0) \subset H \to H$ be densely defined, self-adjoint, negative, and have compact resolvent. There is an orthonormal basis $(e_k)_{k\ge1}$ and eigenvalues $0 < \lambda_1 \le \lambda_2 \le \cdots$ such that

$$A_0 e_k = -\lambda_k e_k, \qquad \lambda_k \asymp k^{2/d}.$$

Let $W_t = \sum_{k\ge1} \beta_k(t) e_k$ be cylindrical Wiener noise and let $Q^{1/2} e_k = q_k e_k$.

Assumption 2.1 (Positive core)**.** *The factual parameter belongs to a compact rectangle*

$$\vartheta = (\theta, \sigma) \in [\underline{\theta}, \overline{\theta}] \times [\underline{\sigma}, \overline{\sigma}] \Subset (0,\infty)^2.$$

The intervention set $\widetilde{\Theta}$ is a compact subset of $(0,\infty)$. The initial state is deterministic. At minimum,

$$\sum_{k\ge1} \frac{q_k^2}{\lambda_k} < \infty.$$

For pathwise Galerkin rates we impose the stronger fractional summability stated in Section 3.

The factual mild solution is

$$U_t^\theta = e^{t\theta A_0} u_0 + \sigma \int_0^t e^{(t-s)\theta A_0} Q^{1/2} dW_s.$$

The intervention replaces only the generator coefficient:

$$dU_t^a = a A_0 U_t^a \, dt + \sigma Q^{1/2} dW_t, \qquad U_0^a = u_0, \qquad a \in \widetilde{\Theta}. \tag{3}$$

All fields in (3) are constructed with the same u_0, Q, σ, and W.

For $w \in L^2(0,T;H)$ define

$$\Psi_w(U) = \int_0^T \langle w_t, U_t\rangle \, dt.$$

Write $w_k(t) = \langle w_t, e_k\rangle$ and $u_{0,k} = \langle u_0, e_k\rangle$. The law of $\Psi_w(U^a)$ is obtained in three displayed steps. Modally, $u_k^a(t) = e^{-a\lambda_k t} u_{0,k} + \sigma q_k \int_0^t e^{-a\lambda_k(t-s)} d\beta_k(s)$, so

$$\Psi_w(U^a) = \sum_{k\ge1} \int_0^T w_k(t) u_k^a(t)\, dt = m_a(w) + \sigma \sum_{k\ge1} q_k \int_0^T w_k(t) \int_0^t e^{-a\lambda_k(t-s)}\, d\beta_k(s)\, dt.$$

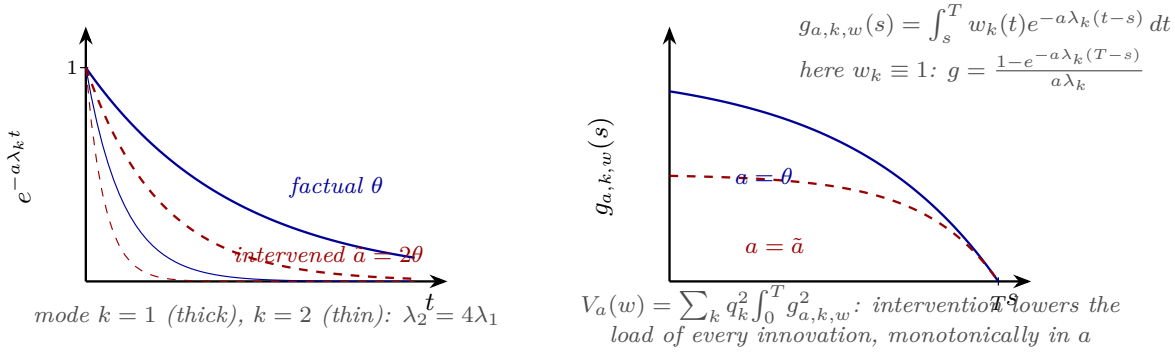


Figure 1: The two ingredients of the post-intervention law. Left: modal decay $e^{-a\lambda_k t}$ under the factual coefficient and under $\mathrm{do}(\theta = \tilde{a})$, for a slow and a fast mode. Right: the backward transfer kernel $g_{a,k,w}$ collecting, for each innovation time s, the future weight it loads; by Itô isometry the target variance is $\sigma^2 \sum_k q_k^2 \, \|g_{a,k,w}\|_{L^2}^2$.

The stochastic Fubini interchange (Part S0, imported stochastic Fubini theorem) applies because

$$\int_0^T \Big(\mathbb{E} \int_0^T w_k(t)^2 e^{-2a\lambda_k (t-s)} \mathbf{1}_{s \le t}\, dt \Big)^{1/2} ds \le \sqrt{T}\, \|w_k\|_{L^2} < \infty \qquad \text{for each } k,$$

and exchanging the two integrals collects, for each innovation time s, the total future weight it loads:

$$\int_0^T w_k(t) \int_0^t e^{-a\lambda_k (t-s)} d\beta_k(s)\, dt = \int_0^T \underbrace{\Big(\int_s^T w_k(t) e^{-a\lambda_k (t-s)}\, dt \Big)}_{= g_{a,k,w}(s)} d\beta_k(s).$$

Hence

$$\Psi_w(U^a) = m_a(w) + \sigma \sum_{k \ge 1} q_k \int_0^T g_{a,k,w}(s)\, d\beta_k(s),$$

where

$$m_a(w) = \sum_{k \ge 1} u_{0,k} \int_0^T w_k(t) e^{-a\lambda_k t}\, dt, \tag{4}$$

$$g_{a,k,w}(s) = \int_s^T w_k(t) e^{-a\lambda_k (t-s)}\, dt. \tag{5}$$

Proposition 2.1 (Gaussian counterfactual law)**.** *Under Assumption 2.1,*

$$\Psi_w(U^a) \sim N\big(m_a(w), \sigma^2 V_a(w)\big),$$

where

$$V_a(w) = \sum_{k \ge 1} q_k^2 \int_0^T g_{a,k,w}(s)^2\, ds. \tag{6}$$

Consequently,

$$F_{a,w}(z; \sigma) = \Phi\left(\frac{z - m_a(w)}{\sigma \sqrt{V_a(w)}} \right).$$

Interpretation. The intervention law is explicit once the invariant structural modules are fixed. The factual drift parameter θ is absent from the fixed absolute-intervention law because it has been overwritten.

The indexed target set is

$$\mathcal{T} = \widetilde{\Theta} \times \mathcal{W} \times \mathcal{Z},$$

where $\mathcal{Z}$ is compact and $\mathcal{W}$ is a compact ellipsoid in a space-time Sobolev basis. For CDF and quantile inference we impose

$$\inf_{(a,w) \in \widetilde{\Theta} \times \mathcal{W}} V_a(w) > 0.$$

Wasserstein and field-functional convergence do not require this non-degeneracy condition.

3 Galerkin structural intervention semantics

Let $H_K = \operatorname{span}\{e_1, \dots, e_K\}$ and let P_K be the orthogonal projection. Fix a partition $0 = t_0 < \cdots < t_J = T$. For mode k and interval j, use the Brownian segment

$$E_{k,j}(r) = \beta_k(t_j + r) - \beta_k(t_j), \qquad 0 \le r \le \Delta_j,$$

as the primitive exogenous variable.

For $a > 0$, define the exact local map

$$G_{a,k,r}(x, e) = e^{-a\lambda_k r} x + \sigma q_k \mathfrak{I}_{a\lambda_k, r}(e),$$

where

$$\mathfrak{I}_{c,r}(e) = e(r) - c \int_0^r e^{-c(r-s)} e(s)\, ds. \tag{7}$$

The map in (7) is continuous on $C_0([0, r])$. For Brownian e it equals the Itô convolution

$$\int_0^r e^{-c(r-s)} de(s)$$

by integration by parts. Thus the structural equation is a measurable, indeed continuous, function of the primitive Brownian segment.

The finite model recursively applies $G_{a,k,r}$ to every mode and interval. The intervention $\mathrm{do}(\theta = a)$ replaces the family $G_{\theta,k,r}$ by $G_{a,k,r}$ while preserving all exogenous segments.

Proposition 3.1 (Exact structural representation)**.** *For every K, partition, and $a \in \widetilde{\Theta}$, the intervened structural recursion equals the exact projected post-intervention SPDE path:*

$$\mathcal{X}_t^{K,\mathrm{do}(\theta=a)} = U_t^{K,a} = P_K U_t^a \quad \text{for all } t \in [0, T] \quad \text{almost surely.}$$

Theorem 3.1 (Galerkin limits of intervened SCMs)**.** *Assume, for some $\alpha \ge 0$ and $\delta > 0$,*

$$u_0 \in D((-A_0)^\alpha), \qquad \sum_{k \ge 1} q_k^2 \lambda_k^{2(\alpha+\delta)-1} < \infty.$$

Then

$$\sup_{a \in \widetilde{\Theta}} \mathbb{E} \sup_{t \le T} \|U_t^a - P_K U_t^a\|_H^2 \le C \lambda_{K+1}^{-2\alpha}.$$

Under only $u_0 \in H$ and $\sum q_k^2/\lambda_k < \infty$,

$$\sup_a \mathbb{E} \int_0^T \|U_t^a - P_K U_t^a\|^2 \, dt \to 0.$$

Finally, for every bounded $\mathcal{W} \subset L^2(0, T; H)$,

$$\sup_{a,w} \mathbb{E}\, |\Psi_w(U^a) - \Psi_w(P_K U^a)|^2 \to 0,$$

and the corresponding laws converge uniformly in quadratic Wasserstein distance.

Interpretation. The post-intervention SPDE is not introduced by notation alone. It is the limit of finite structural models whose operator mechanism is surgically replaced while the primitive shocks are fixed. Figure 2 displays the two-world construction.

Remark 3.1 (Relation to finite-dimensional intervention limits). For SDEs, Sokol and Hansen [33] obtain the post-intervention equation as the limit of intervened Euler-scheme structural models; ODE precursors are Mooij et al. [26], Rubenstein et al. [31]. Two features distinguish the present construction: the finite recursion is *exact* (no scheme error enters the semantics, cf. the transfer principle closing this section), and the limit is taken in the spatial resolution K rather than in a time step, which is what makes the common-noise coupling compatible with the spectral statistical experiment of Section 5.

Remark 3.2 (Regularity margin). The boundary condition

$$\sum_k q_k^2 \lambda_k^{2\alpha-1} < \infty$$

controls fixed-time fractional regularity but does not by itself justify the stated supremum-in-time L^2 maximal estimate. The positive margin δ is a clean factorisation condition in the sense of the factorisation method of Da Prato et al. [15]. For one-dimensional space-time white noise it implies $\alpha + \delta < 1/4$.

The exact recursion has no temporal approximation error. Such an error appears only when $G_{a,k,r}$ is replaced by an Euler or other numerical map. This distinction becomes important in Section 7.

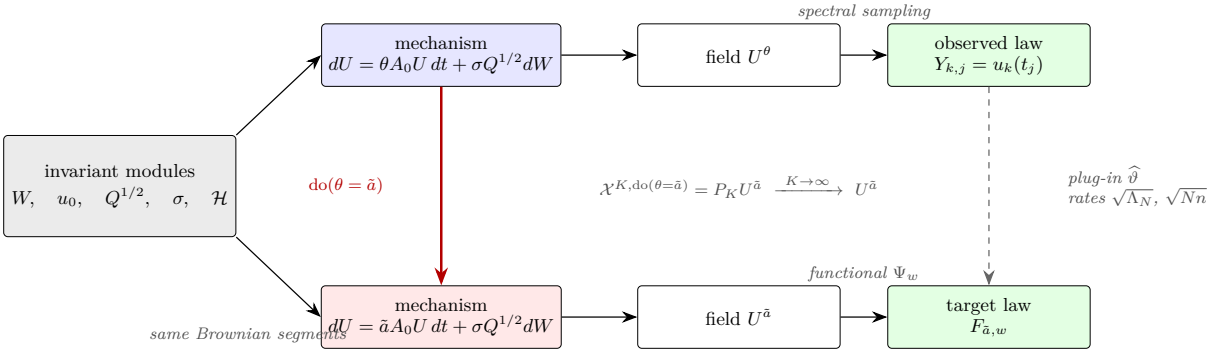


Figure 2: Common-noise structural semantics. The invariant modules (left) feed both worlds; $\mathrm{do}(\theta = \tilde{a})$ replaces only the generator mechanism. The observed spectral law identifies $\widehat{\vartheta}$, which is transported to the target law along the dashed edge at the mixed rates of Section 5.

4 Target identification

Identification is a property of the target, observation experiment, and admissible mechanism class. Let P^Y_ϑ denote the factual observed law and let $T_\tau(\vartheta)$ denote a counterfactual target.

Lemma 4.1 (Fibre criterion)**.** *The target T_τ is identified from P^Y_ϑ if and only if*

$$P^Y_{\vartheta_1} = P^Y_{\vartheta_2} \quad \Longrightarrow \quad T_\tau(\vartheta_1) = T_\tau(\vartheta_2)$$

for every pair ϑ_1, ϑ_2 in the admissible model.

This elementary observation is stated as a lemma because its value is organisational rather than technical: it separates necessary-and-sufficient logic from convenient sufficient conditions. We now specialise it to spectral observations.

Let $t_j = j\Delta$, and observe

$$Y_{k,j} = u_k(t_j), \qquad k = 1, \dots, N.$$

For each mode,

$$Y_{k,j+1} \mid Y_{k,j} \sim N\left(\rho_k(\theta) Y_{k,j}, \eta_k^2(\theta,\sigma)\right), \tag{8}$$

where

$$\rho_k(\theta) = e^{-\theta\lambda_k\Delta}, \qquad \eta_k^2(\theta,\sigma) = \sigma^2 q_k^2 \frac{1 - e^{-2\theta\lambda_k\Delta}}{2\theta\lambda_k}.$$

Theorem 4.1 (Exact finite-sample spectral identification)**.** *Suppose an observed mode satisfies $q_k > 0$ and is observed at $0, \Delta, 2\Delta$. Equality of its finite-dimensional law identifies*

$$\rho_k = \frac{\mathrm{Cov}(Y_{k,1}, Y_{k,2})}{\mathrm{Var}(Y_{k,1})}, \qquad \theta = -\frac{\log \rho_k}{\lambda_k \Delta},$$

and then identifies σ from $\mathrm{Var}(Y_{k,1})$. Hence (θ, σ) and every fixed-intervention target are identified.

With only one positive-time observation, full identification still holds if an observed initial coordinate is non-zero or if at least two distinct eigenvalues are observed. If all observed initial coordinates vanish and only one repeated eigenvalue is present, the data identify the composite scale $\sigma^2 h_\lambda(\theta)$ rather than the two parameters separately.

Remark 4.1 (No aliasing). Identification of θ from $\rho_k = e^{-\theta\lambda_k\Delta}$ exploits positivity and reality of the semigroup spectrum: on the positive core, the discrete transition determines the continuous-time generator uniquely. This contrasts with the aliasing problem for general finite-parameter continuous-time systems observed at equally spaced times [20, 30], in which distinct generators are compatible with the same discrete-time law.

For a fixed absolute intervention, the law in Proposition 2.1 depends on σ but not on the overwritten factual θ. Thus a non-degenerate fixed-intervention target can remain identified even when the full factual parameter is not.

For noisy observations

$$\widetilde{Y}_{k,j} = Y_{k,j} + \varepsilon_{k,j},$$

a known error law with non-vanishing characteristic function identifies the latent finite-dimensional law by deconvolution. In the Gaussian state-space model, two positive observation times identify (θ, σ) when the measurement variance is known; three positive times identify an unknown constant measurement variance as well.

For a general linear sensor $Y = \mathcal{H}U$, the mean and covariance must be treated separately. Let $\mathfrak{M}_\theta$ map an initial direction to its observed mean trajectory and let $\mathfrak{K}_\theta$ map a covariance input to the observed covariance kernel. Let $\mathfrak{L}_{a,w}$ and $\mathfrak{V}_{a,w}$ be the corresponding target maps.

Proposition 4.1 (Partial-observation kernel criterion)**.** *A Gaussian target is identified on a linear nuisance class if and only if*

$$\ker \mathfrak{M}_\theta \subseteq \ker \mathfrak{L}_{a,w} \quad \textit{and} \quad \ker \mathfrak{K}_\theta \subseteq \ker \mathfrak{V}_{a,w}.$$

Ordinary state observability controls the mean component. Covariance identification is a lifted operator problem and does not follow automatically from state observability.

A simple modal support condition becomes necessary only in an enlarged model with free mode-specific amplitudes. In the scalar positive core, known Q links unobserved modes through the common σ, so observing every target mode is not necessary.

5 Exact likelihood, mixed-rate LAN, and efficiency

Let $\mathcal{E}_{N,n}$ be the experiment generated by the transitions in (8). Conditional on the initial values, the log-likelihood is

$$\ell_{N,n}(\theta, \sigma) = -\frac{1}{2} \sum_{k=1}^{N} \sum_{j=0}^{n-1} \left\{ \log \eta_{k,n}^2 + \frac{(Y_{k,j+1} - \rho_{k,n} Y_{k,j})^2}{\eta_{k,n}^2} \right\} + C.$$

The scale parameter can be profiled explicitly for every θ, leaving a one-dimensional compact optimisation.

Set

$$\Lambda_N = \sum_{k=1}^{N} \lambda_k, \qquad A_{N,n} = \operatorname{diag}(\sqrt{\Lambda_N}, \sqrt{Nn}).$$

Assumption 5.1 (Joint spectral and temporal regime)**.** *As $N, n \to \infty$,*

$$\lambda_N \Delta_n \to 0, \qquad \frac{\lambda_N}{\Lambda_N} \to 0,$$

and the deterministic initial energy satisfies

$$\sum_{k \le N} \frac{\lambda_k u_{0,k}^2}{q_k^2} = o(\Lambda_N).$$

Theorem 5.1 (Mixed-rate LAN)**.** *Under Assumptions 2.1 and 5.1, for every bounded $h \in \mathbb{R}^2$,*

$$\ell_{N,n}(\vartheta_0 + A_{N,n}^{-1} h) - \ell_{N,n}(\vartheta_0) = h^\top \Delta_{N,n} - \frac{1}{2} h^\top J(\vartheta_0) h + o_{\mathbb{P}}(1),$$

uniformly on compact h-sets, where

$$\Delta_{N,n} \Rightarrow N(0, J(\vartheta_0)), \qquad J(\vartheta_0) = \begin{pmatrix} T/(2\theta_0) & 0 \\ 0 & 2/\sigma_0^2 \end{pmatrix}.$$

The exact conditional MLE is consistent, asymptotically linear, and efficient:

$$\begin{pmatrix} \sqrt{\Lambda_N}(\widehat{\theta} - \theta_0) \\ \sqrt{Nn}(\widehat{\sigma} - \sigma_0) \end{pmatrix} \Rightarrow N\left(0, \begin{pmatrix} 2\theta_0/T & 0 \\ 0 & \sigma_0^2/2 \end{pmatrix}\right).$$

Polynomial likelihood-ratio bounds imply convergence of polynomial moments.

Interpretation. Spatial replication supplies the drift information that fixed-horizon high-frequency sampling of one diffusion path cannot. The scale accumulates information from every transition and is learned faster.

For the Dirichlet heat equation on $(0, 1)$,

$$\lambda_k = \pi^2 k^2, \qquad \Lambda_N \sim \frac{\pi^2}{3} N^3.$$

The temporal condition is $N^2/n \to 0$, and

$$N^{3/2}(\widehat{\theta} - \theta_0) \Rightarrow N\left(0, \frac{6\theta_0}{\pi^2 T}\right).$$

Remark 5.1 (Where the drift information comes from). The rate $\sqrt{\Lambda_N}$ has a two-line derivation. One transition of mode k carries drift information $(\partial_\theta \rho_k)^2 \mathbb{E} Y_{k,j}^2 / \eta_k^2 \approx (\lambda_k \Delta)^2 \cdot \frac{\sigma^2 q_k^2}{2\theta\lambda_k} \cdot \frac{1}{\sigma^2 q_k^2 \Delta} = \frac{\lambda_k \Delta}{2\theta}$ at stationarity; over n transitions this is $\lambda_k T/(2\theta)$ per mode — *finite*, because a fixed-horizon path of one mode cannot

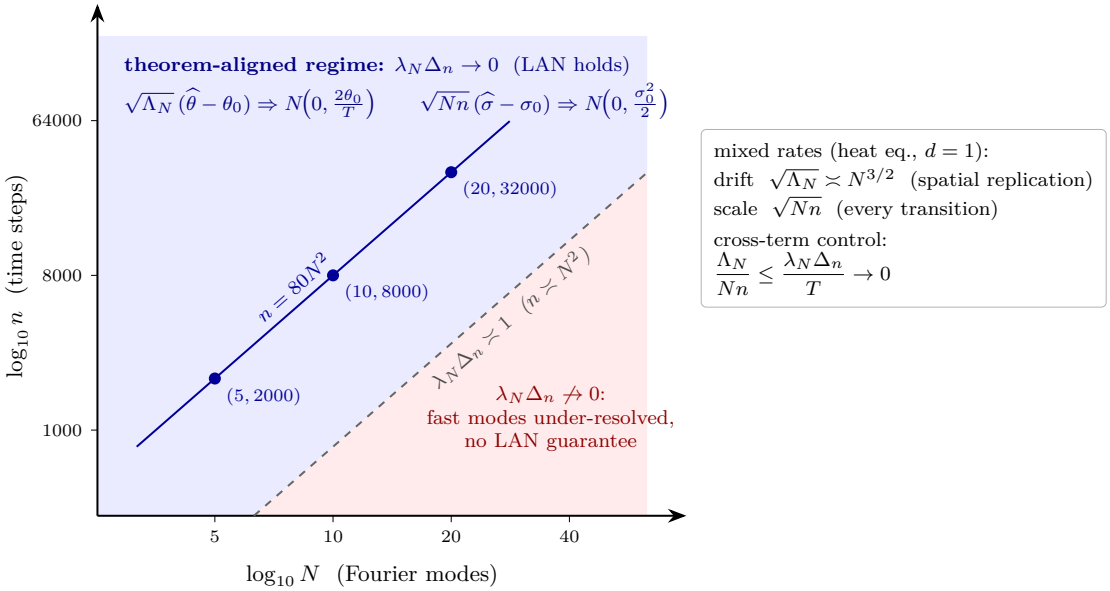


Figure 3: Theorem-aligned designs for the Dirichlet heat equation. The publication designs sit on $n = 80N^2$, inside the regime $\lambda_N\Delta_n \to 0$ where Theorem 5.1 applies; drift and scale are learned at the anisotropic rates shown.

identify θ — and summing across observed modes gives $\Lambda_N T/(2\theta)$. Spatial replication, not time, drives drift information; the scale σ, in contrast, collects $2/\sigma^2$ from every one of the Nn standardised innovations. This is the entire content of the limit $J(\vartheta_0)$ in Theorem 5.1; the proof in part S3 controls the remainder of this arithmetic uniformly.

Remark 5.2 (Rate separation within the joint regime). The normalised drift–scale cross-information vanishes at rate $\sqrt{\Lambda_N/(Nn)} \le \sqrt{\lambda_N\Delta_n/T}$, since $\Lambda_N \le N\lambda_N$ and $n\Delta_n = T$; asymptotic orthogonality of the two coordinates is therefore a consequence of resolving the fastest observed relaxation time, not an additional assumption. Complete proofs of Theorem 5.1 — the score CLT, the uniform quadratic remainder, consistency via exact scale profiling, and a polynomial likelihood-ratio bound yielding moment convergence — are given in Part S3 (Section 9.4).

Continuous spectral observation is not used as the joint regular experiment because quadratic variation reveals σ exactly on a continuously observed path. The discrete experiment retains a regular two-parameter likelihood.

Remark 5.3 (Rate comparison with continuous observation). Under continuous observation of the first N modes on $[0, T]$, the drift MLE attains the same spectral rate $\sqrt{\Lambda_N}$ [22]; see Cialenco [13], Lototsky [25]. The contribution of discreteness here is not a different drift rate but a *regular joint* experiment: σ becomes a genuine parameter learned at $\sqrt{Nn}$, and the LAN structure with diagonal normalisation transfers to every post-intervention functional in Section 6.

6 Uniform counterfactual inference and multiplier bands

Let $T_\tau(\vartheta)$ denote a CDF, quantile, tail, or effect target. The map

$$\vartheta \mapsto T(\vartheta) \in C(\mathcal{T})$$

is uniformly Fréchet differentiable on the compact target class. Because the estimated factual model has two parameters, the limiting processes are finite rank.

For a fixed absolute-intervention CDF,

$$F^{\mathrm{cf}}_{a,w,z}(\sigma) = \Phi\left(\frac{z - m_a(w)}{\sigma\sqrt{V_a(w)}}\right),$$

and

$$d^{\mathrm{cf}}_\sigma(a, w, z) = -\frac{c_{a,w,z}\,\phi(c_{a,w,z})}{\sigma_0}.$$

Theorem 6.1 (Uniform mixed-rate target processes). *Under Assumptions 2.1 and 5.1 and compactness of $\mathcal{T}$,*

$$\sqrt{Nn}\left(\widehat{F}^{\mathrm{cf}} - F^{\mathrm{cf}}\right) \Rightarrow \mathbb{G}_{\mathrm{cf}} \quad \text{in } C(\mathcal{T}),$$

where

$$\mathbb{G}_{\mathrm{cf}}(a,w,z) = d_\sigma^{\mathrm{cf}}(a,w,z)Z_\sigma, \qquad Z_\sigma \sim N(0,\sigma_0^2/2).$$

Factual CDFs satisfy

$$\sqrt{\Lambda_N}(\widehat{F}^{\mathrm{fact}} - F^{\mathrm{fact}}) \Rightarrow d_\theta^{\mathrm{fact}}(w,z)Z_\theta, \qquad Z_\theta \sim N(0, 2\theta_0/T).$$

For the absolute-intervention distributional effect

$$\Delta_{a,w,z} = F^{\mathrm{cf}}_{a,w,z} - F^{\mathrm{fact}}_{w,z},$$

the slow first-order limit is

$$\sqrt{\Lambda_N}(\widehat{\Delta} - \Delta) \Rightarrow -d_\theta^{\mathrm{fact}}(w,z)Z_\theta.$$

Theorem 6.2 (Uniform quantile and tail processes)**.** *Let $\mathcal{P} \Subset (0,1)$ be compact and let $Q_{a,w}(p) = m_a(w) + \sigma\sqrt{V_a(w)}\,\Phi^{-1}(p)$ be the fixed-intervention quantile surface. Under the conditions of Theorem 6.1,*

$$\sqrt{Nn}\big(\widehat{Q} - Q\big) \Rightarrow \sqrt{V_a(w)}\,\Phi^{-1}(p)\,Z_\sigma \qquad \text{in } C(\widetilde{\Theta} \times \mathcal{W} \times \mathcal{P}),$$

and the fixed-intervention tail process $\sqrt{Nn}(\widehat{S} - S)$, $S := 1 - F^{\mathrm{cf}}$, converges to $-\mathbb{G}_{\mathrm{cf}}$. Factual and distributional-effect versions hold at rate $\sqrt{\Lambda_N}$ with the corresponding θ-loadings. Proofs are in Part S4 (Section 9.5).

Remark 6.1 (Counterfactual-distribution parallel). The objects of Theorems 6.1 and 6.2 are the dynamic-field analogues of counterfactual distribution and quantile processes in microeconometrics [11]. The structural difference is the limit geometry: a two-parameter factual experiment makes every target process finite rank, so simultaneous inference reduces to a two-dimensional Gaussian draw (Theorem 6.3) rather than to empirical-process approximations over Donsker classes.

The slow effect limit is constant in the absolute intervention coordinate because the counterfactual part is learned at the faster scale rate. A relative intervention $a = \eta\theta$ yields intervention-indexed slow uncertainty when that is scientifically appropriate.

For feasible inference, let $S_{k,N,n}(\widehat{\vartheta})$ be the complete modal score block and set

$$\widehat{B}_k = A_{N,n}^{-1} S_{k,N,n}(\widehat{\vartheta}), \qquad \widehat{J}_B = \sum_{k=1}^{N} \widehat{B}_k \widehat{B}_k^\top.$$

With independent Gaussian multipliers ξ_k define

$$H^* = \widehat{J}_B^{-1} \sum_{k=1}^{N} \xi_k \widehat{B}_k.$$

Conditionally on the data,

$$H^* \sim N(0, \widehat{J}_B^{-1}).$$

Thus the score multiplier is equivalent to a direct two-dimensional Gaussian draw with estimated inverse-information covariance.

Theorem 6.3 (Simultaneous confidence bands)**.** *The multiplier target processes obtained by multiplying the estimated derivative surfaces by the corresponding coordinates of H^* converge conditionally to the processes in Theorem 6.1. Conditional sup-norm quantiles yield asymptotically valid simultaneous bands.*

At derivative zeros, define

$$\widehat{s}_\kappa(\tau) = \max\{\widehat{s}(\tau), \kappa_{N,n}\,\|\widehat{s}\|_\infty\}.$$

For fast targets require

$$\kappa_{N,n} \downarrow 0, \qquad \sqrt{Nn}\,\kappa_{N,n} \to \infty.$$

For slow targets also require

$$\sqrt{\Lambda_N}\,\kappa_{N,n} \to \infty, \qquad \frac{\sqrt{\Lambda_N}}{\sqrt{Nn}\,\kappa_{N,n}} \to 0.$$

The resulting floor-studentised bands remain valid uniformly, including at first-order derivative zeros.

Equality hypotheses use sup-norm or integrated-square statistics. For one-sided hypotheses $G(\tau) \le 0$, define the estimated contact set

$$\widehat{\mathcal{C}}_{N,n} = \left\{ \tau : \frac{r_{N,n}\widehat{G}(\tau)}{\widehat{s}_\kappa(\tau)} \ge -\eta_{N,n} \right\},$$

with $\eta_{N,n} \to \infty$ and $\eta_{N,n}/r_{N,n} \to 0$. Bootstrapping the positive-part supremum over this set gives asymptotically valid and consistent stochastic-dominance, tail-sign, and quantile-sign tests, following the logic of moment-selection procedures [7, 17, 24].

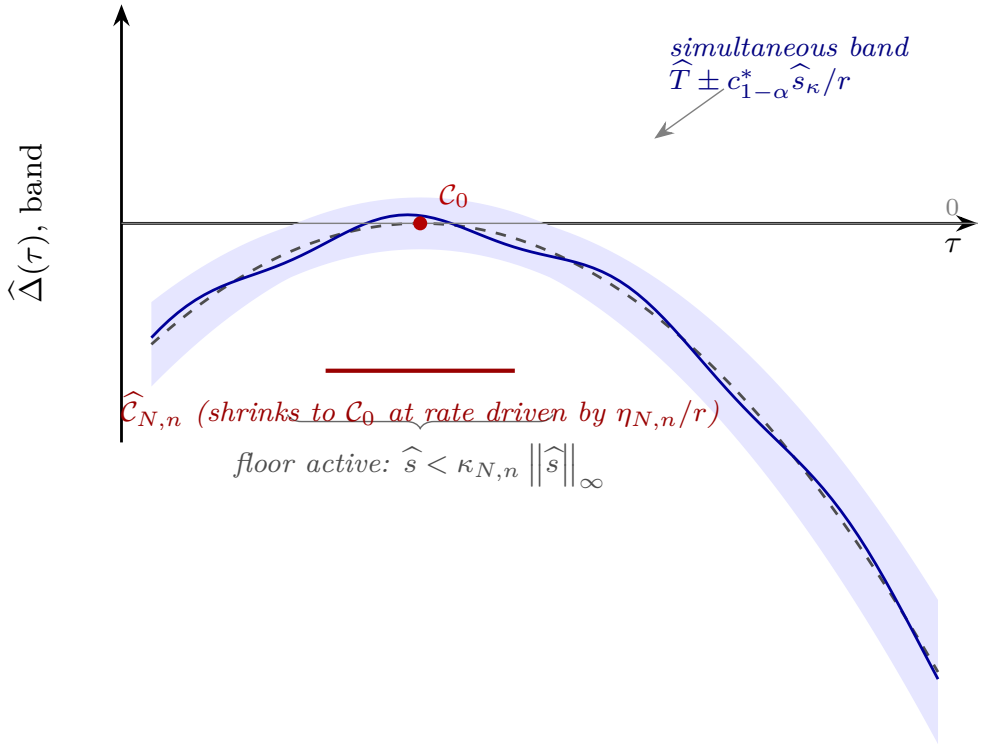


Figure 4: Schematic of the floor-studentised simultaneous band and the estimated contact set for a one-sided effect hypothesis. Near the derivative zero the floor $\kappa_{N,n} \|\widehat{s}\|_\infty$ keeps the band width above the quadratic remainder; the estimated contact set $\widehat{\mathcal{C}}_{N,n}$ retains the true contact set and shrinks onto it because $\eta_{N,n} \to \infty$ while $\eta_{N,n}/r_{N,n} \to 0$.

7 Operator sensitivity and total approximation error

The operator derivative must distinguish bounded perturbations from the paper's unbounded scalar-rescaling direction.

Let A generate a C_0-semigroup $S(t)$ and let $B \in \mathcal{L}(H)$. The bounded perturbation theorem [16, 28] gives

$$D_B S(t) = \int_0^t S(t-s) B S(s)\, ds.$$

Theorem 7.1 (Operator-to-law differentiability)**.** *For bounded B, the map $A \mapsto U^A$ is Fréchet differentiable in*

$$L^2(\Omega; L^2(0,T;H)).$$

The derivative is

$$\dot{U}_t[B] = D_B S(t) u_0 + \sigma \int_0^t D_B S(t-r) Q^{1/2} dW_r = \int_0^t S(t-s) B U_s\, ds.$$

For the Gaussian target,

$$D_A F_{A,w}(z)[B] = \phi(c) \left[-\frac{D_A m_A(w)[B]}{\sqrt{v_A(w)}} - \frac{z - m_A(w)}{2 v_A(w)^{3/2}} D_A v_A(w)[B] \right],$$

with a uniform quadratic remainder on bounded target classes. Moreover,

$$W_2\left(\mathcal{L}(\Psi_w(U^{A+\varepsilon B})), \mathcal{L}(\Psi_w(U^A))\right) \le C|\varepsilon|\, \|B\|\, \|w\|_{L^2}.$$

The primary intervention uses $A_a = aA_0$, whose direction A_0 is unbounded. Analytic-semigroup calculus instead gives

$$\partial_a e^{taA_0} = tA_0 e^{taA_0}, \qquad \partial_a^2 e^{taA_0} = t^2 A_0^2 e^{taA_0},$$

with operator-norm remainder bounds uniform for a in compact positive intervals.

We next combine the inferential and numerical layers. Assume

$$q_k^2 \lesssim \lambda_k^{-\beta}, \qquad u_0 \in D((-A_0)^{r_0/2}),$$

and let the weight ellipsoid have spatial smoothness s. Define

$$\mu_m = \frac{r_0 + 1 + 2s}{2}, \qquad \mu_V = \beta + 2 + 2s.$$

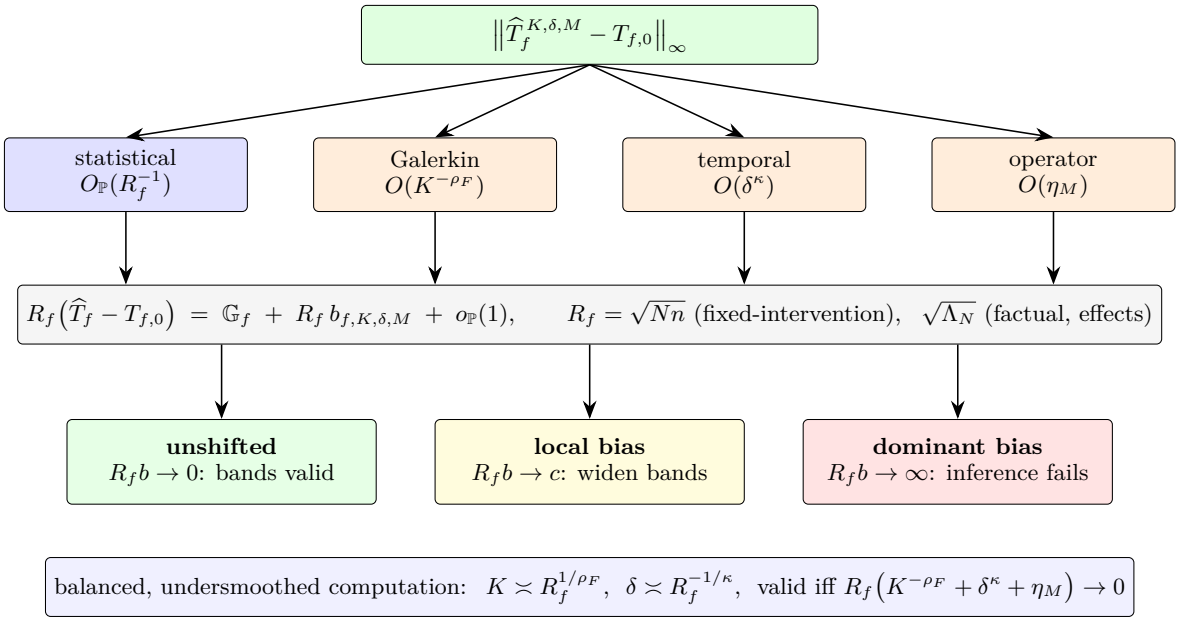


Figure 5: The error budget of Theorem 7.2 at the inferential scale: statistical error at the family rate R_f, three numerical error sources, and the resulting bias trichotomy with the balanced allocation rule.

Then

$$\sup_{a,w} |m_a(w) - m_{a,K}(w)| \lesssim \lambda_{K+1}^{-\mu_m},$$

and

$$\sup_{a,w} |V_a(w) - V_{a,K}(w)| \lesssim \lambda_{K+1}^{-\mu_V}.$$

Under a uniform positive variance lower bound, Gaussian law errors depend linearly on the variance tail, not on its square root. Under Weyl growth the exponent of (9) below follows in two lines: the Gaussian-law error is Lipschitz in (m, V) under the variance floor, so the uniform CDF error is at most $C(\lambda_{K+1}^{-\mu_m} + \lambda_{K+1}^{-\mu_V})$; inserting $\lambda_{K+1} \asymp K^{2/d}$ turns the spectral exponents (μ_m, μ_V) into the truncation exponents $(2\mu_m/d,\, 2\mu_V/d)$, of which the smaller one binds:

$$\rho_F = \min\left\{\frac{2\mu_m}{d}, \frac{2\mu_V}{d}\right\} = \min\left\{\frac{r_0 + 1 + 2s}{d}, \frac{2(\beta + 2 + 2s)}{d}\right\}. \tag{9}$$

Let δ be the functional quadrature step and let η_M bound a bounded operator approximation. Let κ be the proven temporal exponent for the declared implementation.

Theorem 7.2 (Full statistical–numerical error)**.** *For a target family f with statistical rate*

$$R_f = \begin{cases} \sqrt{Nn}, & \textit{fixed-intervention targets}, \\ \sqrt{\Lambda_N}, & \textit{factual and effect targets}, \end{cases}$$

the implemented estimator satisfies

$$\left\|\widehat{T}_f^{K,\delta,M} - T_{f,0}\right\|_\infty = O_{\mathbb{P}}(R_f^{-1}) + O\left(K^{-\rho_F} + \delta^\kappa + \eta_M\right).$$

More precisely,

$$R_f(\widehat{T}_f^{K,\delta,M} - T_{f,0}) = \mathbb{G}_f + R_f b_{f,K,\delta,M} + o_{\mathbb{P}}(1).$$

Thus ordinary bands remain valid under

$$R_f\left(K^{-\rho_F} + \delta^\kappa + \eta_M\right) \to 0.$$

A certified deterministic approximation bound can instead be added to the statistical half-width. A first-order operator-response correction replaces η_M by η_M^2.

The theorem distinguishes an unshifted regime, a local-bias regime with a shifted Gaussian process, and a dominant-bias regime in which naive inference fails. Balanced computation uses

$$K \asymp R_f^{1/\rho_F}, \qquad \delta \asymp R_f^{-1/\kappa},$$

with a slowly diverging factor for strict undersmoothing.

8 Sharp non-identification boundaries

The positive results rely on structural invariance, not merely analytic regularity. We now show what fails when key modules are unrestricted.

Let a general mechanism have drift

$$AU_t + G(t, U_t).$$

For bounded L, define

$$\mathcal{T}_L(A, G) = (A - L, G + L\,\cdot).$$

Theorem 8.1 (Operator–forcing gauge)**.** *The mechanisms (A, G) and $(A - L, G + L\,\cdot)$ generate the same factual stochastic evolution equation and hence the same factual path under common exogenous noise. Under a common operator intervention $A \mapsto \widetilde{A}$, their post-intervention drifts differ by Lu. Unless the selected target annihilates this direction, the post-intervention laws differ.*

This is not merely an adapted-feedback pathology. Let $a_1 \neq a_2$, let $h(0) = 0$, and define an open-loop input

$$F_i(t) = Z(h'(t) + a_i h(t)),$$

where Z is generated ex ante. Both equations

$$dX_t = -a_i X_t dt + F_i(t)dt$$

have the same complete factual path $X_t = Zh(t)$. After a common intervention to $\widetilde{a}$,

$$X_t^{i,do} = Z\left[h(t) + (a_i - \widetilde{a})\int_0^t e^{-\widetilde{a}(t-s)}h(s)ds\right],$$

so the target laws separate.

A positive-noise construction works on every finite observation grid. With a piecewise-constant Gaussian input, the observed vector is

$$Y = L_a Z + \varepsilon_a.$$

For a common covariance C, choosing

$$K_a = L_a^{-1}(C - C_a^W)L_a^{-\top}$$

produces the same factual $N(0, C)$ law under distinct operators, while a common post-intervention operator generally produces different covariance matrices.

Theorem 8.2 (Sharp boundary assembly)**.** *Each of the following enlarged model classes contains observationally equivalent factual mechanisms with different post-intervention target laws:*

- *(i) unrestricted bounded state-feedback forcing;*
- *(ii) unrestricted open-loop forcing law, even when independent of the Wiener noise;*
- *(iii) unknown random initial law under finite observation;*
- *(iv) rank-deficient sensing with a free target-relevant latent direction;*
- *(v) unknown calibration for field-scale targets;*
- *(vi) intervention-dependent measurement;*
- *(vii) non-identified state and measurement noise decomposition.*

Furthermore, identified marginal counterfactual laws do not identify a unit-level contrast without a cross-world coupling convention.

The qualifications are target specific. Unknown calibration can leave sensor-scale counterfactuals identified while field-scale targets fail. Partial sensing can be harmless when known cross-mode structure links latent and observed directions. Marginal CDF and quantile effects do not require identification of cross-world correlation.

The shared-Wiener construction of Section 3 selects a structural coupling for unit-level contrasts. Other couplings can preserve the same marginals and change

$$\mathrm{Var}(X^a - X^b) = v_a + v_b - 2\rho\sqrt{v_a v_b}.$$

This limits individual-effect claims but does not weaken the marginal distribution results.

freed module (nuisance class)	**factual law**	**fixed-interv. marginal $F_{a,w}$**	**distributional effect $\Delta_{a,w,z}$**	**unit-level contrast**
baseline: all modules invariant (Ass. 2.1, §4)	✓	✓	✓	✓†
state-feedback forcing $G + L \cdot$ (gauge, Thm 8.1)	✓	✗	✗	✗
open-loop forcing law (ex-ante Z)	✓	✗	✗	✗
initial law free, finite observation	✓	✗	✗	✗
rank-deficient sensing, free latent direction	✓	✗	✗	✗
rank-deficient sensing, known cross-mode link	✓	✓	✓	✓†
unknown calibration c (sensor $Y = cU$)	✓	∼	∼	✗
intervention-dependent measurement	✓	✗	✗	✗
cross-world coupling free	✓	✓	✓	✗

∼ target-specific: field-scale targets fail, sensor-scale targets remain identified (Thm 8.2(v)). † unit-level contrasts additionally require the cross-world coupling convention (shared Wiener path, §3); marginals never do.
Reading: each row frees one module of the SCM while preserving observational equivalence of the factual law (column 1 always ✓); columns record which post-intervention targets survive (§4, Thm 8.1–8.2).

Figure 6: Identification boundary map. Each row frees one structural module while preserving observational equivalence of the factual law; columns record which post-intervention targets survive (Theorems 8.1 and 8.2).

9 Proofs of the main results

This section proves every result of the paper. Part S0 first restates the external theorems we rely on as attributed local propositions, so that each later invocation is an internal reference with hypotheses verified at the point of use. Parts S1–S8 then follow the dependency order semantics → identification → likelihood → uniform processes → bootstrap → sensitivity → total error → non-identification, beginning, as the finite-dimensional layer must, with the Galerkin approximation. A statement that re-proves a result already announced carries the mark "establishes Theorem $x.y$" in its title; all other numbered statements are supporting lemmas. Notation is local to each part where it deviates from the main text. Figure 7 displays the dependency structure.

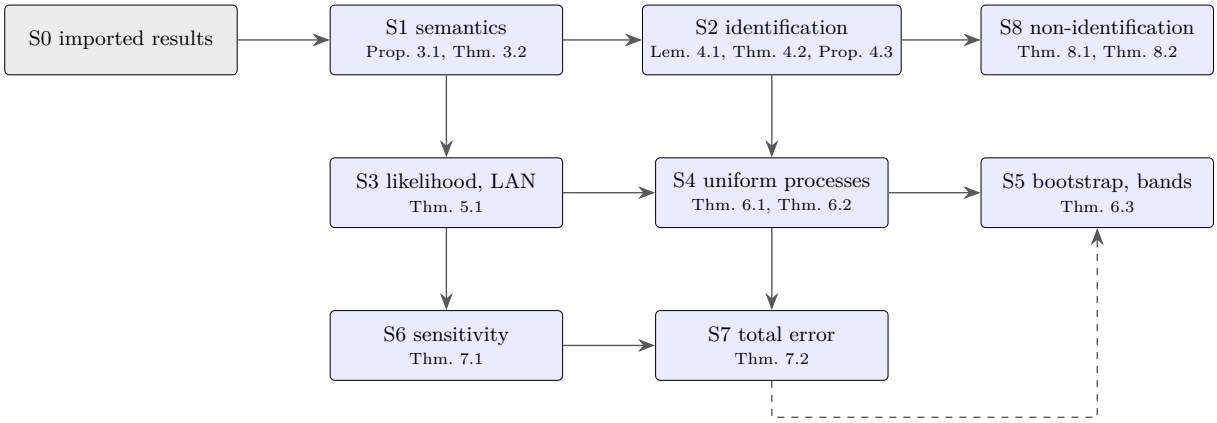


Figure 7: Dependency structure of the proofs. Solid arrows are proof-level dependencies; the dashed arrow is the bias-aware band construction feeding back from the total-error analysis into the inference machinery. Part tags list the main-text statements established in each part.

9.1 Part S0: Imported results

Every external theorem used in Parts S1–S8 is restated here with attribution in the title and hypotheses in the statement. At each point of use, the text verifies the hypotheses explicitly; the pointers below record where.

Proposition 9.1 (Corollary 3.1 of 19; martingale array CLT)**.** *For each (N, n) let $\{\xi_i, \mathcal{G}_i\}_{i \le m}$ be a square-integrable martingale-difference array (indices ordered arbitrarily but fixed), and let $\eta^2 \ge 0$ be a* constant*. If, as $N, n \to \infty$, (i) $\sum_i \mathbb{E}[\xi_i^2 \mid \mathcal{G}_{i-1}] \xrightarrow{\mathbb{P}} \eta^2$, and (ii) the conditional Lindeberg condition holds ($\sum_i \mathbb{E}[\xi_i^2 \mathbf{1}\{|\xi_i| > \epsilon\} \mid \mathcal{G}_{i-1}] \xrightarrow{\mathbb{P}} 0$ for every $\epsilon > 0$; implied by $\sum_i \mathbb{E}\xi_i^4 \to 0$), then $\sum_i \xi_i \Rightarrow \mathcal{N}(0, \eta^2)$. When the limit η^2 is random, the corollary additionally requires nested σ-fields and delivers stable convergence; the constant-limit version used here does not.* Used in: *Part S3 (score central limit theorem, via Cramér–Wold) and Part S5 (conditional multiplier central limit theorem).*

Proposition 9.2 (Theorem 1.1 of 32; Hanson–Wright)**.** *Let $X = (X_1, \dots, X_m)$ have independent centred components with $\max_i \|X_i\|_{\psi_2} \le K$, and let A be a deterministic $m \times m$ matrix. Then for every $t \ge 0$,*

$$\mathbb{P}\big(|X^\top A X - \mathbb{E}X^\top A X| > t\big) \le 2\exp\Big\{ - c \min\Big(\frac{t^2}{K^4 \|A\|_F^2},\ \frac{t}{K^2 \|A\|_{\mathrm{op}}} \Big) \Big\}.$$

Used in: *Part S3 (polynomial likelihood-ratio bound), where X collects the standardised Gaussian innovations (K an absolute constant) and A is the deterministic quadratic-form matrix of the centred log-likelihood field.*

Proposition 9.3 (Theorem 3.9.4 of 5; functional delta method)**.** *Let $\phi : \mathbb{D}_\phi \subseteq \mathbb{D} \to \mathbb{E}$ be Hadamard differentiable at θ tangentially to $\mathbb{D}_0$, with derivative ϕ'_θ. If $r_n(T_n - \theta) \Rightarrow T$ in $\mathbb{D}$ with T taking values in $\mathbb{D}_0$, then $r_n(\phi(T_n) - \phi(\theta)) \Rightarrow \phi'_\theta(T)$.* Used in: *Part S4 (uniform target processes; the target maps are uniformly Fréchet — hence Hadamard — differentiable by the Part S4 differentiability theorems, the domain is a finite-dimensional parameter neighbourhood, and weak convergence of the normalised estimator is supplied by Part S3).*

Proposition 9.4 (Theorem 1.11.1 of 5; extended continuous mapping)**.** *Let g_n, g be maps with the property that $x_n \to x$ (along the relevant domains) implies $g_n(x_n) \to g(x)$, with g continuous on a set containing the support of X. If $X_n \Rightarrow X$, then $g_n(X_n) \Rightarrow g(X)$. In particular this covers deterministic maps $g_n \to g$*

uniformly on compacts with g continuous. Used in: *Part S5 (floor-studentised bands, with $g_n = V_{\kappa_{N,n}}$ equi-Lipschitz and $g = V$).*

Proposition 9.5 (Theorem 4.33 of 3; stochastic Fubini)**.** *Let $(E, \mathcal{E}, \mu)$ be σ-finite, and let $\Phi : (t, \omega, x) \mapsto \Phi(t, x)$ be an appropriately measurable L_2^0-valued family with $\int_E \left(\mathbb{E} \int_0^T \|\Phi(t, x)\|_{L_2^0}^2 \, dt\right)^{1/2} \mu(dx) < \infty$. Then $\int_E \int_0^T \Phi \, dW \, d\mu = \int_0^T \int_E \Phi \, d\mu \, dW$ a.s.* Used in: *Parts S1–S2 (Gaussian law of Ψ_w, mild-solution manipulations); the required square-integrability is displayed at each use.*

Proposition 9.6 (Section 5.3 of 3, after 2; factorisation identity)**.** *For $\gamma \in (0, 1)$ and a stochastic convolution $Z_t = \int_0^t S(t-s)\Phi \, dW_s$ with the γ-integrability displayed in Part S1,*

$$Z_t = \frac{\sin(\pi\gamma)}{\pi} \int_0^t (t-s)^{\gamma-1} S(t-s) Y_\gamma(s) \, ds, \qquad Y_\gamma(s) := \int_0^s (s-r)^{-\gamma} S(s-r)\Phi \, dW_r.$$

Used in: *Part S1 (uniform fractional maximal estimate), where the L^p bound on Y_γ is proved before the identity is invoked.*

Proposition 9.7 (Theorem III.1.3 of 16; bounded perturbation)**.** *Let A generate a C_0-semigroup $T(t)$ with $\|T(t)\| \le Me^{\omega t}$ and let $B \in \mathcal{L}(X)$. Then $A+B$ (same domain) generates a C_0-semigroup $S(t)$ with $\|S(t)\| \le Me^{(\omega + M\|B\|)t}$, given by the Dyson–Phillips series $S(t) = \sum_{j \ge 0} S_j(t)$, $S_0 = T$, $S_{j+1}(t) = \int_0^t T(t-s) B S_j(s) \, ds$.* Used in: *Part S6 (uniform bounded-perturbation estimate; hypotheses verified there: exponentially stable analytic semigroup by the Part S6 standing assumption, εB bounded with $\|\varepsilon B\| \le \varepsilon_0 R$).*

Remark 9.1 (Elementary identities used without citation). The one-dimensional Gaussian optimal-coupling identity $W_2^2(\mathcal{N}(m_1, v_1), \mathcal{N}(m_2, v_2)) = (m_1 - m_2)^2 + (\sqrt{v_1} - \sqrt{v_2})^2$ (see, e.g., 4, Chapter 6) and the characteristic-function division argument for deconvolution are proved inline where used (Parts S7 and S2 respectively).

9.2 Part S1: Intervention semantics and Galerkin limits

9.2.1 Model, notation, and assumptions

Let $H = L^2(D)$, where $D \subset \mathbb{R}^d$ is bounded. Set

$$\Lambda := -A_0.$$

Thus Λ is positive and self-adjoint.

Assumption 9.1 (Spectral generator)**.** *The operator $A_0 : D(A_0) \subset H \to H$ is densely defined, self-adjoint, negative, and has compact resolvent. There are an orthonormal basis $(e_k)_{k \ge 1}$ and eigenvalues*

$$0 < \lambda_1 \le \lambda_2 \le \cdots, \qquad \lambda_k \uparrow \infty,$$

such that

$$A_0 e_k = -\lambda_k e_k, \qquad \Lambda e_k = \lambda_k e_k.$$

When a rate in K is translated into a dimension-dependent power, we additionally assume Weyl growth $\lambda_k \asymp k^{2/d}$.

Assumption 9.2 (Intervention and scale classes)**.** *The intervention set and noise-scale set are compact subsets of the positive half-line:*

$$\widetilde{\Theta} = [\underline{\widetilde{\theta}}, \overline{\widetilde{\theta}}] \Subset (0, \infty), \qquad \Sigma = [\underline{\sigma}, \overline{\sigma}] \Subset (0, \infty).$$

All constants below may depend on $T, \underline{\widetilde{\theta}}, \overline{\widetilde{\theta}}, \underline{\sigma}, \overline{\sigma}$, but not on K or on the particular $\widetilde{\theta} \in \widetilde{\Theta}$.

Let W be a cylindrical Wiener process on H, represented as

$$W_t = \sum_{k \ge 1} \beta_k(t) e_k,$$

where $(\beta_k)_{k \ge 1}$ are independent standard Brownian motions. Let

$$Q^{1/2} e_k = q_k e_k, \qquad q_k \ge 0.$$

For $\widetilde{\theta} \in \widetilde{\Theta}$, define

$$S_{\widetilde{\theta}}(t) := e^{t\widetilde{\theta} A_0} = e^{-t\widetilde{\theta}\Lambda}.$$

The post-intervention field is the mild solution

$$U_t^{\widetilde{\theta}} = S_{\widetilde{\theta}}(t) u_0 + \sigma \int_0^t S_{\widetilde{\theta}}(t-s) Q^{1/2} \, dW_s. \tag{10}$$

Its modal coordinates satisfy

$$u_k^{\widetilde{\theta}}(t) = e^{-\widetilde{\theta}\lambda_k t} u_{0,k} + \sigma q_k \int_0^t e^{-\widetilde{\theta}\lambda_k (t-s)}\, d\beta_k(s). \tag{11}$$

Assumption 9.3 (Baseline Hilbert-space summability)**.** *The initial state belongs to H, and*

$$\sum_{k\geq 1} \frac{q_k^2}{\lambda_k} < \infty.$$

Assumption 9.3 is the minimal condition used in the integrated convergence and field-functional results below. It is not, by itself, used to claim a maximal fractional-domain estimate.

Assumption 9.4 (Fractional maximal-regularity margin)**.** *There exist $\alpha > 0$ and $\delta \in (0, \frac{1}{2})$ such that*

$$u_0 \in D(\Lambda^\alpha), \qquad \sum_{k\geq 1} q_k^2 \lambda_k^{2(\alpha+\delta)-1} < \infty.$$

The positive margin δ is deliberate. The boundary condition $\sum_k q_k^2 \lambda_k^{2\alpha-1} < \infty$ controls fixed-time second moments in $D(\Lambda^\alpha)$, but it does not automatically provide the stated $L^2(\Omega; C([0,T]; D(\Lambda^\alpha)))$ maximal estimate. Assumption 9.4 is a clean sufficient condition obtained from the stochastic-convolution factorisation method.

9.2.2 The exact Brownian-segment structural map

Fix an interval length $h > 0$, and let

$$C_0([0,h];\mathbb{R}^K) := \{e \in C([0,h];\mathbb{R}^K) : e(0) = 0\}.$$

The key point is that the exact Ornstein–Uhlenbeck convolution can be made a continuous functional of an ordinary continuous path segment.

Definition 9.1 (Pathwise convolution functional)**.** *For $a \geq 0$, $r \in [0,h]$, and $e \in C_0([0,h];\mathbb{R})$, define*

$$\mathfrak{I}_{a,r}(e) := e(r) - a\int_0^r e^{-a(r-s)} e(s)\, ds. \tag{12}$$

For $x = (x_1, \ldots, x_K) \in \mathbb{R}^K$, define the local path map

$$\big[\mathfrak{G}_{\widetilde{\theta},K,h}(x,e)(r)\big]_k := e^{-\widetilde{\theta}\lambda_k r} x_k + \sigma q_k \mathfrak{I}_{\widetilde{\theta}\lambda_k, r}(e_k), \qquad 0 \leq r \leq h. \tag{13}$$

Its endpoint map is

$$G_{\widetilde{\theta},K,h}(x,e) := \mathfrak{G}_{\widetilde{\theta},K,h}(x,e)(h).$$

Proposition 9.8 (Continuity and stochastic-integral identity)**.** *For fixed K and h, the map*

$$(\widetilde{\theta}, x, e) \longmapsto \mathfrak{G}_{\widetilde{\theta},K,h}(x,e)$$

is jointly continuous from $\widetilde{\Theta} \times \mathbb{R}^K \times C_0([0,h];\mathbb{R}^K)$ into $C([0,h];\mathbb{R}^K)$. In particular, it is Borel measurable. If $E(r) = \beta(t+r) - \beta(t)$ is a Brownian segment, then, for every $a \geq 0$ and $r \in [0,h]$,

$$\mathfrak{I}_{a,r}(E) = \int_0^r e^{-a(r-s)}\, dE(s) \quad \textit{almost surely.} \tag{14}$$

Consequently, $\mathfrak{G}$ is a rigorous structural map driven by an ordinary path-valued exogenous variable.

Proof. For fixed (a,r), formula (12) is linear and continuous in the supremum norm. On compact sets of (a,r), the kernel $(a,r,s) \mapsto a\mathbf{1}_{\{s\leq r\}} e^{-a(r-s)}$ varies continuously in $L^1(ds)$. This gives joint continuity of $(a,r,e) \mapsto \mathfrak{I}_{a,r}(e)$, uniformly in r, and therefore joint continuity of $\mathfrak{G}$.

For the stochastic identity, fix a, r and set $f_s = e^{-a(r-s)}$ on $[0,r]$. The function f is deterministic and continuously differentiable, with $df_s = a f_s\, ds$. Semimartingale integration by parts yields

$$f_r E(r) - f_0 E(0) = \int_0^r f_s\, dE(s) + \int_0^r E(s)\, df_s.$$

Because $f_r = 1$ and $E(0) = 0$, rearrangement gives exactly (12)–(14). No pathwise Riemann–Stieltjes integral against the Brownian path is being postulated: the right-hand side of (12) is an ordinary Lebesgue integral and is equal to the Itô integral by integration by parts. □

Interpretation. The primitive disturbance is the Brownian segment itself. The intervention changes the deterministic filter through which that same segment enters the state equation. The induced conditional transition variance therefore changes with $\widetilde{\theta}$, although the primitive noise mechanism has not changed.

9.2.3 Galerkin structural causal model and intervention

Let

$$H_K = \operatorname{span}\{e_1, \dots, e_K\}, \qquad P_K : H \to H_K$$

be the orthogonal projection. Let $A_{0,K} = P_K A_0|_{H_K}$. Consider a deterministic partition

$$\pi = \{0 = t_0 < t_1 < \cdots < t_J = T\}, \qquad h_j = t_{j+1} - t_j.$$

For each interval and mode, define the exogenous path segment

$$E_{k,j}(r) = \beta_k(t_j + r) - \beta_k(t_j), \qquad 0 \le r \le h_j,$$

and write $E_j^K = (E_{1,j}, \dots, E_{K,j})$.

Definition 9.2 (Exact Galerkin structural model)**.** *For a mechanism value $\eta > 0$, set*

$$X_0^{K,\eta} = P_K u_0$$

(in modal coordinates) and recursively define

$$X_{j+1}^{K,\eta} = G_{\eta,K,h_j}(X_j^{K,\eta}, E_j^K). \tag{15}$$

For $r \in [0, h_j]$, define the within-interval path by

$$\mathcal{X}_{t_j+r}^{K,\eta,\pi} = \mathfrak{G}_{\eta,K,h_j}(X_j^{K,\eta}, E_j^K)(r). \tag{16}$$

The intervention $do_K(\theta = \widetilde{\theta})$ replaces every local mechanism $\mathfrak{G}_{\theta,K,h_j}$ by $\mathfrak{G}_{\widetilde{\theta},K,h_j}$, while leaving $P_K u_0$, σ, $(q_k)_{k\le K}$, the basis, the partition, and every realised segment E_j^K unchanged.

Proposition 9.9 (Consistency, overwrite, and projection commutation)**.** *The intervention has the following properties.*

(i) Factual consistency:

$$do_K(\theta = \theta)\mathcal{X}^{K,\theta,\pi} = \mathcal{X}^{K,\theta,\pi}.$$

(ii) Overwrite consistency:

$$do_K(\theta = \theta_2) \circ do_K(\theta = \theta_1) = do_K(\theta = \theta_2).$$

(iii) Projection/intervention commutation:

$$P_K\big[do(\theta = \widetilde{\theta})U\big] = do_K(\theta = \widetilde{\theta})[P_K U].$$

Proof. The first two identities follow directly from replacement of the mechanism label. For the third, P_K commutes with $S_{\widetilde{\theta}}(t)$, and

$$P_K \int_0^t S_{\widetilde{\theta}}(t-s)Q^{1/2}\,dW_s = \int_0^t S_{\widetilde{\theta},K}(t-s)P_K Q^{1/2}\,dW_s.$$

Thus the projection of the post-intervention mild solution is the Galerkin post-intervention mild solution. In the present diagonal model this equality is also immediate mode by mode. □

Theorem 9.1 (Exact finite structural representation; establishes Proposition 3.1)**.** *Fix K, a partition π, and $\widetilde{\theta} \in \widetilde{\Theta}$. Then the path generated by the intervened exact Galerkin structural model satisfies*

$$\mathcal{X}_t^{K,do(\theta=\widetilde{\theta}),\pi} = U_t^{K,\widetilde{\theta}} = P_K U_t^{\widetilde{\theta}}, \qquad 0 \le t \le T, \tag{17}$$

almost surely. The equality holds for the entire stitched path, not only at the grid points. Consequently, the exact structural path is independent of the chosen partition.

Proof. At $t_0 = 0$, both sides equal $P_K u_0$. Suppose equality holds at t_j. For $r \in [0, h_j]$, the Galerkin mild flow is

$$U_{t_j+r}^{K,\widetilde{\theta}} = S_{\widetilde{\theta},K}(r)U_{t_j}^{K,\widetilde{\theta}} + \sigma \int_{t_j}^{t_j+r} S_{\widetilde{\theta},K}(t_j + r - s)P_K Q^{1/2}\,dW_s.$$

In coordinate $k \le K$, Proposition 9.8 rewrites the stochastic integral as

$$\sigma q_k \mathfrak{I}_{\widetilde{\theta}\lambda_k,r}(E_{k,j}).$$

The induction hypothesis and Definition 9.2 therefore give equality on the whole interval $[t_j, t_{j+1}]$. Induction proves the first equality in (17); Proposition 9.9 gives the second. □

Meaning. At finite K, there is no temporal approximation. The recursive structural system is exactly the Galerkin SPDE written as a composition of measurable local mechanisms. The partition is only a device for displaying the recursive causal architecture.

9.2.4 Uniform integrated Galerkin convergence

Theorem 9.2 (Integrated convergence under baseline summability)**.** *Under Assumptions 9.1–9.3,*

$$\sup_{\widetilde{\theta}\in\widetilde{\Theta},\,\sigma\in\Sigma} \mathbb{E}\int_0^T \left\|U_t^{\widetilde{\theta}} - P_K U_t^{\widetilde{\theta}}\right\|_H^2 dt \longrightarrow 0. \tag{18}$$

More precisely, with

$$B_K := \frac{1}{2\underline{\widetilde{\theta}}} \sum_{k>K} \frac{|u_{0,k}|^2}{\lambda_k} + \frac{T\overline{\sigma}^2}{2\underline{\widetilde{\theta}}} \sum_{k>K} \frac{q_k^2}{\lambda_k}, \tag{19}$$

we have

$$\sup_{\widetilde{\theta},\sigma} \mathbb{E}\int_0^T \left\|U_t^{\widetilde{\theta}} - P_K U_t^{\widetilde{\theta}}\right\|_H^2 dt \le B_K \downarrow 0.$$

Proof. By orthogonality and the zero mean of the stochastic convolution,

$$\begin{aligned}
&\mathbb{E}\int_0^T \left\|(I-P_K)U_t^{\widetilde{\theta}}\right\|_H^2 dt \\
&= \sum_{k>K} |u_{0,k}|^2 \int_0^T e^{-2\widetilde{\theta}\lambda_k t}\, dt + \sigma^2 \sum_{k>K} q_k^2 \int_0^T \int_0^t e^{-2\widetilde{\theta}\lambda_k (t-s)}\, ds\, dt.
\end{aligned}$$

The first term is bounded by

$$\frac{1}{2\underline{\widetilde{\theta}}} \sum_{k>K} \frac{|u_{0,k}|^2}{\lambda_k}.$$

For the second, Itô isometry and $\int_0^t e^{-2\widetilde{\theta}\lambda_k r}\, dr \le (2\underline{\widetilde{\theta}}\lambda_k)^{-1}$ give

$$\frac{T\overline{\sigma}^2}{2\underline{\widetilde{\theta}}} \sum_{k>K} \frac{q_k^2}{\lambda_k}.$$

Both tails vanish by Assumption 9.3 and $u_0 \in H$. □

Meaning. This is the baseline approximation result needed for integrated field functionals. It does not require a maximal-in-time estimate and therefore uses only the natural covariance trace condition $\sum q_k^2/\lambda_k < \infty$.

9.2.5 Fractional maximal regularity and pathwise convergence

Lemma 9.1 (Uniform fractional maximal estimate)**.** *Under Assumptions 9.1, 9.2, and 9.4, the mild solution has a continuous $D(\Lambda^\alpha)$-valued version and*

$$\sup_{\widetilde{\theta}\in\widetilde{\Theta},\,\sigma\in\Sigma} \mathbb{E} \sup_{0\le t\le T} \left\|\Lambda^\alpha U_t^{\widetilde{\theta}}\right\|_H^2 < \infty. \tag{20}$$

In fact, for a suitable $p > 2$, the supremum in (20) has a uniformly bounded p-th moment.

Proof. Write $U^{\widetilde{\theta}} = S_{\widetilde{\theta}}(\cdot)u_0 + Z^{\widetilde{\theta}}$, where

$$Z_t^{\widetilde{\theta}} = \sigma \int_0^t S_{\widetilde{\theta}}(t-s) Q^{1/2}\, dW_s.$$

The deterministic term satisfies

$$\sup_{t\le T} \left\|\Lambda^\alpha S_{\widetilde{\theta}}(t)u_0\right\|_H \le \|\Lambda^\alpha u_0\|_H$$

because $S_{\widetilde{\theta}}$ is a contraction and commutes with Λ^α.

Choose

$$0 < \gamma < \min\{\delta, \tfrac{1}{2}\}$$

and then choose $p > \max\{2, 1/\gamma\}$. Define the factor process

$$Y_\gamma^{\widetilde{\theta}}(s) := \sigma \int_0^s (s-r)^{-\gamma} \Lambda^\alpha S_{\widetilde{\theta}}(s-r) Q^{1/2}\, dW_r.$$

Itô isometry and diagonalisation give

$$\mathbb{E}\left\|Y_\gamma^{\widetilde{\theta}}(s)\right\|_H^2 = \sigma^2 \sum_{k\geq 1} q_k^2 \lambda_k^{2\alpha} \int_0^s r^{-2\gamma} e^{-2\widetilde{\theta}\lambda_k r}\, dr$$
$$\leq C_{\gamma,\widetilde{\theta},\overline{\sigma}} \sum_{k\geq 1} q_k^2 \lambda_k^{2\alpha+2\gamma-1} < \infty.$$

The last inequality follows from

$$\int_0^\infty r^{-2\gamma} e^{-2\widetilde{\theta}\lambda_k r}\, dr = \Gamma(1-2\gamma)(2\widetilde{\theta}\lambda_k)^{2\gamma-1}$$

and Assumption 9.4. The bound is uniform in s, $\widetilde{\theta}$, and σ. Since $Y_\gamma^{\widetilde{\theta}}(s)$ is a centred Gaussian H-valued random variable, Gaussian moment equivalence yields

$$\sup_{s,\widetilde{\theta},\sigma} \mathbb{E}\left\|Y_\gamma^{\widetilde{\theta}}(s)\right\|_H^p < \infty.$$

Hence

$$\sup_{\widetilde{\theta},\sigma} \mathbb{E}\left\|Y_\gamma^{\widetilde{\theta}}\right\|_{L^p(0,T;H)}^p < \infty.$$

The stochastic-convolution factorisation identity (3, Section 5.3; going back to 2) gives

$$\Lambda^\alpha Z_t^{\widetilde{\theta}} = c_\gamma \int_0^t (t-s)^{\gamma-1} S_{\widetilde{\theta}}(t-s) Y_\gamma^{\widetilde{\theta}}(s)\, ds, \qquad c_\gamma = \frac{\sin(\pi\gamma)}{\pi}. \tag{21}$$

The required stochastic Fubini interchange is justified by the preceding square-integrability estimate, which is exactly the integrability hypothesis of the stochastic Fubini theorem [3, Theorem 4.33]. Let $p' = p/(p-1)$. Since $\gamma > 1/p$,

$$\int_0^T r^{(\gamma-1)p'}\, dr < \infty.$$

Hölder's inequality and the contraction property therefore imply

$$\sup_{t\leq T} \left\|\Lambda^\alpha Z_t^{\widetilde{\theta}}\right\|_H \leq C_{\gamma,p,T} \left\|Y_\gamma^{\widetilde{\theta}}\right\|_{L^p(0,T;H)}.$$

Taking p-th moments proves a uniform $L^p(\Omega; C([0,T];H))$ bound for $\Lambda^\alpha Z^{\widetilde{\theta}}$. The same representation supplies continuity. Adding the deterministic term and using $p > 2$ proves (20). □

Burden-bearing assumption. The extra exponent δ creates room to choose a factorisation exponent $\gamma > 0$. Without such room, fixed-time fractional moments need not upgrade to the maximal path norm used in the projection-rate argument.

Theorem 9.3 (Uniform pathwise Galerkin rate)**.** *Under Assumptions 9.1, 9.2, and 9.4, there is a finite constant C_{path}, independent of K and $\widetilde{\theta}$, such that*

$$\sup_{\widetilde{\theta}\in\widetilde{\Theta},\,\sigma\in\Sigma} \mathbb{E} \sup_{0\leq t\leq T} \left\|U_t^{\widetilde{\theta}} - P_K U_t^{\widetilde{\theta}}\right\|_H^2 \leq C_{\mathrm{path}} \lambda_{K+1}^{-2\alpha}. \tag{22}$$

Consequently,

$$\sup_{\widetilde{\theta},\sigma} \mathbb{E}\left\|U^{\widetilde{\theta}} - P_K U^{\widetilde{\theta}}\right\|_{C([0,T];H)}^2 \longrightarrow 0.$$

Proof. For every $x \in D(\Lambda^\alpha)$, spectral calculus gives

$$\|(I-P_K)x\|_H^2 = \sum_{k>K} |x_k|^2 \leq \lambda_{K+1}^{-2\alpha} \sum_{k>K} \lambda_k^{2\alpha}|x_k|^2 \leq \lambda_{K+1}^{-2\alpha} \|\Lambda^\alpha x\|_H^2.$$

Apply this pointwise to $U_t^{\widetilde{\theta}}$, take the supremum over t, expectation, and then the supremum over the parameter class. Lemma 9.1 supplies the uniform constant. □

9.2.6 Main causal-semantics theorem

Theorem 9.4 (SPDE intervention as the limit of intervened Galerkin SCMs; establishes Theorem 3.1)**.** *Under Assumptions 9.1, 9.2, and 9.4, let π_K be any deterministic partition of $[0,T]$ for each K. Then*

$$\sup_{\widetilde{\theta}\in\widetilde{\Theta},\,\sigma\in\Sigma} \mathbb{E}\left\|\mathcal{X}^{K,do(\theta=\widetilde{\theta}),\pi_K} - U^{\widetilde{\theta}}\right\|_{C([0,T];H)}^2 \leq C_{\mathrm{path}} \lambda_{K+1}^{-2\alpha} \longrightarrow 0. \tag{23}$$

At every finite K, the intervention changes only the operator-dependent structural mechanism and preserves the same realised Brownian segments. Thus the post-intervention SPDE is the $L^2(\Omega; C([0,T]; H))$ limit of modular interventions on finite-dimensional structural models.

Proof. Theorem 9.1 identifies the intervened structural path with $P_K U^{\widetilde{\theta}}$ exactly, for every partition. Theorem 9.3 then gives (23). □

Scope. The theorem establishes intervention semantics, not identification. It explains what the counterfactual object means under a declared invariance of u_0, Q, σ, W, and the observation functional. Whether the law of this object is determined by factual observations is a separate statistical question.

9.2.7 Field-functional and law convergence

For deterministic $w \in L^2(0,T;H)$, define

$$\Psi_w(x) = \int_0^T \langle w_t, x_t \rangle_H \, dt, \qquad \mathcal{W}_R = \{w : \|w\|_{L^2(0,T;H)} \le R\}.$$

Corollary 9.1 (Uniform field-functional convergence)**.** *Under Assumptions 9.1–9.3, for every finite R,*

$$\sup_{\widetilde{\theta}\in\widetilde{\Theta},\, \sigma\in\Sigma,\, w\in\mathcal{W}_R} \mathbb{E}\left|\Psi_w(U^{\widetilde{\theta}}) - \Psi_w(P_K U^{\widetilde{\theta}})\right|^2 \le R^2 B_K \longrightarrow 0, \tag{24}$$

where B_K is defined in (19).

Proof. Cauchy–Schwarz in time gives

$$|\Psi_w(U) - \Psi_w(P_K U)|^2 \le \|w\|^2_{L^2(0,T;H)} \int_0^T \|(I - P_K)U_t\|_H^2 \, dt.$$

Take expectations and apply Theorem 9.2. □

Corollary 9.2 (Uniform quadratic-Wasserstein convergence)**.** *Under the assumptions of Corollary 9.1,*

$$\sup_{\widetilde{\theta},\sigma,w\in\mathcal{W}_R} \mathsf{W}_2\Big(\mathcal{L}(\Psi_w(U^{\widetilde{\theta}})), \mathcal{L}(\Psi_w(P_K U^{\widetilde{\theta}}))\Big) \le R\sqrt{B_K} \longrightarrow 0. \tag{25}$$

Proof. Use the common-Wiener coupling already built into the structural model. For any coupling (X, Y), $\mathsf{W}_2^2(\mathcal{L}(X), \mathcal{L}(Y)) \le \mathbb{E}|X - Y|^2$. Apply Corollary 9.1. □

Corollary 9.3 (Uniform CDF convergence under non-degeneracy)**.** *Let $\mathcal{W}_\star \subset \mathcal{W}_R$ satisfy the uniform non-degeneracy condition*

$$\inf_{\widetilde{\theta}\in\widetilde{\Theta},\, \sigma\in\Sigma,\, w\in\mathcal{W}_\star} \mathrm{Var}\big(\Psi_w(U^{\widetilde{\theta}})\big) \ge v_\star > 0.$$

Then

$$\sup_{\widetilde{\theta},\sigma,w\in\mathcal{W}_\star,z\in\mathbb{R}} \Big|\mathbb{P}\{\Psi_w(P_K U^{\widetilde{\theta}}) \le z\} - \mathbb{P}\{\Psi_w(U^{\widetilde{\theta}}) \le z\}\Big| \longrightarrow 0.$$

Proof. Every full target is Gaussian with density bounded uniformly by $M = (2\pi v_\star)^{-1/2}$. Under the common-noise coupling, for every $\varepsilon > 0$,

$$\sup_z |F_K(z) - F(z)| \le \frac{\mathbb{E}|X_K - X|}{\varepsilon} + 2M\varepsilon.$$

Optimising in ε and using $\mathbb{E}|X_K - X| \le (\mathbb{E}|X_K - X|^2)^{1/2}$, uniformly controlled by (24), proves the claim. □

9.2.8 Exact flow versus approximate numerical flow

The exact structural model above has no time-discretisation error. A practical numerical recursion may replace $\mathfrak{G}$ by an approximate measurable map. The following result isolates the additional assumption needed for such a replacement.

Proposition 9.10 (Numerical-SCM transfer principle)**.** *Let $\widehat{\mathcal{X}}^{K,\Delta,\widetilde{\theta}}$ be an approximate structural path built from the same initial state and the same Brownian segments as the exact Galerkin SCM. Suppose that, for some $\kappa > 0$,*

$$\sup_{K\ge 1} \sup_{\widetilde{\theta}\in\widetilde{\Theta},\, \sigma\in\Sigma} \mathbb{E}\left\|\widehat{\mathcal{X}}^{K,\Delta,\widetilde{\theta}} - \mathcal{X}^{K,do(\theta=\widetilde{\theta}),\pi_\Delta}\right\|^2_{C([0,T];H)} \le C_{\mathrm{num}} \Delta^{2\kappa}. \tag{26}$$

Then, under Assumption 9.4,

$$\sup_{\widetilde{\theta},\sigma} \mathbb{E} \left\| \widehat{\mathcal{X}}^{K,\Delta,\widetilde{\theta}} - U^{\widetilde{\theta}} \right\|^2_{C([0,T];H)} \leq 2C_{\text{path}} \lambda_{K+1}^{-2\alpha} + 2C_{\text{num}} \Delta^{2\kappa}. \tag{27}$$

The analogous field-functional and Wasserstein bounds follow by Cauchy–Schwarz and the common-noise coupling.

Proof. Insert the exact Galerkin structural path between the numerical path and the SPDE path, apply $(a+b)^2 \leq 2a^2 + 2b^2$, then use (26) and Theorem 9.4. □

9.2.9 Heat-equation specialisation

Corollary 9.4 (Dirichlet heat equation on $(0,1)$)**.** *Let $A_0 = \Delta$ with Dirichlet boundary conditions on $(0,1)$, so $\lambda_k = \pi^2 k^2$. Suppose*

$$q_k^2 \lesssim \lambda_k^{-\beta}.$$

Then Assumption 9.4 holds whenever

$$\alpha + \delta < \frac{1}{4} + \frac{\beta}{2}, \qquad u_0 \in D((-\Delta)^\alpha).$$

For space–time white noise, $q_k \equiv 1$, this becomes

$$\alpha + \delta < \frac{1}{4}.$$

Under these conditions, the pathwise Galerkin error obeys

$$\sup_{\widetilde{\theta},\sigma} \mathbb{E} \sup_{t \leq T} \left\| U_t^{\widetilde{\theta}} - P_K U_t^{\widetilde{\theta}} \right\|^2_H \lesssim K^{-4\alpha}.$$

Proof. The summability exponent is

$$q_k^2 \lambda_k^{2(\alpha+\delta)-1} \lesssim k^{2(2(\alpha+\delta)-1-\beta)}.$$

The series converges precisely under $2(2(\alpha+\delta)-1-\beta) < -1$, which is the displayed condition. Since $\lambda_{K+1}^{-2\alpha} \asymp K^{-4\alpha}$, Theorem 9.3 gives the rate. □

9.2.10 Audit conclusions and change log

Relative to the preceding Part S1 note, the theorem package makes the following corrections.

1. **The structural Itô map is now explicit and continuous.** The notation $e \mapsto \int e^{-a(r-s)} de(s)$ has been replaced by the ordinary path functional (12). Semimartingale integration by parts proves equality with the Itô integral on Brownian paths.
2. **Fixed-time and maximal regularity are separated.** The boundary series $\sum q_k^2 \lambda_k^{2\alpha-1}$ is not used to justify a supremum-in-time fractional moment. The main path theorem assumes a positive factorisation margin $\delta > 0$.
3. **The artificial restriction $\alpha < 1/2$ is removed.** The abstract linear theorem permits any $\alpha > 0$ for which the summability condition holds. Concrete noises and spatial dimensions determine the feasible range; white noise in one spatial dimension gives $\alpha + \delta < 1/4$.
4. **Exact and numerical time stepping are no longer conflated.** The exact Brownian-segment recursion equals the Galerkin SPDE for every partition and has no Δ-error. Temporal error appears only after an approximate local map is introduced.
5. **The numerical rate is conditional on a verified scheme-specific bound.** No universal $\Delta^{2\kappa}$ rate is claimed for Euler or exponential Euler without stability and regularity hypotheses.
6. **Field-functional convergence uses the weaker integrated theorem.** The functional and W_2 conclusions do not require the stronger pathwise maximal estimate.
7. **Uniform CDF convergence is qualified by variance non-degeneracy.** A bounded L^2-ball contains zero and arbitrarily small weights, so a uniform Kolmogorov result over that entire ball would be invalid at degenerate limits.
8. **The conceptual boundary is explicit.** Structural semantics, statistical identification, estimation, and numerical approximation remain distinct claims.

Literature position

The factorisation argument follows the classical regularity programme for stochastic convolutions initiated by Da Prato, Kwapień, and Zabczyk. Modern maximal-inequality and pathwise-uniform numerical results provide broader settings and sharper scheme-specific conclusions. The intervention construction is aligned with the finite-SDE principle of relating a post-intervention stochastic equation to limits of intervened structural equation models. The contribution developed here is the explicit Galerkin-to-SPDE operator-intervention link, with primitive common noise and law-level field-functional consequences.

9.3 Part S2: Target identification

9.3.1 Positive model and counterfactual target

Let $H = L^2(D)$ and let A_0 be self-adjoint, negative, and have compact resolvent. Write

$$A_0 e_k = -\lambda_k e_k, \qquad 0 < \lambda_1 \le \lambda_2 \le \cdots,$$

for an orthonormal eigenbasis $(e_k)_{k\ge 1}$. Let

$$Q^{1/2} e_k = q_k e_k, \qquad q_k \ge 0.$$

The factual parameter is

$$\vartheta = (\theta, \sigma) \in [\underline{\theta}, \overline{\theta}] \times [\underline{\sigma}, \overline{\sigma}] \Subset (0,\infty)^2.$$

The factual field satisfies

$$dU_t = \theta A_0 U_t \, dt + \sigma Q^{1/2} \, dW_t, \qquad U_0 = u_0, \tag{28}$$

where $u_0 \in H$ is deterministic and known. Its modal coordinates are independent Ornstein–Uhlenbeck processes,

$$du_k(t) = -\theta \lambda_k u_k(t) \, dt + \sigma q_k \, d\beta_k(t), \qquad u_k(0) = u_{0,k}. \tag{29}$$

For a fixed intervention value $\widetilde{\theta} \in \widetilde{\Theta} \Subset (0,\infty)$, Part S1 defines

$$dU_t^{\widetilde{\theta}} = \widetilde{\theta} A_0 U_t^{\widetilde{\theta}} \, dt + \sigma Q^{1/2} \, dW_t, \qquad U_0^{\widetilde{\theta}} = u_0. \tag{30}$$

For $w \in L^2(0,T;H)$, set

$$\Psi_w(U) = \int_0^T \langle w_t, U_t \rangle_H \, dt, \qquad w_k(t) = \langle w_t, e_k \rangle_H .$$

Assumption 9.5 (Finite target variance)**.** *For every target $(\widetilde{\theta}, w)$ considered below,*

$$V^0_{\widetilde{\theta},w} := \sum_{k\ge 1} q_k^2 \int_0^T \left(\int_s^T e^{-\widetilde{\theta}\lambda_k (t-s)} w_k(t) \, dt \right)^2 ds < \infty.$$

This is implied by the baseline Hilbert-space summability used in Part S1 together with the stated L^2 condition on w.

Proposition 9.11 (Gaussian counterfactual law; establishes Proposition 2.1)**.** *Define*

$$m_{\widetilde{\theta},w} := \sum_{k\ge 1} u_{0,k} \int_0^T e^{-\widetilde{\theta}\lambda_k t} w_k(t) \, dt, \tag{31}$$

$$g^{\widetilde{\theta},w}_{k,s} := \int_s^T e^{-\widetilde{\theta}\lambda_k (t-s)} w_k(t) \, dt. \tag{32}$$

Then

$$\Psi_w(U^{\widetilde{\theta}}) \sim \mathcal{N}\left(m_{\widetilde{\theta},w}, \sigma^2 V^0_{\widetilde{\theta},w} \right). \tag{33}$$

Consequently,

$$F_{\widetilde{\theta},w}(z;\vartheta) = \Phi\left(\frac{z - m_{\widetilde{\theta},w}}{\sigma \sqrt{V^0_{\widetilde{\theta},w}}} \right) \tag{34}$$

when $V^0_{\widetilde{\theta},w} > 0$, with the evident point-mass formula when $V^0_{\widetilde{\theta},w} = 0$.

Proof. Insert the mild modal representation of (30) into Ψ_w, apply stochastic Fubini, and obtain

$$\Psi_w(U^{\widetilde{\theta}}) = m_{\widetilde{\theta},w} + \sigma \sum_{k\ge 1} q_k \int_0^T g^{\widetilde{\theta},w}_{k,s} \, d\beta_k(s).$$

The stochastic term is centred Gaussian. Independence of the Brownian coordinates and Itô isometry give the displayed variance. □

Identification consequence. For a fixed operator intervention $do(\theta = \widetilde{\theta})$, the factual drift parameter θ does not appear in (33). It has been overwritten. The factual data matter through identification of the structurally invariant scale σ. This is different from a factual-versus-counterfactual distributional effect, whose factual component generally depends on both θ and σ.

9.3.2 Identification as constancy on observational fibres

Definition 9.3 (Observational and target equivalence)**.** *Let P^Y_ϑ denote the factual observed-data law under ϑ. Define*

$$\vartheta_1 \sim_{\text{obs}} \vartheta_2 \quad \Longleftrightarrow \quad P^Y_{\vartheta_1} = P^Y_{\vartheta_2}.$$

For a law-valued target $T(\vartheta)$, define

$$\vartheta_1 \sim_T \vartheta_2 \quad \Longleftrightarrow \quad T(\vartheta_1) = T(\vartheta_2).$$

The target is point identified from the observed law if every observational equivalence class is contained in a target equivalence class.

Theorem 9.5 (Necessary and sufficient fibre criterion; establishes Lemma 4.1)**.** *A target T is point identified from the observed law if and only if*

$$\vartheta_1 \sim_{\text{obs}} \vartheta_2 \quad \Longrightarrow \quad \vartheta_1 \sim_T \vartheta_2. \tag{35}$$

Equivalently, T is constant on every fibre of the observation map $\vartheta \mapsto P^Y_\vartheta$.

Proof. If T is identified, two parameter values inducing the same observed law must yield the same uniquely determined target. Conversely, if (35) holds, define the target at an observed law by choosing any parameter value in its fibre. Constancy on fibres makes the definition independent of the choice. □

9.3.3 Exact discrete spectral observations

Fix $\Delta > 0$, observation times $t_j = j\Delta$, $j = 0, \dots, n$, and a finite non-empty observed set $I \subset \mathbb{N}$. The exact spectral data are

$$Y_{k,j} = u_k(t_j), \qquad k \in I, \quad j = 0, \dots, n.$$

For each k, define

$$\rho_k(\theta) := e^{-\theta\lambda_k\Delta}, \tag{36}$$

$$h_k(\theta) := \frac{1 - e^{-2\theta\lambda_k\Delta}}{2\theta\lambda_k}. \tag{37}$$

Then

$$Y_{k,j+1} \mid Y_{k,j} \sim \mathcal{N}\left(\rho_k(\theta)Y_{k,j}, \sigma^2 q_k^2 h_k(\theta)\right). \tag{38}$$

The modes are independent.

Lemma 9.2 (Strict spectral-ratio monotonicity)**.** *Let $0 < \lambda_a < \lambda_b$ and define*

$$R_{a,b}(\theta) := \frac{h_{\lambda_a}(\theta)}{h_{\lambda_b}(\theta)}, \qquad h_\lambda(\theta) := \frac{1 - e^{-2\theta\lambda\Delta}}{2\theta\lambda}.$$

Then $R_{a,b}$ is strictly increasing on $(0, \infty)$.

Proof. Set $x = 2\theta\Delta$ and $f(y) = (1 - e^{-y})/y$, so $h_\lambda(\theta) = \Delta f(x\lambda)$. Direct differentiation gives

$$\frac{d}{dx}\log f(x\lambda) = \frac{\lambda}{e^{x\lambda} - 1} - \frac{1}{x}.$$

Hence

$$\frac{d}{dx}\log R_{a,b} = \frac{\lambda_a}{e^{x\lambda_a} - 1} - \frac{\lambda_b}{e^{x\lambda_b} - 1} > 0,$$

because $\lambda \mapsto \lambda/(e^{x\lambda} - 1)$ is strictly decreasing for every $x > 0$. □

Theorem 9.6 (Exact finite-sample identification; establishes Theorem 4.1)**.** *As stated in Theorem 4.1 of the main text.*

Proof. For part (i), choose any $k \in I$ and write $V_{k,1} = \mathrm{Var}(Y_{k,1})$. Since $q_k, \sigma > 0$, $V_{k,1} > 0$. The Markov recursion gives

$$\mathrm{Cov}(Y_{k,1}, Y_{k,2}) = \rho_k(\theta) V_{k,1}.$$

Thus the joint law identifies

$$\rho_k(\theta) = \frac{\mathrm{Cov}(Y_{k,1}, Y_{k,2})}{\mathrm{Var}(Y_{k,1})}. \tag{39}$$

Strict monotonicity of $\theta \mapsto e^{-\theta\lambda_k\Delta}$ identifies

$$\theta = -\frac{\log \rho_k}{\lambda_k \Delta}.$$

Then

$$\sigma^2 = \frac{\mathrm{Var}(Y_{k,1})}{q_k^2 h_k(\theta)} \tag{40}$$

identifies σ.

For part (ii), the one-step mean and variance are

$$\mathbb{E}Y_{k,1} = \rho_k(\theta) u_{0,k}, \qquad \mathrm{Var}(Y_{k,1}) = \sigma^2 q_k^2 h_k(\theta).$$

If some $u_{0,k} \neq 0$, the mean identifies ρ_k and hence θ; the variance then identifies σ. If all observed initial coefficients vanish but two distinct eigenvalues are observed, the law identifies the variance ratio

$$\frac{\mathrm{Var}(Y_{a,1})/q_a^2}{\mathrm{Var}(Y_{b,1})/q_b^2} = \frac{h_a(\theta)}{h_b(\theta)},$$

and Lemma 9.2 identifies θ, followed by σ.

Conversely, if all observed initial coefficients vanish and all observed eigenvalues equal a common value $\lambda_\star$, the joint law depends on (θ, σ) only through

$$c = \sigma^2 h_{\lambda_\star}(\theta).$$

For any $\theta_2 \neq \theta_1$, choosing

$$\sigma_2^2 = \sigma_1^2 \frac{h_{\lambda_\star}(\theta_1)}{h_{\lambda_\star}(\theta_2)}$$

produces the same observed law. The pair is therefore not identified. □

9.3.4 Target-specific identification in the scalar core

Theorem 9.7 (Sharp identification of the fixed-intervention law)**.** *Fix $(\widetilde{\theta}, w)$ and write $m = m_{\widetilde{\theta},w}$ and $V^0 = V^0_{\widetilde{\theta},w}$.*

(i) *If $V^0 = 0$, the counterfactual law is the known point mass δ_m and is identified without identifying either θ or σ.*

(ii) *If $V^0 > 0$, the full counterfactual law is identified if and only if σ is constant on every observational equivalence class.*

(iii) *If $V^0 > 0$ and only the scalar CDF ordinate $F_{\widetilde{\theta},w}(z)$ is targeted, then:*

  (a) *at $z = m$, the ordinate equals $1/2$ for every $\sigma > 0$ and is therefore identified even when σ is not;*

  (b) *at every $z \neq m$, the ordinate is identified if and only if σ is constant on every observational equivalence class.*

Proof. Part (i) follows from Proposition 9.11. For part (ii), Gaussian laws with the same known mean m are equal exactly when their positive variances $\sigma^2 V^0$ are equal. Because V^0 is known and positive, this is equivalent to equality of σ. For part (iii), use (34). At $z = m$ the argument of Φ is zero. For $z \neq m$, the map $\sigma \mapsto \Phi((z-m)/(\sigma\sqrt{V^0}))$ is strictly monotone on $(0, \infty)$. □

Corollary 9.5 (Identification under exact spectral sampling)**.** *Under any condition in Theorem 9.6 that identifies σ, every fixed-intervention field-functional law satisfying Assumption 9.5 is identified, including targets with Fourier support outside the observed set I.*

9.3.5 Noisy spectral observations

Suppose now that

$$Z_{k,j} = Y_{k,j} + \varepsilon_{k,j}, \tag{41}$$

where the measurement errors are independent of the state process.

Proposition 9.12 (Known-error deconvolution)**.** *Fix a finite observed vector Y and suppose the additive error vector ε has a known distribution, is independent of Y, and has a characteristic function that is nowhere zero. Then the law of Y is identified from the law of $Z = Y + \varepsilon$. Consequently, every identification result for exact spectral observations transfers to this noisy experiment.*

Proof. Independence gives

$$\varphi_Z(t) = \varphi_Y(t)\varphi_\varepsilon(t).$$

Since φ_ε is known and never vanishes, $\varphi_Y = \varphi_Z/\varphi_\varepsilon$ is uniquely determined. Characteristic functions determine probability laws. □

Theorem 9.8 (Explicit Gaussian measurement-error identification)**.** *Observe one mode k with $q_k > 0$ at equally spaced times and let*

$$Z_j = Y_{k,j} + \varepsilon_j, \qquad \varepsilon_j \overset{\text{iid}}{\sim} \mathcal{N}(0, r),$$

independently of the state.

(i) *If r is known and $n \geq 2$, then (θ, σ) is identified.*
(ii) *If r is unknown and $n \geq 3$, then (θ, σ, r) is identified.*

Proof. Let $V_1 = \mathrm{Var}(Y_{k,1}) > 0$ and $\rho = e^{-\theta\lambda_k\Delta}$. If r is known,

$$V_1 = \mathrm{Var}(Z_1) - r, \qquad \mathrm{Cov}(Z_1, Z_2) = \rho V_1,$$

so ρ, then θ, then σ are identified as in (40).

If r is unknown, measurement error does not affect cross-time covariances and

$$\mathrm{Cov}(Z_1, Z_2) = \rho V_1, \qquad \mathrm{Cov}(Z_1, Z_3) = \rho^2 V_1.$$

Therefore

$$\rho = \frac{\mathrm{Cov}(Z_1, Z_3)}{\mathrm{Cov}(Z_1, Z_2)}, \qquad V_1 = \frac{\mathrm{Cov}(Z_1, Z_2)}{\rho}.$$

This identifies θ and σ, and finally $r = \mathrm{Var}(Z_1) - V_1$. □

9.3.6 Operator-level identification under partial observation

This section states the correct infinite-dimensional principle. It shows why a deterministic observable-subspace condition is exact for the mean but is not, by itself, a complete covariance-identification theorem.

Let $S_\theta(t) = e^{t\theta A_0}$ be fixed and let $\mathcal{H} : H \to Y$ be a bounded observation operator into a Hilbert space Y. Consider a Gaussian model indexed by a deterministic initial state u and a positive trace-class noise covariance C:

$$dU_t = \theta A_0 U_t\, dt + C^{1/2}\, dW_t, \qquad U_0 = u, \qquad Y_t = \mathcal{H}U_t.$$

Define the factual mean and covariance maps

$$(\mathfrak{M}_\theta u)(t) := \mathcal{H}S_\theta(t)u, \tag{42}$$

$$(\mathfrak{K}_\theta C)(t, r) := \int_0^{t\wedge r} \mathcal{H}S_\theta(t-s)CS_\theta(r-s)^*\mathcal{H}^*\, ds. \tag{43}$$

For a counterfactual target, define

$$r_0^{\widetilde{\theta},w} := \int_0^T S_{\widetilde{\theta}}(t)^* w_t\, dt, \tag{44}$$

$$g_s^{\widetilde{\theta},w} := \int_s^T S_{\widetilde{\theta}}(t-s)^* w_t\, dt, \tag{45}$$

$$\mathfrak{L}_{\widetilde{\theta},w}(u) := \left\langle r_0^{\widetilde{\theta},w}, u \right\rangle_H, \tag{46}$$

$$\mathfrak{V}_{\widetilde{\theta},w}(C) := \int_0^T \left\langle C g_s^{\widetilde{\theta},w}, g_s^{\widetilde{\theta},w} \right\rangle_H ds. \tag{47}$$

Theorem 9.9 (Kernel characterisation for Gaussian partial observation; establishes Proposition 4.1)**.** *Let the admissible model class be a product $\mathcal{U} \times \mathcal{C}$, where $\mathcal{U} \subset H$ and $\mathcal{C}$ is a convex class of positive trace-class operators. The counterfactual Gaussian law is identified from the complete factual output law if and only if*

$$\mathrm{Ker}(\mathfrak{M}_\theta) \cap (\mathcal{U} - \mathcal{U}) \subseteq \mathrm{Ker}(\mathfrak{L}_{\widetilde{\theta},w}), \tag{48}$$

$$\mathrm{Ker}(\mathfrak{K}_\theta) \cap (\mathcal{C} - \mathcal{C}) \subseteq \mathrm{Ker}(\mathfrak{V}_{\widetilde{\theta},w}). \tag{49}$$

Moreover, if

$$\mathcal{O}_\theta := \overline{\operatorname{span}}\{S_\theta(t)^* \mathcal{H}^* y : t \in [0,T],\ y \in Y\},$$

then the unrestricted mean condition (48) is equivalent to

$$\widetilde{r_0^{\theta,w}} \in \mathcal{O}_\theta. \tag{50}$$

Proof. A Gaussian output law is determined by its mean and covariance. Equality of output means for u_1, u_2 is equivalent to $u_1 - u_2 \in \operatorname{Ker}(\mathfrak{M}_\theta)$; equality of output covariances for C_1, C_2 is equivalent to $C_1 - C_2 \in \operatorname{Ker}(\mathfrak{K}_\theta)$. The counterfactual Gaussian laws agree exactly when the target means and variances agree. This proves the two kernel conditions.

For the final statement,

$$\operatorname{Ker}(\mathfrak{M}_\theta) = \mathcal{O}_\theta^\perp.$$

A continuous linear functional $u \mapsto \langle r_0, u \rangle_H$ vanishes on $\mathcal{O}_\theta^\perp$ if and only if $r_0 \in \mathcal{O}_\theta$. □

9.3.7 Exact modal-support theorem in a nuisance enlargement

To make the observable-support boundary substantive, enlarge the scalar core to

$$du_k(t) = -a_k u_k(t)\, dt + b_k\, d\beta_k(t), \qquad b_k \geq 0, \tag{51}$$

with mode-specific invariant amplitudes b_k. Suppose exact spectral data identify (a_k, b_k) for every observed mode $k \in I_{\mathrm{obs}}$. Let J_{free} be the set of unobserved modes whose amplitudes are not otherwise known. Under intervention, the drift of mode k is replaced by a known $\widetilde{a}_k > 0$, while b_k is preserved.

Define

$$B_k(w) := \int_0^T \left(\int_s^T e^{-\widetilde{a}_k (t-s)} w_k(t)\, dt \right)^2 ds. \tag{52}$$

The counterfactual variance is $\sum_k b_k^2 B_k(w)$.

Lemma 9.3 (Effective support)**.** *For $\widetilde{a}_k > 0$, $B_k(w) = 0$ if and only if $w_k = 0$ in $L^2(0,T)$.*

Proof. Let $g(s) = \int_s^T e^{-\widetilde{a}_k (t-s)} w_k(t)\, dt$. Then g is absolutely continuous, $g(T) = 0$, and $g'(s) = \widetilde{a}_k g(s) - w_k(s)$ almost everywhere. If $B_k(w) = \|g\|_{L^2}^2 = 0$, then $g = 0$ almost everywhere and therefore $w_k = 0$ almost everywhere. The converse is immediate. □

Theorem 9.10 (Sharp diagonal target-support condition)**.** *In model (51), assume the admissible class permits each b_k^2, $k \in J_{\mathrm{free}}$, to vary over a non-degenerate interval independently of the observed parameters. Then the counterfactual field-functional law is identified if and only if*

$$B_k(w) = 0 \quad \text{for every } k \in J_{\mathrm{free}} \setminus I_{\mathrm{obs}}. \tag{53}$$

Equivalently,

$$I_{\mathrm{tar}}(w) \cap J_{\mathrm{free}} \subseteq I_{\mathrm{obs}}, \qquad I_{\mathrm{tar}}(w) := \{k : w_k \neq 0 \text{ in } L^2(0,T)\}.$$

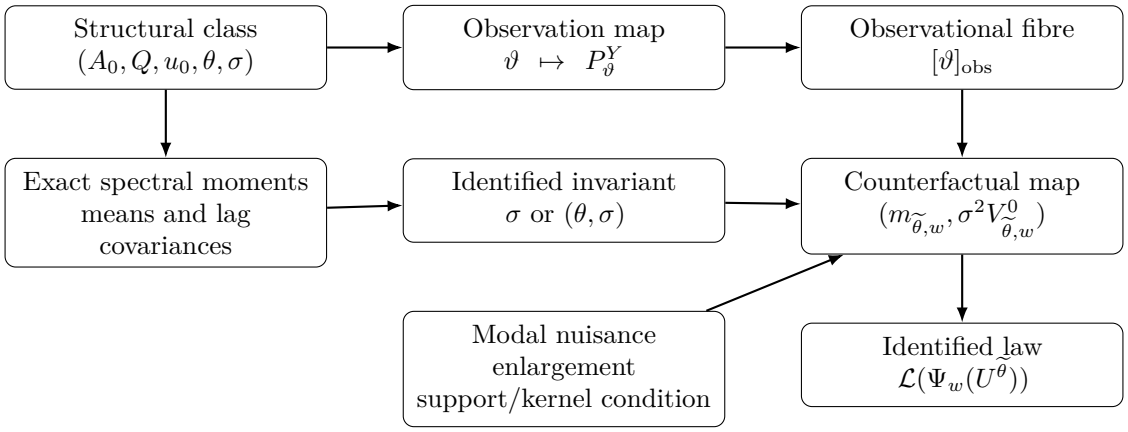
Unobserved modes with structurally known amplitudes need not belong to I_{obs}.

Proof. For observed modes and structurally known unobserved modes, every term $b_k^2 B_k(w)$ is identified. If (53) holds, all remaining free unobserved terms vanish, so the target variance and hence the Gaussian law are identified.

Conversely, suppose $B_\ell(w) > 0$ for some free unobserved mode ℓ. Change b_ℓ^2 while holding every observed parameter and every other structural component fixed. The factual observed law is unchanged because mode ℓ is not observed, while the target variance changes by $\Delta b_\ell^2 B_\ell(w) \neq 0$. The target law is not identified. □

9.3.8 Sharp counterexamples and corrected failure catalogue

9.3.9 Identification DAG



9.3.10 Assumptions, conclusions, and failure modes

Setting	Identification conclusion	Principal failure mode
Exact, one excited mode, $n \geq 2$	(θ, σ) and all fixed-intervention laws are identified	No excited observed mode
Exact, one step	Identified iff a non-zero observed initial coefficient or two distinct observed eigenvalues are available	Zero starts and one repeated eigenvalue reveal only $\sigma^2 h_\lambda(\theta)$
Known additive error, non-vanishing characteristic function	Latent finite-dimensional law is recovered by deconvolution	Unknown or vanishing error characteristic function
Gaussian error, known variance, $n \geq 2$	(θ, σ) identified from variance and lag covariance	Correlated or misspecified error
Gaussian error, unknown variance, $n \geq 3$	(θ, σ, r) identified from two off-diagonal covariances	Too few lags
Scalar parametric core	Unobserved target modes are identified once σ is identified	Unknown Q shape or broken common-scale restriction
Mode-specific nuisance amplitudes	Target identified iff every free unobserved target coefficient vanishes	Invisible free modal amplitude
General partial observation	Kernel inclusions (48)–(49) are necessary and sufficient	Covariance map has target-relevant null directions

9.4 Part S3: Exact likelihood, mixed-rate LAN, and efficient estimation

9.4.1 Statistical experiment

Let $H = L^2(D)$, and let A_0 be self-adjoint, negative, and have compact resolvent. Write

$$A_0 e_k = -\lambda_k e_k, \qquad 0 < \lambda_1 \leq \lambda_2 \leq \cdots,$$

for an orthonormal eigenbasis. The factual SPDE is

$$dU_t = \theta A_0 U_t \, dt + \sigma Q^{1/2} dW_t, \qquad U_0 = u_0, \tag{54}$$

where

$$Q^{1/2} e_k = q_k e_k, \qquad q_k > 0,$$

and u_0 and (q_k) are known. The unknown parameter is

$$\vartheta = (\theta, \sigma) \in \Theta := [\underline{\theta}, \overline{\theta}] \times [\underline{\sigma}, \overline{\sigma}] \Subset (0, \infty)^2.$$

The true value $\vartheta_0 = (\theta_0, \sigma_0)$ lies in the interior of Θ.

The Fourier coordinates solve independent Ornstein–Uhlenbeck equations,

$$du_k(t) = -\theta \lambda_k u_k(t) \, dt + \sigma q_k d\beta_k(t), \qquad u_k(0) = u_{0,k}. \tag{55}$$

Fix $T > 0$, let $\Delta_n = T/n$, and observe

$$Y_{k,j} = u_k(j\Delta_n), \qquad k = 1, \dots, N, \quad j = 0, \dots, n.$$

The initial values are deterministic and therefore contribute no density. The experiment is

$$\mathcal{E}_{N,n} = \left(\mathcal{X}_{N,n}, \mathcal{A}_{N,n}, \{P_\vartheta^{N,n} : \vartheta \in \Theta\}\right).$$

For $k \le N$, define

$$\rho_{k,n}(\theta) := e^{-\theta\lambda_k\Delta_n}, \tag{56}$$

$$h_{k,n}(\theta) := \frac{1 - e^{-2\theta\lambda_k\Delta_n}}{2\theta\lambda_k}, \tag{57}$$

$$v_{k,n}(\vartheta) := \sigma^2 q_k^2 h_{k,n}(\theta). \tag{58}$$

Then

$$Y_{k,j+1} \mid \mathcal{F}_{k,j} \sim \mathcal{N}(\rho_{k,n}(\theta)Y_{k,j}, v_{k,n}(\vartheta)), \tag{59}$$

where $\mathcal{F}_{k,j}$ contains the past of mode k. The standardised innovations

$$\varepsilon_{k,j}(\vartheta) := \frac{Y_{k,j+1} - \rho_{k,n}(\theta)Y_{k,j}}{\sigma q_k \sqrt{h_{k,n}(\theta)}} \tag{60}$$

are independent standard normals at the true parameter.

9.4.2 Exact likelihood and profile MLE

Set

$$r_{k,j}(\theta) := Y_{k,j+1} - \rho_{k,n}(\theta)Y_{k,j}.$$

Up to an additive constant, the exact conditional log-likelihood is

$$\ell_{N,n}(\theta, \sigma) = -\frac{1}{2}\sum_{k=1}^{N}\sum_{j=0}^{n-1}\left\{\log\left(\sigma^2 q_k^2 h_{k,n}(\theta)\right) + \frac{r_{k,j}(\theta)^2}{\sigma^2 q_k^2 h_{k,n}(\theta)}\right\}. \tag{61}$$

Proposition 9.13 (Exact profile likelihood)**.** *For fixed θ, the unconstrained maximiser in σ^2 is*

$$\widehat{\sigma}_{N,n}^2(\theta) = \frac{1}{Nn}\sum_{k=1}^{N}\sum_{j=0}^{n-1}\frac{r_{k,j}(\theta)^2}{q_k^2 h_{k,n}(\theta)}. \tag{62}$$

Consequently, an exact conditional MLE is obtained from

$$\widehat{\theta}_{N,n} \in \arg\min_{\theta\in[\underline{\theta},\overline{\theta}]}\left\{Nn\log\widehat{\sigma}_{N,n}^2(\theta) + n\sum_{k=1}^{N}\log h_{k,n}(\theta)\right\}, \tag{63}$$

$$\widehat{\sigma}_{N,n} = \widehat{\sigma}_{N,n}(\widehat{\theta}_{N,n}), \tag{64}$$

with projection to the compact scale interval when necessary. Since ϑ_0 is interior, that projection is inactive with probability tending to one.

Proof. Differentiate (61) with respect to σ^2. The score equation is linear in $1/\sigma^2$ and gives (62). Substitution into (61) removes constants and terms independent of θ, yielding (63). □

Why exact transitions matter. An Euler least-squares estimator replaces $\rho_{k,n}(\theta)$ by $1 - \theta\lambda_k\Delta_n$. Its deterministic centring error is typically of order $\Delta_n \sum_{k\le N}\lambda_k^2 / \sum_{k\le N}\lambda_k$, hence of order $\lambda_N\Delta_n$. Root-information centring would then require a stronger undersmoothing condition. The exact likelihood has no such approximation bias; $\lambda_N\Delta_n \to 0$ is used only to identify the continuous-time information limit and asymptotic orthogonality.

9.4.3 Score and exact Fisher information

Define

$$g_{k,n}(\theta) := \partial_\theta \log h_{k,n}(\theta) = \frac{2\lambda_k\Delta_n}{e^{2\theta\lambda_k\Delta_n} - 1} - \frac{1}{\theta}. \tag{65}$$

Also,

$$\dot{\rho}_{k,n}(\theta) = -\lambda_k\Delta_n\rho_{k,n}(\theta). \tag{66}$$

For one transition, evaluated at a generic parameter,

$$s_{\theta,k,j}(\vartheta) = \frac{r_{k,j}(\theta)\dot{\rho}_{k,n}(\theta)Y_{k,j}}{v_{k,n}(\vartheta)} + \frac{1}{2}\left(\frac{r_{k,j}(\theta)^2}{v_{k,n}(\vartheta)} - 1\right) g_{k,n}(\theta), \tag{67}$$

$$s_{\sigma,k,j}(\vartheta) = \frac{1}{\sigma}\left(\frac{r_{k,j}(\theta)^2}{v_{k,n}(\vartheta)} - 1\right). \tag{68}$$

At ϑ_0, these are martingale differences. The term in the first line proportional to $\varepsilon_{k,j}$ is the transition-mean score; the centred-square term is the transition-variance score.

Proposition 9.14 (Exact information matrix)**.** *Let*

$$M_{N,n}(\vartheta) := \sum_{k=1}^{N}\sum_{j=0}^{n-1} \frac{\dot{\rho}_{k,n}(\theta)^2 \mathbb{E}_\vartheta[Y_{k,j}^2]}{\sigma^2 q_k^2 h_{k,n}(\theta)}. \tag{69}$$

Then the Fisher information based on the conditional likelihood is

$$I_{N,n}(\vartheta) = \begin{pmatrix} M_{N,n}(\vartheta) + \dfrac{n}{2}\sum_{k=1}^{N} g_{k,n}(\theta)^2 & \dfrac{n}{\sigma}\sum_{k=1}^{N} g_{k,n}(\theta) \\ \dfrac{n}{\sigma}\sum_{k=1}^{N} g_{k,n}(\theta) & \dfrac{2Nn}{\sigma^2} \end{pmatrix}. \tag{70}$$

Moreover, with $\rho_k = \rho_{k,n}(\theta)$ and

$$A_{k,n} := \sum_{j=0}^{n-1} \rho_k^{2j} = \frac{1-\rho_k^{2n}}{1-\rho_k^2},$$

we have the closed form

$$M_{N,n}(\vartheta) = \sum_{k=1}^{N} \frac{\lambda_k^2 \Delta_n^2 \rho_k^2}{\sigma^2 q_k^2 h_{k,n}(\theta)} \left\{ u_{0,k}^2 A_{k,n} + \frac{\sigma^2 q_k^2}{2\theta\lambda_k}(n - A_{k,n}) \right\}. \tag{71}$$

Proof. For a Gaussian observation with parameter-dependent mean μ and variance v, conditional Fisher information is

$$\frac{\partial_a \mu\, \partial_b \mu}{v} + \frac{1}{2}\partial_a \log v\, \partial_b \log v.$$

Here $\partial_\theta \mu = \dot{\rho}_{k,n} Y_{k,j}$, $\partial_\sigma \mu = 0$, $\partial_\theta \log v = g_{k,n}$, and $\partial_\sigma \log v = 2/\sigma$, which gives (70). The second moment

$$\mathbb{E}_\vartheta Y_{k,j}^2 = \rho_k^{2j} u_{0,k}^2 + \frac{\sigma^2 q_k^2}{2\theta\lambda_k}(1 - \rho_k^{2j})$$

then yields (71) after summing the geometric series. □

9.4.4 Asymptotic regime and information geometry

Let

$$\Lambda_N := \sum_{k=1}^{N} \lambda_k.$$

The local scale matrix and its inverse are

$$D_{N,n} := \operatorname{diag}\left(\Lambda_N^{-1/2}, (Nn)^{-1/2}\right), \qquad A_{N,n} := D_{N,n}^{-1}. \tag{72}$$

Assumption 9.6 (Joint spectral high-frequency regime)**.** *As $N, n \to \infty$ with $\Delta_n = T/n$:*

(i) $\Lambda_N \to \infty$ *and* $\lambda_N/\Lambda_N \to 0$*;*
(ii) $\lambda_N \Delta_n \to 0$*;*
(iii) the known initial condition satisfies

$$R_N(u_0, Q) := \sum_{k=1}^{N} \frac{\lambda_k u_{0,k}^2}{q_k^2} = o(\Lambda_N). \tag{73}$$

Condition (i) prevents one Fourier coordinate from carrying a non-negligible fraction of total drift information. Condition (ii) resolves the fastest observed relaxation time. Condition (iii) excludes an initial deterministic signal whose information is of the same or larger order than the stochastic equilibrium contribution. It holds for $u_0 = 0$ and under many weighted smoothness conditions. If R_N/Λ_N has a positive finite limit, the same analysis holds with an additional deterministic term in the drift information; see Remark 9.2.

Lemma 9.4 (Uniform small-mesh expansions)**.** *Under Assumption 9.6, uniformly in $k \le N$ and $\theta \in [\underline{\theta}, \overline{\theta}]$,*

$$h_{k,n}(\theta) = \Delta_n\{1 - \theta\lambda_k\Delta_n + O((\lambda_k\Delta_n)^2)\}, \tag{74}$$

$$g_{k,n}(\theta) = -\lambda_k\Delta_n + \frac{1}{3}\theta\lambda_k^2\Delta_n^2 + O(\lambda_k^4\Delta_n^4). \tag{75}$$

In particular,

$$\frac{n}{2}\sum_{k=1}^{N} g_{k,n}(\theta)^2 = O\left(\Delta_n\sum_{k=1}^{N}\lambda_k^2\right) = o(\Lambda_N), \tag{76}$$

$$\frac{n}{\sigma}\sum_{k=1}^{N} g_{k,n}(\theta) = -\frac{T}{\sigma}\Lambda_N\{1 + o(1)\}. \tag{77}$$

Proof. Apply Taylor expansion to $e^{-2\theta\lambda_k\Delta_n}$ and to $2x/(e^{2x}-1)$. Uniformity follows from $\lambda_N\Delta_n \to 0$. Since $\sum\lambda_k^2 \le \lambda_N\Lambda_N$, the first information remainder is $o(\Lambda_N)$. □

Theorem 9.11 (Mixed Fisher-information limit)**.** *Under Assumption 9.6,*

$$D_{N,n}I_{N,n}(\vartheta_0)D_{N,n} \longrightarrow J(\vartheta_0) := \begin{pmatrix} \dfrac{T}{2\theta_0} & 0 \\ 0 & \dfrac{2}{\sigma_0^2} \end{pmatrix}. \tag{78}$$

Thus the drift and scale information rates are, respectively,

$$\Lambda_N \qquad \text{and} \qquad Nn.$$

The asymptotic correlation vanishes because

$$\frac{\Lambda_N}{\sqrt{\Lambda_N}\sqrt{Nn}} = \sqrt{\frac{\Lambda_N}{Nn}} \le \sqrt{\frac{\lambda_N\Delta_n}{T}} \longrightarrow 0. \tag{79}$$

Proof. By (71), Riemann-sum approximation of each exact OU transition gives

$$M_{N,n}(\vartheta_0) = \sum_{k=1}^{N}\left\{\frac{T\lambda_k}{2\theta_0} - \frac{1-e^{-2\theta_0\lambda_k T}}{4\theta_0^2} + \frac{\lambda_k u_{0,k}^2}{2\theta_0\sigma_0^2 q_k^2}(1-e^{-2\theta_0\lambda_k T})\right\} + o(\Lambda_N).$$

The middle sum is $O(N) = o(\Lambda_N)$ because $\lambda_k \to \infty$ implies $\Lambda_N/N \to \infty$ by Cesàro averaging. The initial term is $o(\Lambda_N)$ by (73). Hence $M_{N,n}/\Lambda_N \to T/(2\theta_0)$. Lemma 9.4 makes the variance contribution to $I_{\theta\theta}$ negligible. The scale entry is exact, and (79) removes the normalised cross entry. □

Remark 9.2 (Non-negligible known initial signal)*.* If

$$\Lambda_N^{-1}\sum_{k\le N}\frac{\lambda_k u_{0,k}^2}{q_k^2} \longrightarrow \chi_0 \in (0,\infty),$$

then the upper-left entry of J becomes

$$\frac{T}{2\theta_0} + \frac{\chi_0}{2\theta_0\sigma_0^2}.$$

The initial condition adds information because it is known and its deterministic decay depends on θ. The zero-start or negligible-start formulation is retained as the main theorem because it isolates information generated by the stochastic field itself.

9.4.5 Central sequence and LAN

Let

$$S_{N,n}(\vartheta_0) = \nabla_\vartheta\ell_{N,n}(\vartheta_0), \qquad \Delta_{N,n} := D_{N,n}S_{N,n}(\vartheta_0).$$

Lemma 9.5 (Score central limit theorem)**.** *Under Assumption 9.6,*

$$\Delta_{N,n} \Rightarrow \mathcal{N}_2(0, J(\vartheta_0)). \tag{80}$$

Moreover, the observed normalised information satisfies

$$-D_{N,n}\nabla_\vartheta^2\ell_{N,n}(\vartheta_0)D_{N,n} \xrightarrow{\mathbb{P}} J(\vartheta_0). \tag{81}$$

Proof. Throughout, expectations are under $P_{\vartheta_0}^{N,n}$ and C denotes a finite constant depending only on Θ, T, and the constant in $\lambda_N\Delta_n \le x_0$; it may change between displays. Write $\rho_k = \rho_{k,n}(\theta_0)$, $v_k = v_{k,n}(\vartheta_0)$, $g_k = g_{k,n}(\theta_0)$, $s_k^2 = \sigma_0^2 q_k^2/(2\theta_0\lambda_k)$ (the stationary modal variance), and let

$$\varepsilon_{k,j} = \frac{Y_{k,j+1} - \rho_k Y_{k,j}}{\sqrt{v_k}}$$

be the standardised innovations, which are i.i.d. $\mathcal{N}(0,1)$ and independent of $\mathcal{F}_{k,j}$.

Step 1: martingale-difference array. By (67)–(68), the per-transition scores at ϑ_0 are

$$s_{\theta,k,j} = -\frac{\lambda_k\Delta_n\rho_k\, Y_{k,j}}{\sqrt{v_k}}\,\varepsilon_{k,j} + \frac{g_k}{2}\,(\varepsilon_{k,j}^2 - 1), \qquad s_{\sigma,k,j} = \frac{1}{\sigma_0}(\varepsilon_{k,j}^2 - 1). \tag{82}$$

Since $Y_{k,j}$ is $\mathcal{F}_{k,j}$-measurable and $\mathbb{E}[\varepsilon_{k,j} \mid \mathcal{F}_{k,j}] = \mathbb{E}[\varepsilon_{k,j}^2 - 1 \mid \mathcal{F}_{k,j}] = 0$, the array $\xi_{k,j} := D_{N,n}(s_{\theta,k,j}, s_{\sigma,k,j})^\top$, ordered lexicographically in (k,j), is a square-integrable martingale-difference array with respect to the filtration generated by all modes up to the current cell (modes are independent, so enlarging the per-mode filtration preserves the martingale property).

Step 2: predictable quadratic variation. Using $\mathbb{E}[\varepsilon^2] = 1$, $\mathbb{E}[\varepsilon^3] = 0$, $\mathbb{E}[(\varepsilon^2-1)^2] = 2$, the summed conditional covariance matrix $V_{N,n} := \sum_{k,j}\mathbb{E}[\xi_{k,j}\xi_{k,j}^\top \mid \mathcal{F}_{k,j}]$ has entries

$$V_{N,n}^{\theta\theta} = \frac{1}{\Lambda_N}\sum_{k\le N}\sum_{j<n}\left\{\frac{\lambda_k^2\Delta_n^2\rho_k^2 Y_{k,j}^2}{v_k} + \frac{g_k^2}{2}\right\}, \qquad V_{N,n}^{\sigma\sigma} = \frac{2}{\sigma_0^2} \quad \text{(deterministic)}, \tag{83}$$

$$V_{N,n}^{\theta\sigma} = \frac{1}{\sqrt{\Lambda_N N n}}\,\frac{n}{\sigma_0}\sum_{k\le N} g_k \quad \text{(deterministic)}.$$

The cross term satisfies, by (77) and (79), $|V_{N,n}^{\theta\sigma}| \le C\sqrt{\Lambda_N/(Nn)} \le C\sqrt{\lambda_N\Delta_n/T} \to 0$. The g^2 contribution to $V^{\theta\theta}$ is $o(1)$ by (76). For the random part $W_N := \Lambda_N^{-1}\sum_{k,j}\lambda_k^2\Delta_n^2\rho_k^2 Y_{k,j}^2/v_k$, Proposition 9.14 (closed form (71)) and the Riemann-sum argument of Theorem 9.11 give $\mathbb{E}W_N \to T/(2\theta_0)$. For its concentration, decompose $Y_{k,j} = \rho_k^j u_{0,k} + Z_{k,j}$ with $(Z_{k,j})_j$ centred Gaussian, $\mathrm{Cov}(Z_{k,i}, Z_{k,j}) \le s_k^2\rho_k^{|i-j|}$. Gaussian second-moment algebra gives $\mathrm{Cov}(Y_{k,i}^2, Y_{k,j}^2) \le 2s_k^4\rho_k^{2|i-j|} + 4u_{0,k}^2 s_k^2\rho_k^{|i-j|}$, and since $\sum_{i,j<n}\rho_k^{|i-j|} \le 2n/(1-\rho_k) \le Cn/(\theta_0\lambda_k\Delta_n)$ (using $1-e^{-x} \ge cx$ on $[0,x_0]$),

$$\mathrm{Var}\Big(\sum_{j<n} Y_{k,j}^2\Big) \le C\Big(\frac{s_k^4\, n}{\lambda_k\Delta_n} + \frac{u_{0,k}^2 s_k^2\, n}{\lambda_k\Delta_n}\Big).$$

Multiplying by the squared coefficient $(\lambda_k^2\Delta_n^2\rho_k^2/v_k)^2 \le C\lambda_k^4\Delta_n^2/(\sigma_0^4 q_k^4)$ (since $v_k \ge c\sigma_0^2 q_k^2\Delta_n$ by (74)) and using $s_k^2 = \sigma_0^2 q_k^2/(2\theta_0\lambda_k)$,

$$\mathrm{Var}\Big(\text{mode-}k\text{ term of }\Lambda_N W_N\Big) \le C\Big(\lambda_k T + \lambda_k T\cdot\frac{\lambda_k u_{0,k}^2}{q_k^2}\Big).$$

By independence across modes,

$$\mathrm{Var}(W_N) \le \frac{CT}{\Lambda_N^2}\Big(\Lambda_N + \lambda_N\sum_{k\le N}\frac{\lambda_k u_{0,k}^2}{q_k^2}\Big) = \frac{CT}{\Lambda_N} + CT\,\frac{\lambda_N}{\Lambda_N}\cdot\frac{R_N(u_0,Q)}{\Lambda_N} \longrightarrow 0$$

by Assumption 9.6(i),(iii). Hence $V_{N,n} \xrightarrow{\mathbb{P}} J(\vartheta_0)$.

Step 3: conditional Lyapunov condition. It suffices that $\sum_{k,j}\mathbb{E}\,\|\xi_{k,j}\|^4 \to 0$. Using $(a+b)^4 \le 8(a^4+b^4)$, $\mathbb{E}\varepsilon^8 < \infty$, $\mathbb{E}Y_{k,j}^4 \le C(s_k^4 + u_{0,k}^4\rho_k^{4j})$, and $v_k \ge c\sigma_0^2 q_k^2\Delta_n$:

$$\begin{aligned}
\sum_{k,j}\mathbb{E}\,\xi_{\theta,k,j}^4 &\le \frac{C}{\Lambda_N^2}\sum_{k\le N}\Big\{n\,\frac{\lambda_k^4\Delta_n^4 s_k^4}{v_k^2} + \frac{\lambda_k^4\Delta_n^4 u_{0,k}^4}{v_k^2}\sum_{j<n}\rho_k^{4j} + n\,g_k^4\Big\}\\
&\le \frac{C}{\Lambda_N^2}\Big\{T\Delta_n\sum_k\lambda_k^2 + \lambda_N\Delta_n\Big(\sum_k\frac{\lambda_k u_{0,k}^2}{q_k^2}\Big)^2 + T\Delta_n^3\sum_k\lambda_k^4\Big\}\\
&\le C\Big\{\frac{T\lambda_N\Delta_n}{\Lambda_N} + \lambda_N\Delta_n\Big(\frac{R_N}{\Lambda_N}\Big)^2 + \frac{T(\lambda_N\Delta_n)^3}{\Lambda_N}\Big\} \longrightarrow 0,
\end{aligned}$$

where the middle term used $\sum_k\lambda_k^2 u_{0,k}^4/q_k^4 \le (\sum_k\lambda_k u_{0,k}^2/q_k^2)^2$ and $\sum_j\rho_k^{4j} \le C/(\lambda_k\Delta_n)$. Similarly $\sum_{k,j}\mathbb{E}\,\xi_{\sigma,k,j}^4 \le C(Nn)^{-2}\cdot Nn = C/(Nn) \to 0$, and mixed fourth moments are controlled by Cauchy–Schwarz. The martingale central limit theorem applies in the form of Hall and Heyde [19, Corollary 3.1]; its three conditions are verified above: (i) $(\xi_{k,j})$ is a square-integrable martingale-difference array with respect to the nested per-(N,n) filtrations (Step 1); (ii) the summed conditional covariance converges in probability to the *deterministic*

matrix $J(\vartheta_0)$ (Step 2), so the nesting condition needed only for stable convergence with random limits is not required; (iii) the conditional Lindeberg condition holds, being implied by the Lyapunov fourth-moment bound of Step 3. (For vector arrays, apply the corollary to $a^\top \xi_{k,j}$ for each $a \in \mathbb{R}^2$ and use Cramér–Wold.) This yields $\Delta_{N,n} \Rightarrow \mathcal{N}_2(0, J(\vartheta_0))$, which is (80).

Step 4: observed information. Direct differentiation of (67)–(68) shows that each entry of $-\nabla^2_\vartheta \ell_{N,n}(\vartheta_0)$ equals the corresponding entry of the summed conditional information plus a centred term of the form $\sum_{k,j}\{a_{k,n} Y_{k,j}\varepsilon_{k,j} + b_{k,n}(\varepsilon^2_{k,j} - 1)\}$ with deterministic coefficients satisfying $|a_{k,n}| \le C\lambda_k^2 \Delta_n^2/\sqrt{v_k}$ and $|b_{k,n}| \le C(\lambda_k\Delta_n)^2 + C\lambda_k\Delta_n/\sigma_0$ (the largest coefficients arise in the $\theta\theta$ entry; the $\theta\sigma$ and $\sigma\sigma$ entries carry one factor σ_0^{-1} or σ_0^{-2} and at most one factor g_k). These are martingale sums, and after the $D_{N,n} \cdot D_{N,n}$ normalisation their second moments are bounded by the same quantities as in Steps 2–3, hence vanish. Combining with Step 2 gives (81). □

Lemma 9.6 (Uniform local quadratic remainder)**.** *For every compact $K \subset \mathbb{R}^2$,*

$$\sup_{h\in K} \left| \ell_{N,n}(\vartheta_0 + D_{N,n}h) - \ell_{N,n}(\vartheta_0) - h^\top \Delta_{N,n} + \frac{1}{2} h^\top J(\vartheta_0) h \right| \xrightarrow{\mathbb{P}} 0. \tag{84}$$

Proof. Fix a compact $K \subset \mathbb{R}^2$ and let $R_K := \sup_{h\in K} \|h\|$. For N, n large, $\vartheta_0 + D_{N,n}h$ lies in the interior of Θ for all $h \in K$ because $D_{N,n} \to 0$. A third-order Taylor expansion of $h \mapsto \ell_{N,n}(\vartheta_0 + D_{N,n}h)$ around $h = 0$ gives, for every $h \in K$,

$$\ell_{N,n}(\vartheta_0 + D_{N,n}h) - \ell_{N,n}(\vartheta_0) = h^\top \Delta_{N,n} + \tfrac{1}{2} h^\top \big[D_{N,n} \nabla^2_\vartheta \ell_{N,n}(\vartheta_0) D_{N,n}\big] h + \tfrac{1}{6}\, \mathrm{T}_{N,n}(\bar h)[h,h,h], \tag{85}$$

where $\bar h$ lies on the segment $[0, h]$ and $\mathrm{T}_{N,n}(\bar h)$ is the trilinear form with entries $D^{(a)}_{N,n} D^{(b)}_{N,n} D^{(c)}_{N,n}\, \partial^3_{abc} \ell_{N,n}(\vartheta_0 + D_{N,n}\bar h)$, $a, b, c \in \{\theta, \sigma\}$, $D^{(\theta)}_{N,n} = \Lambda_N^{-1/2}$, $D^{(\sigma)}_{N,n} = (Nn)^{-1/2}$. By (81) (Step 4 of the proof of Lemma 9.5), the quadratic term is $-\frac{1}{2} h^\top J(\vartheta_0) h + o_{\mathbb{P}}(1)$ uniformly on K (a fixed quadratic form applied to a compact set). It therefore suffices to show

$$\sup_{\|\bar h\| \le R_K} \max_{a,b,c} \left| D^{(a)}_{N,n} D^{(b)}_{N,n} D^{(c)}_{N,n}\, \partial^3_{abc} \ell_{N,n}(\vartheta_0 + D_{N,n}\bar h) \right| = o_{\mathbb{P}}(1). \tag{86}$$

Step 1: structure of third derivatives. Per transition the log-density is $-\frac{1}{2}\log v(\vartheta) - r(\theta)^2/(2v(\vartheta)) + \text{const}$ with $r(\theta) = Y_{k,j+1} - \rho_{k,n}(\theta) Y_{k,j}$ and $v = \sigma^2 q_k^2 h_{k,n}(\theta)$. All θ-derivatives of r obey $|\partial^m_\theta r| = (\lambda_k\Delta_n)^m \rho_{k,n}(\theta) |Y_{k,j}|$, and, uniformly on Θ for $\lambda_k\Delta_n \le x_0$,

$$|\partial^m_\theta \log v| \le C(\lambda_k\Delta_n)^m \quad (m = 1,2,3), \qquad |\partial_\sigma \log v| = 2/\sigma, \quad \partial^2_\sigma \log v = -2/\sigma^2, \ \ldots \tag{87}$$

(the first bound extends (75) to two more derivatives of $x \mapsto \log\{(1 - e^{-2\theta x})/(2\theta x)\}$, which is analytic with all derivatives $O(x^m)$ at 0, uniformly for θ compact). Expanding ∂^3_{abc} of the per-transition density by the chain rule, every resulting term is a product of factors from (87) and of r^2/v, $\partial_\theta r \cdot r/v$, $(\partial_\theta r)^2/v$, $\partial^2_\theta r \cdot r/v$, etc. Collecting powers, one obtains the per-transition envelope, valid uniformly on Θ:

$$|\partial^3_{abc} \ell_{k,j}(\vartheta)| \le C\,(\lambda_k\Delta_n)^{m_\theta(a,b,c)}\, \sigma^{-m_\sigma(a,b,c)} \Big(1 + \frac{r(\theta)^2}{v(\vartheta)} + \frac{\lambda_k\Delta_n\, Y^2_{k,j}}{v(\vartheta)/\Delta_n}\Big), \tag{88}$$

where m_θ counts θ-indices and m_σ counts σ-indices among $\{a, b, c\}$. (For the mean part, each θ-index contributes one factor $\lambda_k\Delta_n$; terms with $(\partial_\theta r)^2/v = \lambda_k^2\Delta_n^2\rho^2 Y^2/v$ are absorbed in the last bracket using $v \ge c\sigma^2 q_k^2 \Delta_n$ and $Y^2/(q_k^2) \cdot \lambda_k\Delta_n \le$ the stated form.)

Step 2: uniformity over the shrinking box. For $\vartheta = \vartheta_0 + D_{N,n}\bar h$, $\|\bar h\| \le R_K$: $|\theta - \theta_0| \le R_K \Lambda_N^{-1/2}$ and $|\sigma - \sigma_0| \le R_K (Nn)^{-1/2}$, so $\sigma^{-1} \le C$, and for every $k \le N$,

$$\frac{v(\vartheta_0)}{v(\vartheta)} \le C, \qquad \frac{r(\theta)^2}{v(\vartheta)} \le C\Big(\varepsilon^2_{k,j} + \frac{(\rho_{k,n}(\theta_0) - \rho_{k,n}(\theta))^2 Y^2_{k,j}}{v(\vartheta_0)}\Big) \le C\Big(\varepsilon^2_{k,j} + \frac{\lambda_k^2\Delta_n^2 Y^2_{k,j}}{\Lambda_N\, v(\vartheta_0)} R_K^2\Big),$$

using the mean value theorem for $\rho_{k,n}$ and $e^{\pm\lambda_k\Delta_n R_K \Lambda_N^{-1/2}} \le C$. Hence the envelope (88) holds with ϑ-free right-hand side

$$E_{k,j} := C(\lambda_k\Delta_n)^{m_\theta}\Big(1 + \varepsilon^2_{k,j} + \frac{\lambda_k\Delta_n\, Y^2_{k,j}}{\sigma_0^2 q_k^2}\Big), \qquad \mathbb{E}E_{k,j} \le C(\lambda_k\Delta_n)^{m_\theta}\Big(1 + \frac{u^2_{0,k}\rho_k^{2j}}{s_k^2}\Big) \tag{89}$$

(using $\mathbb{E}Y^2_{k,j} \le u^2_{0,k}\rho_k^{2j} + s_k^2$ and $\lambda_k\Delta_n s_k^2/(\sigma_0^2 q_k^2) = \Delta_n/(2\theta_0) \cdot \lambda_k \le C$ after multiplying by $\lambda_k\Delta_n \le x_0$; the initial-condition part sums over j to $Cu^2_{0,k}/(s_k^2\lambda_k\Delta_n) = C\lambda_k u^2_{0,k}/(q_k^2) \cdot (\sigma_0^2\lambda_k\Delta_n)^{-1} \cdot 2\theta_0\lambda_k\Delta_n \cdot (\text{const}) \le C\lambda_k u^2_{0,k}/q_k^2$).

Step 3: rate bookkeeping. Summing (89) over $k \le N$, $j < n$ and applying the scale factors, Markov's inequality reduces (86) to the vanishing of the deterministic quantities

$$D_{abc} := D^{(a)}_{N,n} D^{(b)}_{N,n} D^{(c)}_{N,n} \Big\{ n\Delta_n^{m_\theta} \sum_{k\le N} \lambda_k^{m_\theta} + \Delta_n^{m_\theta} \sum_{k\le N} \lambda_k^{m_\theta} \cdot \frac{\lambda_k u^2_{0,k}}{q_k^2} \Big\}.$$

Using $\sum_k \lambda_k^m \le \lambda_N^{m-1}\Lambda_N$ and $R_N = \sum_k \lambda_k u_{0,k}^2/q_k^2 = o(\Lambda_N)$:

$$
\begin{aligned}
(\theta\theta\theta): &\quad \Lambda_N^{-3/2}\, T\Delta_n^2 \textstyle\sum \lambda_k^3 \le T(\lambda_N\Delta_n)^2\Lambda_N^{-1/2} \to 0;\\
(\theta\theta\sigma): &\quad \Lambda_N^{-1}(Nn)^{-1/2}\, T\Delta_n \textstyle\sum \lambda_k^2 \le T\lambda_N\Delta_n\,(Nn)^{-1/2} \to 0;\\
(\theta\sigma\sigma): &\quad \Lambda_N^{-1/2}(Nn)^{-1}\, T\textstyle\sum\lambda_k = T\sqrt{\Lambda_N/(Nn)}\,(Nn)^{-1/2} \le T\sqrt{\lambda_N\Delta_n/T}\,(Nn)^{-1/2} \to 0;\\
(\sigma\sigma\sigma): &\quad (Nn)^{-3/2}\, CNn = C(Nn)^{-1/2} \to 0,
\end{aligned}
$$

and the u_0-terms are smaller by the same comparisons with $R_N = o(\Lambda_N)$ in place of one factor $T\Lambda_N/\Delta_n^{-1}$ (e.g. for $(\theta\theta\theta)$: $\Lambda_N^{-3/2}\Delta_n^3\lambda_N^2 R_N \cdot \Delta_n^{-1}\cdot(\text{const}) \le (\lambda_N\Delta_n)^2\Lambda_N^{-3/2}R_N \to 0$). Since the bounds are monotone in $\|\bar h\| \le R_K$ and polynomial in R_K, the convergence is uniform on K; (84) follows from (85). □

Theorem 9.12 (Local asymptotic normality; establishes Theorem 5.1)**.** *Under Assumption 9.6, the sequence of experiments is LAN at every interior ϑ_0. For each fixed $h \in \mathbb{R}^2$ such that $\vartheta_0 + D_{N,n}h \in \Theta$ eventually,*

$$
\log \frac{dP^{N,n}_{\vartheta_0 + D_{N,n}h}}{dP^{N,n}_{\vartheta_0}} = h^\top \Delta_{N,n} - \frac{1}{2} h^\top J(\vartheta_0) h + o_{P^{N,n}_{\vartheta_0}}(1), \tag{90}
$$

where

$$
\Delta_{N,n} \Rightarrow \mathcal{N}_2(0, J(\vartheta_0)).
$$

The expansion is uniform for h in compact subsets of $\mathbb{R}^2$.

Interpretation. The exact discrete experiment converges locally to a two-coordinate Gaussian shift experiment. The drift coordinate aggregates relaxation information across modes, while the scale coordinate aggregates innovation squares across every mode–time cell. Their limiting orthogonality is not assumed; it is a consequence of resolving the highest observed frequency.

9.4.6 Consistency of the exact MLE

Lemma 9.7 (Global Kullback–Leibler separation)**.** *For every $\varepsilon > 0$, there exist constants $c_\varepsilon > 0$ and N_ε such that, for $N, n \ge N_\varepsilon$,*

$$
\inf_{|\theta-\theta_0|\ge\varepsilon,\ \sigma\in[\underline\sigma,\overline\sigma]} \mathbb{E}_{\vartheta_0}\big[\ell_{N,n}(\vartheta_0) - \ell_{N,n}(\theta,\sigma)\big] \ge c_\varepsilon \Lambda_N, \tag{91}
$$

$$
\inf_{|\sigma-\sigma_0|\ge\varepsilon,\ \theta\in[\underline\theta,\overline\theta]} \mathbb{E}_{\vartheta_0}\big[\ell_{N,n}(\vartheta_0) - \ell_{N,n}(\theta,\sigma)\big] \ge c_\varepsilon Nn. \tag{92}
$$

Proof. Write $\rho_k(\theta) = \rho_{k,n}(\theta)$, $v_k(\vartheta) = v_{k,n}(\vartheta)$ and abbreviate the true values ρ_k^0, v_k^0. Taking expectations of the per-transition Gaussian log-likelihood difference and using $\mathbb{E}[(Y_{k,j+1} - \rho_k(\theta)Y_{k,j})^2 \mid \mathcal{F}_{k,j}] = v_k^0 + (\rho_k^0 - \rho_k(\theta))^2 Y_{k,j}^2$,

$$
\mathbb{E}_{\vartheta_0}\big[\ell_{N,n}(\vartheta_0) - \ell_{N,n}(\vartheta)\big] = \underbrace{\frac{Nn}{2}\,\overline{\psi}_{N,n}(\vartheta)}_{\text{scale part}} + \underbrace{\sum_{k\le N} \frac{(\rho_k^0 - \rho_k(\theta))^2}{2v_k(\vartheta)} \sum_{j<n} \mathbb{E}Y_{k,j}^2}_{\text{drift part}}, \tag{93}
$$

where $\overline{\psi}_{N,n}(\vartheta) := (Nn)^{-1}\sum_{k,j}\psi\big(v_k^0/v_k(\vartheta)\big)$ and $\psi(x) := x - 1 - \log x \ge 0$. Both parts are non-negative, so each lower bound may be established separately with the other parameter unrestricted.

Scale separation (92). By (74), uniformly in $k \le N$ and $\theta \in [\underline\theta, \overline\theta]$, $h_{k,n}(\theta_0)/h_{k,n}(\theta) = 1 + O(\lambda_N\Delta_n)$, hence

$$
\frac{v_k^0}{v_k(\vartheta)} = \frac{\sigma_0^2}{\sigma^2}\big(1 + O(\lambda_N\Delta_n)\big) \qquad \text{uniformly.}
$$

If $|\sigma - \sigma_0| \ge \varepsilon$ then $|\sigma_0^2/\sigma^2 - 1| \ge \delta_\varepsilon > 0$ with δ_ε determined by ε and the compact $[\underline\sigma, \overline\sigma]$; consequently, for N, n large enough that the $O(\lambda_N\Delta_n)$ correction is at most $\delta_\varepsilon/2$ in relative size, $|v_k^0/v_k - 1| \ge \delta_\varepsilon/2$ for every (k, j). Since $\psi(x) \ge c\min\{(x-1)^2, |x-1|\}$ and v_k^0/v_k ranges in a fixed compact of $(0,\infty)$, $\psi(v_k^0/v_k) \ge c_\varepsilon$, and the scale part of (93) is at least $c_\varepsilon Nn/2$. This proves (92) with the infimum over all $\theta \in [\underline\theta, \overline\theta]$, because the drift part is non-negative.

Drift separation (91). By the mean value theorem, $\rho_k^0 - \rho_k(\theta) = \lambda_k\Delta_n e^{-\tilde\theta\lambda_k\Delta_n}(\theta - \theta_0)$ for some $\tilde\theta$ between θ and θ_0; since $\lambda_k\Delta_n \le \lambda_N\Delta_n \le x_0$, $e^{-\tilde\theta\lambda_k\Delta_n} \ge c > 0$, so $|\theta - \theta_0| \ge \varepsilon$ implies $(\rho_k^0 - \rho_k(\theta))^2 \ge c\varepsilon^2\lambda_k^2\Delta_n^2$. For the second moments, with $s_k^2 = \sigma_0^2 q_k^2/(2\theta_0\lambda_k)$,

$$
\mathbb{E}Y_{k,j}^2 \ge s_k^2\big(1 - e^{-2\theta_0\lambda_k j\Delta_n}\big) \ge \tfrac{1}{2}s_k^2 \qquad \text{whenever } j \ge j_k := \left\lceil \frac{\log 2}{2\theta_0\lambda_k\Delta_n} \right\rceil.
$$

Fix λ^* with $\lambda^* T \geq 2\log 2/\theta_0$; compact resolvent implies only finitely many eigenvalues lie below λ^*, so $\sum_{k:\lambda_k\geq\lambda^*,\,k\leq N}\lambda_k \geq \Lambda_N - C^* \geq \frac{1}{2}\Lambda_N$ for N large. For such k, $j_k \leq n/2$, hence $\sum_{j<n}\mathbb{E}Y_{k,j}^2 \geq \frac{1}{4}ns_k^2$. Using $v_k(\vartheta) \leq \overline{\sigma}^2 q_k^2\Delta_n(1+o(1)) \leq C\overline{\sigma}^2 q_k^2\Delta_n$,

$$\text{drift part} \;\geq\; \sum_{\substack{k\leq N\\ \lambda_k\geq\lambda^*}} \frac{c\varepsilon^2\lambda_k^2\Delta_n^2}{C\overline{\sigma}^2 q_k^2\Delta_n}\cdot\frac{n\sigma_0^2 q_k^2}{8\theta_0\lambda_k} \;=\; \frac{c\,\sigma_0^2}{8C\overline{\sigma}^2\theta_0}\,\varepsilon^2\,T\sum_{\substack{k\leq N\\ \lambda_k\geq\lambda^*}}\lambda_k \;\geq\; c_\varepsilon\,\Lambda_N$$

for $N, n \geq N_\varepsilon$, uniformly over $\sigma \in [\underline{\sigma}, \overline{\sigma}]$, since the scale part is non-negative. This proves (91). □

Theorem 9.13 (Consistency)**.** *Under Assumption 9.6, every exact conditional MLE defined by Proposition 9.13 satisfies*

$$(\widehat{\theta}_{N,n}, \widehat{\sigma}_{N,n}) \xrightarrow{\mathbb{P}} (\theta_0, \sigma_0). \tag{94}$$

Proof. In addition to Assumption 9.6 we use the spectral-growth condition

$$N\sqrt{\lambda_N} = o(\Lambda_N), \tag{95}$$

which is automatic in the model class of the main text: under Weyl growth $\lambda_k \asymp k^{2/d}$ one has $\Lambda_N \asymp N^{1+2/d}$ while $N\sqrt{\lambda_N} \asymp N^{1+1/d}$, and $1/d < 2/d$. It enters only through the two displays marked below. By Proposition 9.13 it suffices to study the profile criterion

$$\Phi_{N,n}(\theta) := Nn\log\widehat{\sigma}_{N,n}^2(\theta) + n\sum_{k\leq N}\log h_{k,n}(\theta), \qquad \widehat{\theta}_{N,n} \in \arg\min_\theta \Phi_{N,n}(\theta).$$

Write $h_k^0 = h_{k,n}(\theta_0)$, $\rho_k^0 = \rho_{k,n}(\theta_0)$, $v_k^0 = \sigma_0^2 q_k^2 h_k^0$, and decompose $r_{k,j}(\theta) = \sqrt{v_k^0}\,\varepsilon_{k,j} + (\rho_k^0 - \rho_{k,n}(\theta))Y_{k,j}$, so that

$$\widehat{\sigma}_{N,n}^2(\theta) = A(\theta) + B(\theta) + C(\theta), \tag{96}$$

with

$$A(\theta) = \frac{\sigma_0^2}{Nn}\sum_{k,j}\frac{h_k^0}{h_{k,n}(\theta)}\varepsilon_{k,j}^2, \quad B(\theta) = \frac{2}{Nn}\sum_{k,j}\frac{(\rho_k^0-\rho_{k,n}(\theta))\sqrt{v_k^0}}{q_k^2 h_{k,n}(\theta)}\varepsilon_{k,j}Y_{k,j}, \quad C(\theta) = \frac{1}{Nn}\sum_{k,j}\frac{(\rho_k^0-\rho_{k,n}(\theta))^2 Y_{k,j}^2}{q_k^2 h_{k,n}(\theta)} \geq 0.$$

Let $m^2(\theta) := \mathbb{E}\widehat{\sigma}^2(\theta) = \sigma_0^2 H(\theta) + D(\theta)$, where $H(\theta) := N^{-1}\sum_k h_k^0/h_{k,n}(\theta)$ and $D(\theta) := \mathbb{E}C(\theta)$. Note $H(\theta_0) = 1$, $D(\theta_0) = 0$, and by (65), $|\log(h_k^0/h_{k,n}(\theta))| \leq C\lambda_k\Delta_n$ on Θ, so $c \leq H \leq C$ and, with $\beta_{N,n} := \Lambda_N/(Nn)$,

$$\sup_\theta |H(\theta) - 1| \leq CT\beta_{N,n}, \qquad \sup_\theta D(\theta) \leq C\beta_{N,n}, \tag{97}$$

the second from $(\rho_k^0-\rho_k)^2\mathbb{E}Y_{k,j}^2/(q_k^2 h_k) \leq C\lambda_k^2\Delta_n^2(s_k^2 + u_{0,k}^2\rho_k^{2j})/(q_k^2\Delta_n)$ and Assumption 9.6(iii).

Step 1: deterministic profile separation. Let $\bar{\Phi}(\theta) := Nn\log m^2(\theta) + n\sum_k \log h_{k,n}(\theta)$. By concavity of log, $Nn\log H(\theta) \geq n\sum_k \log(h_k^0/h_{k,n}(\theta))$, hence

$$\bar{\Phi}(\theta) - \bar{\Phi}(\theta_0) = Nn\log H(\theta) + n\sum_k \log\frac{h_{k,n}(\theta)}{h_k^0} + Nn\log\Big(1 + \frac{D(\theta)}{\sigma_0^2 H(\theta)}\Big) \;\geq\; c\,Nn\,D(\theta),$$

using $\log(1+u) \geq u/2$ for $0 \leq u \leq 1$ (valid for large N, n by (97)). The drift computation in the proof of Lemma 9.7 gives, for $|\theta - \theta_0| \geq \varepsilon$, $NnD(\theta) \geq c_\varepsilon\Lambda_N$; hence

$$\inf_{|\theta-\theta_0|\geq\varepsilon}\big[\bar{\Phi}(\theta) - \bar{\Phi}(\theta_0)\big] \;\geq\; c_\varepsilon\Lambda_N. \tag{98}$$

Step 2: uniform relative fluctuations at scale $\beta_{N,n}$. Let $\zeta(\theta) := \widehat{\sigma}^2(\theta) - m^2(\theta)$. We claim

$$\sup_\theta |\zeta(\theta) - \zeta(\theta_0)| = o_{\mathbb{P}}(\beta_{N,n}), \qquad \zeta(\theta_0) = O_{\mathbb{P}}\big((Nn)^{-1/2}\big). \tag{99}$$

The second statement is immediate: $\zeta(\theta_0) = \sigma_0^2(Nn)^{-1}\sum_{k,j}(\varepsilon_{k,j}^2 - 1)$ has variance $2\sigma_0^4/(Nn)$. For the first, treat the three parts of (96). *(A)* With $\omega_k := \sum_j(\varepsilon_{k,j}^2 - 1)$, $\mathbb{E}|\omega_k| \leq \sqrt{2n}$, and $|h_k^0/h_{k,n}(\theta) - 1| \leq C\lambda_k\Delta_n$,

$$\mathbb{E}\sup_\theta |A(\theta) - \mathbb{E}A(\theta) - (A(\theta_0) - \mathbb{E}A(\theta_0))| \leq \frac{C}{Nn}\sum_k \lambda_k\Delta_n\sqrt{2n} = \frac{C\Delta_n\Lambda_N}{N\sqrt{n}} = C\beta_{N,n}\cdot\frac{T}{\sqrt{n}} = o(\beta_{N,n}).$$

(B) $B(\theta_0) = 0$, and pointwise-in-θ the coefficient of $M_k := \sum_j \varepsilon_{k,j}Y_{k,j}$ is bounded by $C\sigma_0\lambda_k\sqrt{\Delta_n}/q_k$; with $\mathbb{E}M_k^2 = \sum_j \mathbb{E}Y_{k,j}^2 \leq ns_k^2 + Cu_{0,k}^2/(\lambda_k\Delta_n)$,

$$\mathbb{E}\sup_\theta |B(\theta)| \leq \frac{C}{Nn}\Big\{\sum_k \sigma_0^2\sqrt{\lambda_k T} + \sum_k \frac{\sigma_0\sqrt{\lambda_k}\,u_{0,k}}{q_k}\Big\} \leq \frac{C}{Nn}\Big\{N\sqrt{\lambda_N T} + \sqrt{NR_N}\Big\},$$

by Cauchy–Schwarz. Both terms are $o(\beta_{N,n})$: dividing by $\beta_{N,n} = \Lambda_N/(Nn)$ leaves $N\sqrt{\lambda_N T}/\Lambda_N \to 0$ by (95), and $\sqrt{NR_N}/\Lambda_N \le \sqrt{N\,o(\Lambda_N)}/\Lambda_N \to 0$ since $N \le C\Lambda_N$ ($\lambda_k \ge \lambda_1 > 0$). *(C)* The coefficient of $\sum_j (Y_{k,j}^2 - \mathbb{E}Y_{k,j}^2)$ is bounded by $C\lambda_k^2\Delta_n/q_k^2$ uniformly in θ, and (variance bound in Step 2 of the proof of Lemma 9.5) $\mathrm{sd}\big(\sum_j Y_{k,j}^2\big) \le C(s_k^2 + u_{0,k}s_k)\sqrt{n/(\lambda_k\Delta_n)}$; hence

$$\mathbb{E}\sup_\theta |C(\theta) - D(\theta)| \le \frac{C}{Nn}\Big\{N\sqrt{\lambda_N T} + \sqrt{T}\sqrt{\Lambda_N R_N}\Big\} = o(\beta_{N,n}),$$

the last comparison as in (B) plus $\sqrt{\Lambda_N R_N}/\Lambda_N = \sqrt{R_N/\Lambda_N} \to 0$. This proves (99).

Step 3: assembly. On $|\theta - \theta_0| \ge \varepsilon$, using $m^2, \widehat{\sigma}^2(\theta_0)$ bounded away from 0 and ∞ in probability,

$$\begin{aligned}
\Phi(\theta) - \Phi(\theta_0) &= \bar{\Phi}(\theta) - \bar{\Phi}(\theta_0) + Nn\Big\{\log\frac{\widehat{\sigma}^2(\theta)}{m^2(\theta)} - \log\frac{\widehat{\sigma}^2(\theta_0)}{m^2(\theta_0)}\Big\} \\
&\ge c_\varepsilon\Lambda_N - CNn\Big\{|\zeta(\theta) - \zeta(\theta_0)| + |\zeta(\theta_0)|\,\big|m^2(\theta) - m^2(\theta_0)\big|\Big\} \\
&\ge c_\varepsilon\Lambda_N - Nn\cdot o_{\mathbb{P}}(\beta_{N,n}) - CNn\cdot O_{\mathbb{P}}\big((Nn)^{-1/2}\big)\cdot C\beta_{N,n} \;=\; c_\varepsilon\Lambda_N - o_{\mathbb{P}}(\Lambda_N),
\end{aligned}$$

uniformly in θ, using (97) for the m^2-difference and (99). Hence $\mathbb{P}(|\widehat{\theta}_{N,n} - \theta_0| \ge \varepsilon) \to 0$. Finally, $\widehat{\sigma}_{N,n}^2 = \widehat{\sigma}^2(\widehat{\theta}_{N,n}) = m^2(\widehat{\theta}_{N,n}) + \zeta(\widehat{\theta}_{N,n}) = \sigma_0^2 + O(\beta_{N,n}) + o_{\mathbb{P}}(1) \to \sigma_0^2$ in probability, by (97) and (99). □

9.4.7 Efficient estimation

Theorem 9.14 (Asymptotic linearity and mixed normal limit; establishes Theorem 5.1)**.** *Under Assumption 9.6,*

$$A_{N,n}(\widehat{\vartheta}_{N,n} - \vartheta_0) = J(\vartheta_0)^{-1}\Delta_{N,n} + \mathrm{o}_{\mathbb{P}}(1). \tag{100}$$

Consequently,

$$\begin{pmatrix}\sqrt{\Lambda_N}(\widehat{\theta}_{N,n} - \theta_0)\\ \sqrt{Nn}(\widehat{\sigma}_{N,n} - \sigma_0)\end{pmatrix} \Rightarrow \mathcal{N}_2\left(0, \begin{pmatrix}\dfrac{2\theta_0}{T} & 0\\ 0 & \dfrac{\sigma_0^2}{2}\end{pmatrix}\right). \tag{101}$$

The two normalised estimators are asymptotically independent.

Proof. Consistency places the MLE in an interior shrinking neighbourhood with probability tending to one. Expand the score equation around ϑ_0:

$$0 = S_{N,n}(\vartheta_0) + \nabla_\vartheta^2\ell_{N,n}(\widetilde{\vartheta}_{N,n})(\widehat{\vartheta}_{N,n} - \vartheta_0).$$

Premultiply and postmultiply by the mixed scale matrices. Lemmas 9.5 and 9.6 imply convergence of the normalised Hessian to $-J(\vartheta_0)$ uniformly in the local neighbourhood. Solving gives (100); (101) follows from the score CLT. □

Theorem 9.15 (LAN efficiency; establishes Theorem 5.1)**.** *As stated in Theorem 5.1 of the main text.*

Proof. The LAN expansion identifies the limiting Gaussian shift experiment. Equation (100) shows that the MLE has the efficient influence $J^{-1}\Delta$. The convolution and local minimax conclusions are the standard decision-theoretic consequences of LAN for regular estimators. □

9.4.8 Polynomial likelihood tails and moment convergence

Theorem 9.16 (Polynomial-type likelihood-ratio bound)**.** *Assume the conditions of Theorem 9.12. For every $L > 0$, there exists $C_L < \infty$ such that, for all $r \ge 1$ and all sufficiently large (N, n),*

$$\mathbb{P}_{\vartheta_0}\Big(\Big\|A_{N,n}(\widehat{\vartheta}_{N,n} - \vartheta_0)\Big\| > r\Big) \le C_L r^{-L}. \tag{102}$$

Consequently, for every $p > 0$,

$$\sup_{N,n}\mathbb{E}_{\vartheta_0}\Big[\Big\|A_{N,n}(\widehat{\vartheta}_{N,n} - \vartheta_0)\Big\|^p\Big] < \infty, \tag{103}$$

where finitely many initial indices may be absorbed into the constant.

Lemma 9.8 (Two-dimensional sub-exponential chaining)**.** *Let $\{X(u) : u \in \mathcal{B}\}$ be a separable stochastic process indexed by a ball $\mathcal{B} \subset \mathbb{R}^2$ of radius $R > 0$, with $X(u_*) = 0$ for some $u_* \in \mathcal{B}$ and increments satisfying, for all $u, u' \in \mathcal{B}$ and $t > 0$,*

$$\mathbb{P}\big(|X(u) - X(u')| \ge t\big) \le 2\exp\Big\{-c_0\min\Big(\frac{t^2}{L^2\,\|u - u'\|^2},\ \frac{t}{L\,\|u - u'\|}\Big)\Big\} \tag{104}$$

with constants $c_0, L > 0$. Then there are $C_1, c_1 > 0$ depending only on c_0 such that, for all $x \geq C_1$,

$$\mathbb{P}\Big(\sup_{u \in \mathcal{B}} |X(u)| \geq x\, L\, R \Big) \leq C_1 \exp\{-c_1 x\}.$$

Proof. For $j \geq 0$ let $\mathcal{N}_j \subset \mathcal{B}$ be a $2^{-j}R$-net with $\#\mathcal{N}_j \leq C4^j$ (possible in $\mathbb{R}^2$), π_j a nearest-net-point map, $\pi_0 \equiv u_*$. By separability it suffices to bound the supremum over $\bigcup_j \mathcal{N}_j$, and along the chain $X(u) = \sum_{j \geq 1}\{X(\pi_j u) - X(\pi_{j-1}u)\}$ with $\|\pi_j u - \pi_{j-1}u\| \leq 3 \cdot 2^{-j}R$. Allocate $t_j := \eta\, xLR\, j 2^{-j/2}$, $\eta := (\sum_{j \geq 1} j 2^{-j/2})^{-1}$. At stage j there are at most $C\, 16^j$ increment pairs, so by (104),

$$\mathbb{P}\Big(\sup_{\mathcal{B}} |X| \geq xLR \Big) \leq \sum_{j \geq 1} C\, 16^j \exp\Big\{ - c_0 \min\Big(\tfrac{\eta^2}{9} x^2 j^2 4^{j/2},\ \tfrac{\eta}{3} x\, j\, 2^{j/2} \Big) \Big\} \leq \sum_{j \geq 1} C\, 16^j e^{-c\, x\, j\, 2^{j/2}} \leq C_1 e^{-c_1 x},$$

where the middle inequality uses $x \geq 1$ (so the second entry of the minimum is active up to constants) and the last holds for $x \geq C_1$ with C_1 large enough that $c\, x\, j\, 2^{j/2} \geq 2j \log 16 + c_1 x(1 + j)$ for all $j \geq 1$. □

Proof of the theorem. Throughout $u := A_{N,n}(\vartheta - \vartheta_0) \in \mathbb{R}^2$, and c, C depend only on Θ, T, x_0.

Step 1: global quadratic KL minorant. We claim that for N, n large,

$$K(\vartheta) := \mathbb{E}_{\vartheta_0}\big[\ell_{N,n}(\vartheta_0) - \ell_{N,n}(\vartheta)\big] \ \geq\ c\, \|u\|^2 \qquad \text{for all } \vartheta \in \Theta. \tag{105}$$

Start from the exact decomposition (93) in the proof of Lemma 9.7. *Drift part:* by the mean value theorem bound in the proof of Lemma 9.7, uniformly on Θ, drift $\geq c(\theta - \theta_0)^2 \Lambda_N = c\, u_\theta^2$. *Scale part:* write $v_k^0 / v_k(\vartheta) = x\,(1 + e_k)$ with $x := \sigma_0^2/\sigma^2$ and $e_k := h_k^0/h_{k,n}(\theta) - 1$, $|e_k| \leq C\lambda_k \Delta_n |\theta - \theta_0|$ (Lipschitz in θ via (65)). Since $\psi'(y) = 1 - 1/y$ satisfies $|\psi'(y)| \leq C(|x - 1| + |e_k|)$ for y between x and $x(1 + e_k)$ on the compact range,

$$\big|\psi\big(x(1 + e_k)\big) - \psi(x)\big| \leq C\big(|\sigma - \sigma_0| + \lambda_k \Delta_n |\theta - \theta_0|\big) \lambda_k \Delta_n |\theta - \theta_0|.$$

Summing over (k, j) and using $\psi(x) \geq c(\sigma - \sigma_0)^2$ on compacts, $n\Delta_n \sum_k \lambda_k = T\Lambda_N$ and $n\Delta_n^2 \sum_k \lambda_k^2 \leq T\lambda_N \Delta_n \Lambda_N$:

$$\text{scale} \ \geq\ cNn(\sigma - \sigma_0)^2 - CT\Lambda_N |\sigma - \sigma_0||\theta - \theta_0| - CT(\lambda_N \Delta_n)\Lambda_N (\theta - \theta_0)^2.$$

The last term is $o(1) \cdot u_\theta^2$. For the middle term, AM–GM gives $\Lambda_N |\sigma - \sigma_0||\theta - \theta_0| \leq \frac{\epsilon}{2}\Lambda_N(\theta - \theta_0)^2 + \frac{1}{2\epsilon}\Lambda_N(\sigma - \sigma_0)^2$, and $\Lambda_N/(Nn) \leq \lambda_N \Delta_n / T \to 0$ makes the second piece $o(1) \cdot Nn(\sigma - \sigma_0)^2$. Choosing ϵ small and combining with the drift part yields (105).

Step 2: Gaussian-quadratic representation. Let $\xi = (\varepsilon_{k,j})$ collect the i.i.d. standard normal innovations at ϑ_0. Each $Y_{k,j}$ is affine in $(\varepsilon_{k,0}, \ldots, \varepsilon_{k,j-1})$ with deterministic coefficients, so the centred field $G(\vartheta) := [\ell_{N,n}(\vartheta) - \ell_{N,n}(\vartheta_0)] - \mathbb{E}[\cdot]$ is, for each ϑ, a centred quadratic polynomial $G = \xi^\top M(\vartheta)\xi - \mathbb{E}[\xi^\top M \xi] + b(\vartheta)^\top \xi$ with block-diagonal $M = \oplus_k M_k$. The entries of M_k, b are smooth in ϑ, vanish at ϑ_0, and carry the coefficient profiles $\gamma_k := v_k^0/v_k - 1$ (diagonal), $\delta_k \sqrt{v_k^0}\phi_{k,\cdot}/v_k$ (cross), $\delta_k^2 \phi\phi'/v_k$ (drift), where $\delta_k = \rho_k^0 - \rho_k(\theta)$ and $|\phi_{k,i}| \leq \sqrt{v_k^0}(\rho_k^0)^i$, plus u_0-shift terms in b.

Step 3: second-moment and operator bounds. The computations of Step 2 in the proof of Theorem 9.13, keeping the $(\sigma - \sigma_0)$-dependence, give for all $\vartheta, \vartheta' \in \Theta$:

$$\operatorname{Var} G(\vartheta) = 2\,\|M(\vartheta)\|_F^2 + \|b(\vartheta)\|^2 \ \leq\ C\,\|u\|^2, \tag{106}$$

$$2\,\|M(\vartheta) - M(\vartheta')\|_F^2 + \|b(\vartheta) - b(\vartheta')\|^2 \ \leq\ C\,\|u - u'\|^2, \tag{107}$$

$$\|M(\vartheta)\|_{\mathrm{op}} \ \leq\ C, \qquad \|M(\vartheta) - M(\vartheta')\|_{\mathrm{op}} \ \leq\ C\,\|u - u'\| \cdot \Xi_{N,n}, \qquad \Xi_{N,n} \leq C, \tag{108}$$

where (106) collects the four component classes (γ-diagonal: $Nn(\sigma - \sigma_0)^2 + T\lambda_N \Delta_n \Lambda_N (\theta - \theta_0)^2$; cross: $T\Lambda_N(\theta - \theta_0)^2$; drift-quartic: $T\Lambda_N(\theta - \theta_0)^4 \leq C\, u_\theta^2 \cdot \mathrm{diam}^2$; u_0-terms controlled by (73)), (107) follows from the same computations applied to the coefficient *differences*, which are products of the previous profiles with $\lambda_k \Delta_n \Lambda_N^{-1/2}|u_\theta - u'_\theta|$- and $(Nn)^{-1/2}|u_\sigma - u'_\sigma|$-factors by the mean value theorem in ϑ (each ϑ-derivative of γ_k, δ_k produces one factor $\lambda_k \Delta_n$ resp. σ^{-1}, matching the information normalisation), and (108) follows from the Schur test on the geometric row profiles: $\|M_k\|_{\mathrm{op}} \leq C(|\gamma_k| + \delta_k^2 s_k^2/(v_k(1 - \rho_k^0)) + |\delta_k|\sqrt{v_k^0 s_k^2}/(v_k\sqrt{1 - \rho_k^0})) \leq C$ on Θ, using $1 - \rho_k^0 \geq c\lambda_k \Delta_n$.

Step 4: pointwise and increment tails. By the Hanson–Wright inequality [32, Theorem 1.1] — applicable since the $\varepsilon_{k,j}$ are independent with unit sub-Gaussian norm and $M(\vartheta)$ is deterministic — together with the standard Gaussian tail for $b^\top \xi$ and (106)–(108):

$$\mathbb{P}\big(|G(\vartheta)| \geq t\big) \leq 4 \exp\Big\{ - c \min\Big(\frac{t^2}{\|u\|^2},\ t \Big) \Big\}, \tag{109}$$

$$\mathbb{P}\big(|G(\vartheta) - G(\vartheta')| \geq t\big) \leq 4 \exp\Big\{ - c \min\Big(\frac{t^2}{\|u - u'\|^2},\ \frac{t}{\|u - u'\|} \Big) \Big\}, \tag{110}$$

the increment version because $G(\vartheta) - G(\vartheta')$ is again a Gaussian quadratic with matrix $M(\vartheta) - M(\vartheta')$ and vector $b(\vartheta) - b(\vartheta')$, using (107)–(108).

Step 5: peeling. Let $\mathcal{A}_m := \{\vartheta \in \Theta : 2^m r \le \|u\| < 2^{m+1} r\}$, $m \ge 0$, $r \ge C_1$. On $\{\left\|A_{N,n}(\widehat{\vartheta} - \vartheta_0)\right\| > r\}$ the maximiser satisfies $\ell_{N,n}(\widehat{\vartheta}) \ge \ell_{N,n}(\vartheta_0)$, hence by (105) $\sup_{\mathcal{A}_m} G \ge c4^m r^2$ for some $m \ge 0$. The restriction of G to the ball $\{\|u\| \le 2^{m+1} r\}$ (radius $R_m = 2^{m+1} r$) satisfies the hypotheses of Lemma 9.8 with $u_* = 0$ and $L = C$, by (110) (the field is a polynomial in finitely many Gaussians, hence continuous and separable). Applying the lemma at level $xLR_m = c4^m r^2$, i.e. $x = c' 2^m r \ge C_1$ for $r \ge C_1/c'$:

$$\mathbb{P}\Big(\sup_{\mathcal{A}_m} G \ge c4^m r^2 \Big) \le C \exp\{-c\, 2^m r\}.$$

Summing over $m \ge 0$:

$$\mathbb{P}\big(\left\|A_{N,n}(\widehat{\vartheta}_{N,n} - \vartheta_0)\right\| > r \big) \le C\, e^{-cr} \le C_L r^{-L} \quad \text{for every } L, \text{ uniformly in large } (N,n),$$

which is (102); (103) follows by tail integration, absorbing finitely many initial (N,n) into the constant via the crude bound $\left\|A_{N,n}(\widehat{\vartheta} - \vartheta_0)\right\| \le C(\sqrt{\Lambda_N} + \sqrt{Nn})$ on the compact Θ and the exact Gaussian moment bounds at fixed (N,n). □

Theorem 9.17 (Polynomial moment convergence; establishes Theorem 5.1)**.** *Let $f : \mathbb{R}^2 \to \mathbb{R}$ be continuous and satisfy*

$$|f(x)| \le C(1 + \|x\|^p)$$

for some finite C, p. Then

$$\mathbb{E}_{\vartheta_0}\left[f\Big(A_{N,n}(\widehat{\vartheta}_{N,n} - \vartheta_0) \Big) \right] \longrightarrow \mathbb{E}\left[f\Big(J(\vartheta_0)^{-1/2} Z \Big) \right], \qquad Z \sim \mathcal{N}_2(0, I_2). \tag{111}$$

In particular, all polynomial moments of the normalised MLE converge to the corresponding Gaussian moments.

Proof. Weak convergence follows from Theorem 9.14. The polynomial tail bound in Theorem 9.16 gives uniform integrability of every required order. Truncation and dominated convergence then yield (111). □

9.4.9 Assumption and failure audit

Assumption	Role in the theorem	Failure when removed
Known diagonal A_0, Q and exact spectral modes	Makes the transition law explicit and links all modes to common (θ, σ)	Unknown modal amplitudes or basis require a larger nuisance experiment
$q_k > 0$ on observed modes	Gives a non-degenerate innovation density	Deterministic observed modes contribute only decay information
$\lambda_N / \Lambda_N \to 0$	Lindeberg/no-dominating-mode condition	One mode can dominate and prevent the Gaussian spectral aggregation
$\lambda_N \Delta_n \to 0$	Resolves the fastest observed relaxation and diagonalises information	Drift–scale cross-information need not vanish; limiting rates and covariance change
$R_N = o(\Lambda_N)$	Removes leading deterministic-start information	A non-negligible known start changes, but does not invalidate, the drift-information constant
Compact interior parameter	Uniform smoothness, global separation, inactive boundary	Boundary limits are non-regular and require separate theory
Exact transition likelihood	Eliminates discretisation bias	Euler likelihood may require stronger root-information undersmoothing
Increasing N	Supplies diverging fixed-horizon drift information	With fixed N and fixed T, high-frequency sampling alone does not consistently estimate drift

9.5 Part S4: Uniform post-intervention target processes

9.5.1 Model, target operators, and mixed-rate input

Let $H = L^2(D)$ and let A_0 be self-adjoint, negative, and have compact resolvent,

$$A_0 e_k = -\lambda_k e_k, \qquad 0 < \lambda_1 \le \lambda_2 \le \cdots,$$

for an orthonormal basis (e_k). For $a > 0$, write

$$S_a(t) = e^{taA_0}.$$

Let $Q^{1/2}e_k = q_k e_k$, and suppose the stochastic convolution is H-valued. For an absolute operator intervention $do(\theta = a)$, the counterfactual field solves

$$dU_t^a = aA_0U_t^a\,dt + \sigma Q^{1/2}dW_t, \qquad U_0^a = u_0. \tag{112}$$

The factual parameter is (θ_0, σ_0), but the intervention overwrites θ_0 by a.

Set

$$\mathcal{H} := L^2(0,T;H), \qquad \Psi_w(U) := \int_0^T \langle w_t, U_t\rangle_H\,dt.$$

For $w \in \mathcal{H}$, define the mean and noise-transfer operator

$$m_a(w) := \int_0^T \langle w_t, S_a(t)u_0\rangle_H\,dt, \tag{113}$$

$$(K_a w)_k(s) := q_k \int_s^T e^{-a\lambda_k(t-s)}w_k(t)\,dt, \qquad w_k(t) := \langle w_t, e_k\rangle_H\,. \tag{114}$$

Then

$$\Psi_w(U^a) \sim \mathcal{N}\big(m_a(w), \sigma^2 V_a(w)\big), \qquad V_a(w) := \|K_a w\|_{\mathcal{H}}^2\,. \tag{115}$$

The fixed-intervention CDF is

$$F^{\mathrm{cf}}_{a,w,z}(\sigma) := \Phi(c_{a,w,z}(\sigma))\,, \qquad c_{a,w,z}(\sigma) := \frac{z - m_a(w)}{\sigma\sqrt{V_a(w)}}. \tag{116}$$

The factual CDF is the diagonal restriction

$$F^{\mathrm{fact}}_{w,z}(\theta,\sigma) := F^{\mathrm{cf}}_{\theta,w,z}(\sigma). \tag{117}$$

Part S3 supplies estimators satisfying

$$\Big(a_{N,n}(\widehat{\theta}_{N,n} - \theta_0), b_{N,n}(\widehat{\sigma}_{N,n} - \sigma_0)\Big) \Rightarrow (Z_\theta, Z_\sigma), \tag{118}$$

where

$$a_{N,n} := \sqrt{\Lambda_N}, \quad \Lambda_N := \sum_{k=1}^N \lambda_k, \quad b_{N,n} := \sqrt{Nn}, \quad \frac{a_{N,n}}{b_{N,n}} \to 0, \tag{119}$$

and

$$Z_\theta \sim \mathcal{N}\left(0, \frac{2\theta_0}{T}\right), \qquad Z_\sigma \sim \mathcal{N}\left(0, \frac{\sigma_0^2}{2}\right), \qquad Z_\theta \perp Z_\sigma. \tag{120}$$

9.5.2 A non-trivial compact class of field weights

Let $(\varphi_\ell)_{\ell\ge1}$ be an orthonormal basis of $L^2(0,T)$ and let $\nu_\ell \uparrow \infty$. Expand

$$w(t) = \sum_{k,\ell\ge1} w_{k\ell}\varphi_\ell(t)e_k.$$

For $r, s > 0$ and $R < \infty$, define the ellipsoid

$$\mathcal{W}_R^{r,s} := \left\{ w \in \mathcal{H} : \sum_{k,\ell\ge1} (1+\lambda_k)^{2s}(1+\nu_\ell)^{2r}\,|w_{k\ell}|^2 \le R^2 \right\}. \tag{121}$$

Fix compact intervals

$$\widetilde{\Theta} = [\underline{a}, \overline{a}] \Subset (0,\infty), \qquad \mathcal{Z} = [z_-, z_+], \qquad \mathcal{P} = [p_-, p_+] \Subset (0,1).$$

For constants $0 < v_- < v_+ < \infty$, set

$$\mathcal{W} := \Big\{ w \in \mathcal{W}_R^{r,s} : v_- \le V_a(w) \le v_+ \text{ for every } a \in \widetilde{\Theta} \Big\}. \tag{122}$$

The principal index sets are

$$\mathcal{T}_{\mathrm{cf}} := \widetilde{\Theta} \times \mathcal{W} \times \mathcal{Z}, \tag{123}$$

$$\mathcal{T}_{\mathrm{fact}} := \mathcal{W} \times \mathcal{Z}, \tag{124}$$

$$\mathcal{T}_Q := \widetilde{\Theta} \times \mathcal{W} \times \mathcal{P}. \tag{125}$$

Lemma 9.9 (Compactness of the weight class)**.** *The ellipsoid $\mathcal{W}_R^{r,s}$ is compact in $\mathcal{H}$. If $a \mapsto K_a$ is continuous in operator norm, then $\mathcal{W}$ is a compact subset of $\mathcal{H}$ and the index sets in (123)–(125) are compact metric spaces.*

Proof. The embedding of the weighted coefficient space in $\ell^2(\mathbb{N}^2)$ is diagonal. Since

$$(1+\lambda_k)^{2s}(1+\nu_\ell)^{2r} \to \infty \quad \text{as } k+\ell \to \infty,$$

the image of its unit ball has uniformly vanishing coefficient tails and is relatively compact in $\ell^2(\mathbb{N}^2) \simeq \mathcal{H}$. It is closed, hence compact. Continuity of $(a, w) \mapsto V_a(w) = \|K_a w\|^2$ makes the variance restrictions closed on the compact product $\widetilde{\Theta} \times \mathcal{W}_R^{r,s}$. □

9.5.3 Uniform smoothness of the intervention map

The spectral calculus gives, for $j = 1, 2$,

$$\sup_{a\in\widetilde{\Theta}} \sup_{t\geq 0} \left\| t^j A_0^j S_a(t) \right\|_{\mathcal{L}(H)} < \infty. \tag{126}$$

Define

$$\dot{m}_a(w) := \int_0^T \langle w_t, tA_0 S_a(t) u_0 \rangle_H \, dt, \tag{127}$$

$$(\dot{K}_a w)_k(s) := -q_k \int_s^T \lambda_k (t-s) e^{-a\lambda_k(t-s)} w_k(t)\, dt, \tag{128}$$

$$\dot{V}_a(w) := 2 \left\langle K_a w, \dot{K}_a w \right\rangle_{\mathcal{H}}. \tag{129}$$

Second derivatives are obtained by replacing the kernels in (127) and (128) by $t^2 A_0^2 S_a(t)$ and $q_k \lambda_k^2 (t-s)^2 e^{-a\lambda_k(t-s)}$, respectively.

Assumption 9.7 (Noise-transfer regularity)**.** *The modal noise sequence satisfies*

$$\sup_{k\geq 1} \frac{q_k}{\lambda_k} < \infty.$$

This follows, in particular, from the standing H-valued convolution condition $\sum_k q_k^2/\lambda_k < \infty$.

Lemma 9.10 (Twice continuous differentiability)**.** *Under Assumption 9.7, the maps*

$$a \mapsto m_a \in \mathcal{H}^*, \qquad a \mapsto K_a \in \mathcal{L}(\mathcal{H}), \qquad (a, w) \mapsto V_a(w)$$

are twice continuously differentiable on $\widetilde{\Theta}$. Their derivatives through order two are uniformly bounded on $\widetilde{\Theta} \times \mathcal{W}$.

Proof. For each mode, Young's inequality for the backward Volterra convolution yields

$$\|(K_a w)_k\|_{L^2(0,T)} \leq \frac{q_k}{a\lambda_k} \|w_k\|_{L^2(0,T)}.$$

The L^1 norms of the first and second derivative kernels are bounded by $q_k/(a^2\lambda_k)$ and $2q_k/(a^3\lambda_k)$. Assumption 9.7 and $a \geq \underline{a}$ therefore give uniform operator bounds for K_a, $\dot{K}_a$, and $\ddot{K}_a$. Dominated convergence proves operator-norm continuity. The assertions for m_a follow from (126) and Cauchy–Schwarz. The formulas for V_a follow by the product rule. □

For $c = c_{a,w,z}(\sigma)$, differentiation of (116) gives

$$D_\sigma F^{\mathrm{cf}}_{a,w,z}(\sigma) = -\frac{c\phi(c)}{\sigma}. \tag{130}$$

On the factual diagonal, write

$$c^{\mathrm{fact}}_{w,z}(\theta, \sigma) := c_{\theta,w,z}(\sigma).$$

Then

$$D_\theta F^{\mathrm{fact}}_{w,z}(\theta, \sigma) = \phi(c^{\mathrm{fact}}) \left[-\frac{\dot{m}_\theta(w)}{\sigma\sqrt{V_\theta(w)}} - \frac{c^{\mathrm{fact}} \dot{V}_\theta(w)}{2V_\theta(w)} \right], \tag{131}$$

$$D_\sigma F^{\mathrm{fact}}_{w,z}(\theta, \sigma) = -\frac{c^{\mathrm{fact}} \phi(c^{\mathrm{fact}})}{\sigma}. \tag{132}$$

Theorem 9.18 (Uniform Fréchet differentiability)**.** *Let the parameter rectangle be a compact subset of $(0, \infty)^2$ containing (θ_0, σ_0) in its interior. Under Lemmas 9.9 and 9.10,*

$$\sigma \longmapsto F^{\mathrm{cf}}(\sigma) := \left(F^{\mathrm{cf}}_{a,w,z}(\sigma)\right)_{(a,w,z)\in\mathcal{T}_{\mathrm{cf}}} \in C(\mathcal{T}_{\mathrm{cf}}),$$

$$(\theta, \sigma) \longmapsto F^{\mathrm{fact}}(\theta, \sigma) := \left(F^{\mathrm{fact}}_{w,z}(\theta, \sigma)\right)_{(w,z)\in\mathcal{T}_{\mathrm{fact}}} \in C(\mathcal{T}_{\mathrm{fact}})$$

are twice continuously Fréchet differentiable. Their first derivatives are given by (130)–(132)*. There is a constant $C < \infty$ such that the uniform Taylor remainders satisfy*

$$\left\|F^{\mathrm{cf}}(\sigma+h) - F^{\mathrm{cf}}(\sigma) - hD_\sigma F^{\mathrm{cf}}(\sigma)\right\|_\infty \le Ch^2, \tag{133}$$

$$\left\|F^{\mathrm{fact}}(\vartheta+h) - F^{\mathrm{fact}}(\vartheta) - DF^{\mathrm{fact}}(\vartheta)[h]\right\|_\infty \le C\left\|h\right\|_{\mathbb{R}^2}^2. \tag{134}$$

Proof. Lemma 9.10, compactness, the lower variance bound v_-, and the positive lower bound on σ imply uniform bounds on the first two derivatives of the standardised Gaussian argument. The standard normal CDF has bounded derivatives of every finite order. The Banach-space chain rule gives continuous Fréchet differentiability into the spaces of continuous functions, and Taylor's theorem with the uniform second-derivative bound gives (133)–(134). □

9.5.4 Fast fixed-intervention counterfactual process

Define

$$\widehat{F}^{\mathrm{cf}}_{N,n}(a,w,z) := F^{\mathrm{cf}}_{a,w,z}(\widehat{\sigma}_{N,n}), \qquad F^{\mathrm{cf}}_0(a,w,z) := F^{\mathrm{cf}}_{a,w,z}(\sigma_0).$$

Theorem 9.19 (Uniform fast counterfactual process; establishes Theorem 6.1)**.** *Under* (118)–(120) *and Theorem 9.18,*

$$b_{N,n}\big(\widehat{F}^{\mathrm{cf}}_{N,n} - F^{\mathrm{cf}}_0\big) \Rightarrow \mathbb{G}_{\mathrm{cf}} \quad \text{in } C(\mathcal{T}_{\mathrm{cf}}), \tag{135}$$

where

$$\mathbb{G}_{\mathrm{cf}}(a,w,z) = d^{\mathrm{cf}}_\sigma(a,w,z)Z_\sigma, \qquad d^{\mathrm{cf}}_\sigma(a,w,z) := -\frac{c_{a,w,z}(\sigma_0)\phi(c_{a,w,z}(\sigma_0))}{\sigma_0}. \tag{136}$$

The limit is a continuous rank-one Gaussian process with covariance kernel

$$\begin{aligned}\Gamma_{\mathrm{cf}}(\tau,\tau') &= \frac{\sigma_0^2}{2} d^{\mathrm{cf}}_\sigma(\tau) d^{\mathrm{cf}}_\sigma(\tau') \\ &= \frac{1}{2} c_\tau \phi(c_\tau) c_{\tau'} \phi(c_{\tau'}),\end{aligned} \tag{137}$$

where $c_\tau = c_{a,w,z}(\sigma_0)$.

Proof. Apply (133) with $h = \widehat{\sigma}_{N,n} - \sigma_0$. Since $b_{N,n}h = \mathrm{O}_{\mathbb{P}}(1)$ and $b_{N,n}h^2 = \mathrm{O}_{\mathbb{P}}(b_{N,n}^{-1}) = \mathrm{o}_{\mathbb{P}}(1)$, the uniform remainder is negligible. The derivative is a continuous element of $C(\mathcal{T}_{\mathrm{cf}})$, so (118) and the continuous mapping theorem give (135). The covariance follows from (120). □

Corollary 9.6 (Policy-to-policy contrasts)**.** *Fix $a_0 \in \widetilde{\Theta}$ and define*

$$C_{a,a_0,w,z}(\sigma) := F^{\mathrm{cf}}_{a,w,z}(\sigma) - F^{\mathrm{cf}}_{a_0,w,z}(\sigma).$$

Uniformly on $\widetilde{\Theta} \times \mathcal{W} \times \mathcal{Z}$,

$$b_{N,n}\big(\widehat{C}_{a,a_0,w,z} - C_{a,a_0,w,z}(\sigma_0)\big) \Rightarrow \big(d^{\mathrm{cf}}_\sigma(a,w,z) - d^{\mathrm{cf}}_\sigma(a_0,w,z)\big)Z_\sigma.$$

This contrast retains non-trivial intervention-index variation at the fast rate.

9.5.5 Slow factual and factual-versus-counterfactual processes

Define

$$\widehat{F}^{\mathrm{fact}}_{N,n}(w,z) := F^{\mathrm{fact}}_{w,z}(\widehat{\theta}_{N,n}, \widehat{\sigma}_{N,n}), \qquad F^{\mathrm{fact}}_0(w,z) := F^{\mathrm{fact}}_{w,z}(\theta_0,\sigma_0).$$

Let

$$d^{\mathrm{fact}}_\theta(w,z) := D_\theta F^{\mathrm{fact}}_{w,z}(\theta_0,\sigma_0). \tag{138}$$

Theorem 9.20 (Uniform slow factual process; establishes Theorem 6.1)**.** *Under the assumptions of Theorem 9.19,*

$$a_{N,n}\big(\widehat{F}^{\mathrm{fact}}_{N,n} - F^{\mathrm{fact}}_0\big) \Rightarrow \mathbb{G}_{\mathrm{fact}} \quad \text{in } C(\mathcal{T}_{\mathrm{fact}}), \tag{139}$$

where

$$\mathbb{G}_{\mathrm{fact}}(w,z) = d^{\mathrm{fact}}_\theta(w,z)Z_\theta. \tag{140}$$

Its covariance kernel is

$$\Gamma_{\mathrm{fact}}(\tau,\tau') = \frac{2\theta_0}{T} d^{\mathrm{fact}}_\theta(\tau) d^{\mathrm{fact}}_\theta(\tau'). \tag{141}$$

The scale-estimation contribution vanishes because $a_{N,n}/b_{N,n} \to 0$.

Proof. Use (134). The linear term is

$$d_\theta^{\mathrm{fact}} a_{N,n}(\widehat{\theta}_{N,n} - \theta_0) + D_\sigma F^{\mathrm{fact}} \frac{a_{N,n}}{b_{N,n}} b_{N,n}(\widehat{\sigma}_{N,n} - \sigma_0).$$

The second term is $\mathrm{o}_{\mathbb{P}}(1)$ uniformly. For the remainder,

$$a_{N,n} \left\| (\widehat{\theta}_{N,n} - \theta_0, \widehat{\sigma}_{N,n} - \sigma_0) \right\|^2 = \mathrm{O}_{\mathbb{P}}(a_{N,n}^{-1}) + \mathrm{O}_{\mathbb{P}}(a_{N,n}/b_{N,n}^2) + \mathrm{O}_{\mathbb{P}}(b_{N,n}^{-1}) = \mathrm{o}_{\mathbb{P}}(1).$$

The result follows from (118). □

For $a \in \widetilde{\Theta}$, define the factual-versus-counterfactual distributional effect

$$\Delta_{a,w,z} := F^{\mathrm{cf}}_{a,w,z}(\sigma_0) - F^{\mathrm{fact}}_{w,z}(\theta_0, \sigma_0), \tag{142}$$

and its plug-in estimator

$$\widehat{\Delta}_{a,w,z} := F^{\mathrm{cf}}_{a,w,z}(\widehat{\sigma}_{N,n}) - F^{\mathrm{fact}}_{w,z}(\widehat{\theta}_{N,n}, \widehat{\sigma}_{N,n}).$$

Corollary 9.7 (Uniform distributional-effect process)**.** *In* $C(\widetilde{\Theta} \times \mathcal{W} \times \mathcal{Z})$,

$$a_{N,n}(\widehat{\Delta} - \Delta) \Rightarrow \mathbb{G}_\Delta, \qquad \mathbb{G}_\Delta(a, w, z) = -d_\theta^{\mathrm{fact}}(w, z) Z_\theta. \tag{143}$$

For absolute interventions, the first-order slow-rate uncertainty is constant in a*: the post-intervention term is learned at the faster scale and disappears from the slow limit.*

9.5.6 Quantile and tail processes

Let $z_p := \Phi^{-1}(p)$. The fixed-intervention Gaussian quantile is

$$Q^{\mathrm{cf}}_{a,w}(p; \sigma) = m_a(w) + \sigma \sqrt{V_a(w)} z_p. \tag{144}$$

The factual quantile is

$$Q^{\mathrm{fact}}_{w}(p; \theta, \sigma) = m_\theta(w) + \sigma \sqrt{V_\theta(w)} z_p.$$

Theorem 9.21 (Uniform quantile processes; establishes Theorem 6.2)**.** *Uniformly over* $\mathcal{T}_Q$,

$$b_{N,n}\big(Q^{\mathrm{cf}}_{a,w}(p; \widehat{\sigma}_{N,n}) - Q^{\mathrm{cf}}_{a,w}(p; \sigma_0)\big) \Rightarrow \mathbb{Q}_{\mathrm{cf}}(a, w, p) := \sqrt{V_a(w)} z_p Z_\sigma. \tag{145}$$

Uniformly over $\mathcal{W} \times \mathcal{P}$,

$$a_{N,n}\big(Q^{\mathrm{fact}}_{w}(p; \widehat{\theta}_{N,n}, \widehat{\sigma}_{N,n}) - Q^{\mathrm{fact}}_{w}(p; \theta_0, \sigma_0)\big) \Rightarrow \mathbb{Q}_{\mathrm{fact}}(w, p), \tag{146}$$

where

$$\mathbb{Q}_{\mathrm{fact}}(w, p) = \left[\dot{m}_{\theta_0}(w) + \sigma_0 \frac{\dot{V}_{\theta_0}(w)}{2\sqrt{V_{\theta_0}(w)}} z_p \right] Z_\theta. \tag{147}$$

Consequently, the factual-versus-counterfactual quantile-effect process has slow limit $-\mathbb{Q}_{\mathrm{fact}}$.

Proof. The Gaussian formula (144) is continuously differentiable uniformly on the compact index sets; $p \in \mathcal{P} \Subset (0, 1)$ makes z_p bounded. Apply the same fast and slow Taylor arguments as in Theorems 9.19 and 9.20. Formula (147) is the derivative with respect to θ. □

For a threshold $c \in \mathcal{Z}$, define the tail probability

$$R^{\mathrm{cf}}_{a,w}(c; \sigma) := 1 - F^{\mathrm{cf}}_{a,w,c}(\sigma).$$

Corollary 9.8 (Uniform tail processes)**.** *Every CDF process theorem above transfers to the corresponding tail process by multiplication by* -1*. In particular,*

$$b_{N,n}(\widehat{R}^{\mathrm{cf}} - R^{\mathrm{cf}}_0) \Rightarrow -\mathbb{G}_{\mathrm{cf}}, \qquad a_{N,n}(\widehat{R}^{\mathrm{fact}} - R^{\mathrm{fact}}_0) \Rightarrow -\mathbb{G}_{\mathrm{fact}}.$$

No separate tail empirical-process argument is required for thresholds in a fixed compact set.

9.5.7 Uniform causal hypotheses and supremum statistics

Let Δ be the distributional effect in (142). For equality testing over a compact subset $\mathcal{T}_0 \subseteq \widetilde{\Theta} \times \mathcal{W} \times \mathcal{Z}$, define

$$S^{=}_{N,n} := a_{N,n} \left\| \widehat{\Delta} \right\|_{\mathcal{T}_0, \infty}. \tag{148}$$

Corollary 9.9 (Equality null)**.** *Under*

$$H_0^{=} : \Delta(\tau) = 0 \quad \forall \tau \in \mathcal{T}_0,$$

$$S^{=}_{N,n} \Rightarrow \sup_{\tau\in\mathcal{T}_0} |\mathbb{G}_\Delta(\tau)|\,.$$

Because $\mathbb{G}_\Delta$ is rank one, the limit equals

$$|Z_\theta| \sup_{(w,z):\exists a,(a,w,z)\in\mathcal{T}_0} \left|d^{\mathrm{fact}}_\theta(w,z)\right|.$$

For first-order stochastic dominance in the direction $F^{\mathrm{cf}} \le F^{\mathrm{fact}}$, define

$$S^{\le}_{N,n} := a_{N,n} \sup_{\tau\in\mathcal{T}_0} [\widehat{\Delta}(\tau)]_+. \tag{149}$$

Let the contact set be

$$\mathcal{C}_0 := \{\tau \in \mathcal{T}_0 : \Delta(\tau) = 0\}.$$

Theorem 9.22 (One-sided composite null and contact set)**.** *Under*

$$H^{\le}_0 : \Delta(\tau) \le 0 \quad \forall \tau \in \mathcal{T}_0,$$

the following statements hold.

(i) If $\sup_{\tau\in\mathcal{T}_0} \Delta(\tau) < 0$, then $S^{\le}_{N,n} \to 0$ in probability.

(ii) If $\mathcal{C}_0 \neq \varnothing$ and the usual local separation condition holds away from $\mathcal{C}_0$, then

$$S^{\le}_{N,n} \Rightarrow \sup_{\tau\in\mathcal{C}_0} [\mathbb{G}_\Delta(\tau)]_+.$$

The same contact-set logic applies to non-positive tail effects and quantile-effect sign restrictions.

Proof. Write $Z_{N,n} := a_{N,n}(\widehat{\Delta} - \Delta)$, so that $a_{N,n}\widehat{\Delta} = a_{N,n}\Delta + Z_{N,n}$ and, by Corollary 9.7, $Z_{N,n} \Rightarrow \mathbb{G}_\Delta$ in $C(\mathcal{T}_0)$; in particular $\|Z_{N,n}\|_\infty = O_{\mathbb{P}}(1)$ and the limit process has uniformly continuous sample paths on the compact $\mathcal{T}_0$.

(i) If $\sup_{\mathcal{T}_0} \Delta =: -c_0 < 0$, then $\sup_{\mathcal{T}_0} a_{N,n}\widehat{\Delta} \le -a_{N,n}c_0 + O_{\mathbb{P}}(1) \to -\infty$, so $S^{\le}_{N,n} = \sup_{\mathcal{T}_0}[a_{N,n}\widehat{\Delta}]_+ = 0$ with probability tending to one, and in particular $S^{\le}_{N,n} \to 0$ in probability.

(ii) Let $\mathcal{C}_0 \neq \varnothing$ and assume the local separation condition: for every $\eta > 0$ there is $c(\eta) > 0$ with $\Delta(\tau) \le -c(\eta)$ whenever $d(\tau, \mathcal{C}_0) \ge \eta$. Fix $\eta > 0$ and split $\mathcal{T}_0 = \mathcal{C}_0^\eta \cup (\mathcal{T}_0 \setminus \mathcal{C}_0^\eta)$, $\mathcal{C}_0^\eta := \{\tau : d(\tau, \mathcal{C}_0) < \eta\}$. On the complement, $a_{N,n}\widehat{\Delta} \le -a_{N,n}c(\eta) + \|Z_{N,n}\|_\infty$, whose positive part vanishes with probability tending to one; hence

$$S^{\le}_{N,n} = \sup_{\mathcal{C}_0^\eta} \left[a_{N,n}\Delta + Z_{N,n}\right]_+ + o_{\mathbb{P}}(1) \le \sup_{\mathcal{C}_0^\eta} \left[Z_{N,n}\right]_+ + o_{\mathbb{P}}(1),$$

using $\Delta \le 0$ on $\mathcal{T}_0$; and, restricting the supremum to $\mathcal{C}_0 \subseteq \mathcal{C}_0^\eta$ where $\Delta = 0$, $S^{\le}_{N,n} \ge \sup_{\mathcal{C}_0}[Z_{N,n}]_+$. The map $z \mapsto \sup_A [z]_+$ is continuous on $C(\mathcal{T}_0)$ for compact A, so the continuous mapping theorem gives

$$\sup_{\mathcal{C}_0}[Z_{N,n}]_+ \Rightarrow \sup_{\mathcal{C}_0}[\mathbb{G}_\Delta]_+, \qquad \sup_{\mathcal{C}_0^\eta}[Z_{N,n}]_+ \Rightarrow \sup_{\mathcal{C}_0^\eta}[\mathbb{G}_\Delta]_+.$$

Finally, $\sup_{\mathcal{C}_0^\eta}[\mathbb{G}_\Delta]_+ \downarrow \sup_{\mathcal{C}_0}[\mathbb{G}_\Delta]_+$ almost surely as $\eta \downarrow 0$, by uniform continuity of $\mathbb{G}_\Delta$ and compactness ($\mathcal{C}_0^\eta \downarrow \mathcal{C}_0$). A standard sandwich (portmanteau along $\eta \downarrow 0$) then yields $S^{\le}_{N,n} \Rightarrow \sup_{\mathcal{C}_0}[\mathbb{G}_\Delta]_+$, as claimed; the argument is the compact-index case of the directional delta method for supremum functionals [cf. 1, 17], proved here directly to keep the chain self-contained. The same argument applies verbatim to tail- and quantile-effect surfaces with their loadings. □

9.5.8 Galerkin and temporal approximation interface

Suppose numerical target primitives $m_a^{K,\delta}(w)$ and $V_a^{K,\delta}(w)$ are used, where K is a spatial truncation level and δ a numerical time step. Define

$$\varepsilon_{K,\delta} := \sup_{a\in\widetilde{\Theta},w\in\mathcal{W}} \left(\left|m_a^{K,\delta}(w) - m_a(w)\right| + \left|V_a^{K,\delta}(w) - V_a(w)\right|\right). \tag{150}$$

Assume the approximate variances remain above $v_-/2$ for large K and small δ.

Proposition 9.15 (Transfer of numerical error to target laws)**.** *There is $C < \infty$ such that*

$$\left\|F^{K,\delta} - F\right\|_\infty \le C\varepsilon_{K,\delta}, \tag{151}$$

$$\left\|Q^{K,\delta} - Q\right\|_\infty \le C\varepsilon_{K,\delta}. \tag{152}$$

Consequently:

(i) the fast fixed-intervention limits are unchanged if

$$b_{N,n}\varepsilon_{K,\delta} \to 0;$$

(ii) the slow factual and effect limits are unchanged if

$$a_{N,n}\varepsilon_{K,\delta} \to 0.$$

Consistent plug-in covariance kernels additionally require uniform convergence of the corresponding first derivatives.

Proof. On the region $V \geq v_-/2$ and $\sigma \geq \underline{\sigma}$, the Gaussian CDF and quantile formulas are uniformly Lipschitz in (m, V). This gives (151)–(152). Multiplication by the statistical rate proves the two transfer statements. □

If Part S1 supplies

$$\varepsilon_{K,\delta} \lesssim \lambda_{K+1}^{-\alpha} + \delta^{\kappa},$$

then the sufficient undersmoothing conditions are

$$b_{N,n}\big(\lambda_{K+1}^{-\alpha} + \delta^{\kappa}\big) \to 0 \qquad \text{for fast counterfactual targets,}$$
$$a_{N,n}\big(\lambda_{K+1}^{-\alpha} + \delta^{\kappa}\big) \to 0 \qquad \text{for factual and effect targets.}$$

9.6 Part S5: Multiplier bootstrap and simultaneous inference

9.6.1 Mixed-rate statistical input

Let $\vartheta = (\theta, \sigma)$ and let $S_{k,N,n}(\vartheta) \in \mathbb{R}^2$ be the exact conditional score contributed by the complete observed path of mode k:

$$S_{k,N,n}(\vartheta) := \sum_{j=0}^{n-1} s_{k,j,n}(\vartheta). \tag{153}$$

The modes are independent, while temporal dependence remains inside each block. Define

$$a_{N,n} := \sqrt{\Lambda_N}, \qquad \Lambda_N := \sum_{k=1}^{N} \lambda_k, \qquad b_{N,n} := \sqrt{Nn}, \qquad A_{N,n} := \operatorname{diag}(a_{N,n}, b_{N,n}), \qquad D_{N,n} := A_{N,n}^{-1}. \tag{154}$$

Part S3 proves, under its exact-likelihood regime,

$$A_{N,n}(\widehat{\vartheta}_{N,n} - \vartheta_0) = J(\vartheta_0)^{-1} D_{N,n} \sum_{k=1}^{N} S_{k,N,n}(\vartheta_0) + \mathrm{o}_{\mathbb{P}}(1), \tag{155}$$

where

$$J(\vartheta_0) = \begin{pmatrix} T/(2\theta_0) & 0 \\ 0 & 2/\sigma_0^2 \end{pmatrix}. \tag{156}$$

In particular,

$$A_{N,n}(\widehat{\vartheta}_{N,n} - \vartheta_0) \Rightarrow H \sim \mathcal{N}_2(0, J(\vartheta_0)^{-1}). \tag{157}$$

Assumption 9.8 (Part S3 regime)**.** *The exact spectral experiment satisfies*

$$N \to \infty, \qquad \lambda_N \Delta_n \to 0, \qquad \frac{\lambda_N}{\Lambda_N} \to 0, \qquad \frac{a_{N,n}}{b_{N,n}} \to 0,$$

together with the compact-parameter, known-spectrum, and negligible-initial-energy assumptions of Part S3. The MLE is interior with probability tending to one and satisfies (155)*.*

Lemma 9.11 (Mode-block no-dominance)**.** *Under Assumption 9.8, for some constant $C < \infty$,*

$$\sum_{k=1}^{N} \mathbb{E}_{\vartheta_0} \left\| D_{N,n} S_{k,N,n}(\vartheta_0) \right\|^4 \leq C \left(\frac{\lambda_N}{\Lambda_N} + \frac{1}{N} + (\lambda_N \Delta_n)^2 \right) + o(1). \tag{158}$$

Consequently,

$$\max_{k \leq N} \left\| D_{N,n} S_{k,N,n}(\vartheta_0) \right\| \xrightarrow{\mathbb{P}} 0, \tag{159}$$

$$\sum_{k=1}^{N} D_{N,n} S_{k,N,n}(\vartheta_0) S_{k,N,n}(\vartheta_0)^{\top} D_{N,n} \xrightarrow{\mathbb{P}} J(\vartheta_0). \tag{160}$$

Proof. Fix $k \le N$ and write the mode-k score block at the truth as $S_k = \sum_{j<n} d_{k,j}$, $d_{k,j} = (s_{\theta,k,j}, s_{\sigma,k,j})^\top$, a two-dimensional martingale difference sequence with respect to the mode-k filtration, with (Step 1 of the proof of Lemma 9.5)

$$s_{\theta,k,j} = -\frac{\lambda_k \Delta_n \rho_k Y_{k,j}}{\sqrt{v_k}} \varepsilon_{k,j} + \frac{g_k}{2}(\varepsilon_{k,j}^2 - 1), \qquad s_{\sigma,k,j} = \frac{1}{\sigma_0}(\varepsilon_{k,j}^2 - 1).$$

Fourth moments per block. By Rosenthal's inequality for martingales, for each coordinate $a \in \{\theta, \sigma\}$,

$$\mathbb{E}\, S_{k,a}^4 \le C\Big[\Big(\sum_{j<n} \mathbb{E}\, d_{k,j,a}^2\Big)^2 + \sum_{j<n} \mathbb{E}\, d_{k,j,a}^4\Big].$$

For $a = \theta$: $\sum_j \mathbb{E}d^2 \le CT\lambda_k$ (information computation, Part S3) and, by the moment bounds in the proof of Lemma 9.5, $\sum_j \mathbb{E}d^4 \le C\big(n\lambda_k^2\Delta_n^2 + \lambda_k^2\Delta_n(\lambda_k u_{0,k}^2/q_k^2)^2 \cdot \Delta_n\big) \le CT\lambda_k \cdot \lambda_k\Delta_n(1 + (\lambda_k u_{0,k}^2/q_k^2)^2/n)$, so $\mathbb{E}S_{k,\theta}^4 \le C\lambda_k^2\big(1 + o(1) + u\text{-terms}\big)$ with the u_0-terms summable to $o(\Lambda_N^2)$ exactly as in the score-CLT closure. For $a = \sigma$: $\sum_j \mathbb{E}d^2 = 2n/\sigma_0^2$ and $\sum_j \mathbb{E}d^4 \le Cn$, so $\mathbb{E}S_{k,\sigma}^4 \le Cn^2$. Hence, with $D_{N,n} = \mathrm{diag}(\Lambda_N^{-1/2}, (Nn)^{-1/2})$ and Cauchy–Schwarz for the mixed coordinates,

$$\sum_{k\le N} \mathbb{E}\,\|D_{N,n}S_k\|^4 \le C\Big\{\frac{\sum_k \lambda_k^2}{\Lambda_N^2} + \frac{Nn^2}{(Nn)^2} + \Big(\frac{\sum_k \lambda_k^2}{\Lambda_N^2}\Big)^{1/2}\Big(\frac{Nn^2}{(Nn)^2}\Big)^{1/2}\Big\} + o(1) \le C\Big(\frac{\lambda_N}{\Lambda_N} + \frac{1}{N} + (\lambda_N\Delta_n)^2\Big) + o(1),$$

using $\sum_k \lambda_k^2 \le \lambda_N\Lambda_N$ and absorbing the cross term with $2ab \le a^2 + b^2$; this is (158).

Maximum. For $\delta > 0$, $\mathbb{P}(\max_k \|D_{N,n}S_k\| \ge \delta) \le \delta^{-4}\sum_k \mathbb{E}\,\|D_{N,n}S_k\|^4 \to 0$, giving (159).

Outer product. The blocks are independent across k with $\sum_k \mathbb{E}[D_{N,n}S_kS_k^\top D_{N,n}] = D_{N,n}I_{N,n}(\vartheta_0)D_{N,n} \to J(\vartheta_0)$ (Theorem 9.11). Each entry of the centred sum $\sum_k\{D_{N,n}S_kS_k^\top D_{N,n} - \mathbb{E}[\cdot]\}$ has variance bounded by $\sum_k \mathbb{E}\,\|D_{N,n}S_k\|^4 \to 0$, so (160) follows by Chebyshev. □

9.6.2 Feasible mode-block multiplier

Define the estimated scaled modal score blocks

$$\widehat{B}_{k,N,n} := D_{N,n}S_{k,N,n}(\widehat{\vartheta}_{N,n}), \tag{161}$$

and the block outer-product information estimator

$$\widehat{J}_{B,N,n} := \sum_{k=1}^N \widehat{B}_{k,N,n}\widehat{B}_{k,N,n}^\top. \tag{162}$$

Lemma 9.12 (Feasible information consistency)**.** *Under Assumption 9.8,*

$$\widehat{J}_{B,N,n} \xrightarrow{\mathbb{P}} J(\vartheta_0). \tag{163}$$

Moreover,

$$\max_{k\le N} \Big\|\widehat{B}_{k,N,n}\Big\| \xrightarrow{\mathbb{P}} 0.$$

Proof. Taylor-expand each modal score between ϑ_0 and $\widehat{\vartheta}_{N,n}$. Consistency and the mixed local rate imply that the sum of the scaled plug-in differences is $o_{\mathbb{P}}(1)$ in the outer-product norm. Uniform Gaussian moment bounds control the quadratic remainder. Combine this with Lemma 9.11. □

Let $\xi_1, \dots, \xi_N$ be iid mean-zero, variance-one multipliers independent of the data. Define

$$\Delta_{N,n}^* := \sum_{k=1}^N \xi_k \widehat{B}_{k,N,n}, \qquad H_{N,n}^* := \widehat{J}_{B,N,n}^{-1}\Delta_{N,n}^*. \tag{164}$$

If necessary, replace $\widehat{J}_{B,N,n}^{-1}$ by an eigenvalue-floored inverse with a floor converging to zero.

Proposition 9.16 (Gaussian multiplier shortcut)**.** *If $\xi_k \sim \mathcal{N}(0,1)$, then conditionally on the data,*

$$\Delta_{N,n}^* \sim \mathcal{N}_2(0, \widehat{J}_{B,N,n}), \qquad H_{N,n}^* \sim \mathcal{N}_2(0, \widehat{J}_{B,N,n}^{-1}). \tag{165}$$

Therefore the mode-block score multiplier is conditionally identical in law to a direct draw

$$H_{N,n}^* = \widehat{J}_{B,N,n}^{-1/2} Z^*, \qquad Z^* \sim \mathcal{N}_2(0, I_2).$$

Proof. A linear combination of independent Gaussian multipliers is Gaussian, with conditional covariance

$$\sum_{k=1}^N \widehat{B}_{k,N,n}\widehat{B}_{k,N,n}^\top = \widehat{J}_{B,N,n}.$$

Premultiplication by $\widehat{J}_{B,N,n}^{-1}$ gives (165). □

Theorem 9.23 (Conditional estimator bootstrap)**.** *Under Assumption 9.8,*

$$H^*_{N,n} \Rightarrow_{\mathbb{P}} H \sim \mathcal{N}_2(0, J(\vartheta_0)^{-1}). \tag{166}$$

Equivalently,

$$\sup_{f \in \mathrm{BL}_1} \left| \mathbb{E}^* f(H^*_{N,n}) - \mathbb{E} f(H) \right| \xrightarrow{\mathbb{P}} 0. \tag{167}$$

The result holds for Gaussian multipliers. It also holds for Rademacher multipliers under the conditional Lindeberg condition supplied by Lemmas 9.11–9.12.

Proof. For Gaussian multipliers, Proposition 9.16 and Lemma 9.12 reduce the claim to continuity of the Gaussian law in its positive-definite covariance matrix. For Rademacher multipliers, condition on the data and apply the triangular-array Lindeberg–Feller theorem to $\sum_k \xi_k \widehat{B}_{k,N,n}$; the conditional covariance converges to $J(\vartheta_0)$ and the largest summand vanishes. □

9.6.3 Bootstrap target processes

Use the notation of Part S4. A fixed absolute intervention $do(\theta = a)$ gives

$$F^{\mathrm{cf}}_{a,w,z}(\sigma) = \Phi\left(\frac{z - m_a(w)}{\sigma \sqrt{V_a(w)}} \right).$$

Let

$$d^{\mathrm{cf}}_{\sigma}(a,w,z) := -\frac{c_{a,w,z}(\sigma_0)\phi(c_{a,w,z}(\sigma_0))}{\sigma_0}. \tag{168}$$

Let $d^{\mathrm{fact}}_{\theta}(w,z)$ be the factual CDF derivative defined in Part S4. Write $H^*_{N,n,\theta}$ and $H^*_{N,n,\sigma}$ for the coordinates of $H^*_{N,n}$.

Define

$$\mathbb{Z}^*_{\mathrm{cf},N,n}(a,w,z) := \widehat{d}^{\mathrm{cf}}_{\sigma}(a,w,z) H^*_{N,n,\sigma}, \tag{169}$$

$$\mathbb{Z}^*_{\mathrm{fact},N,n}(w,z) := \widehat{d}^{\mathrm{fact}}_{\theta}(w,z) H^*_{N,n,\theta}, \tag{170}$$

$$\mathbb{Z}^*_{\Delta,N,n}(a,w,z) := -\widehat{d}^{\mathrm{fact}}_{\theta}(w,z) H^*_{N,n,\theta}. \tag{171}$$

For $z_p = \Phi^{-1}(p)$, define

$$\mathbb{Z}^*_{Q,\mathrm{cf},N,n}(a,w,p) := \sqrt{\widehat{V}_a(w)}\, z_p H^*_{N,n,\sigma}, \tag{172}$$

$$\mathbb{Z}^*_{Q,\mathrm{fact},N,n}(w,p) := \left[\widehat{m}_{\widehat{\theta}_{N,n}}(w) + \widehat{\sigma}_{N,n} \frac{\dot{\widehat{V}}_{\widehat{\theta}_{N,n}}(w)}{2\sqrt{\widehat{V}_{\widehat{\theta}_{N,n}}(w)}} z_p \right] H^*_{N,n,\theta}. \tag{173}$$

Tail bootstrap processes are the negatives of their CDF counterparts.

Assumption 9.9 (Uniform derivative consistency)**.** *Every derivative surface appearing in (169)–(173) is continuous on its compact index set and its plug-in estimator converges uniformly in probability to the true derivative surface.*

Theorem 9.24 (Joint conditional target-process convergence)**.** *Under Assumptions 9.8 and 9.9,*

$$\left(\mathbb{Z}^*_{\mathrm{cf},N,n}, \mathbb{Z}^*_{\mathrm{fact},N,n}, \mathbb{Z}^*_{\Delta,N,n}, \mathbb{Z}^*_{Q,\mathrm{cf},N,n}, \mathbb{Z}^*_{Q,\mathrm{fact},N,n}\right) \Rightarrow_{\mathbb{P}} \left(\mathbb{G}_{\mathrm{cf}}, \mathbb{G}_{\mathrm{fact}}, \mathbb{G}_{\Delta}, \mathbb{Q}_{\mathrm{cf}}, \mathbb{Q}_{\mathrm{fact}}\right)$$

in the product of the continuous-function spaces used in Part S4.

Proof. The map sending (h_θ, h_σ) and the derivative surfaces to the five process coordinates is continuous from a finite-dimensional Euclidean space times a product of uniform function spaces into the target product space. Apply Theorem 9.23, Assumption 9.9, and the conditional continuous mapping theorem. □

9.6.4 Simultaneous confidence bands

Let $T_0 \in C(\mathcal{T})$ be one target surface, let $\widehat{T}$ be its estimator, let $r_{N,n}$ be its Part S4 rate, and let $\mathbb{Z}^*_{N,n}$ be its bootstrap process. Assume

$$r_{N,n}(\widehat{T} - T_0) \Rightarrow \mathbb{G} \quad \text{in } C(\mathcal{T}), \qquad \mathbb{Z}^*_{N,n} \Rightarrow_{\mathbb{P}} \mathbb{G}. \tag{174}$$

Raw sup-norm bands Let

$$c_{1-\alpha}^{*,\mathrm{raw}} := q_{1-\alpha}^{*}\left(\left\|\mathbb{Z}_{N,n}^{*}\right\|_{\infty}\right). \tag{175}$$

Define

$$\mathcal{C}_{N,n}^{\mathrm{raw}}(\tau) := \left[\widehat{T}(\tau) - \frac{c_{1-\alpha}^{*,\mathrm{raw}}}{r_{N,n}}, \widehat{T}(\tau) + \frac{c_{1-\alpha}^{*,\mathrm{raw}}}{r_{N,n}}\right]. \tag{176}$$

Theorem 9.25 (Raw simultaneous coverage; establishes Theorem 6.3)**.** *Suppose the distribution function of* $\|\mathbb{G}\|_{\infty}$ *is continuous at its* $(1-\alpha)$ *quantile. Then*

$$\mathbb{P}\left(T_0(\tau) \in \mathcal{C}_{N,n}^{\mathrm{raw}}(\tau) \text{ for every } \tau \in \mathcal{T}\right) \longrightarrow 1-\alpha. \tag{177}$$

For CDF and tail targets, intersecting (176) *with* $[0,1]$ *preserves coverage.*

Proof. Conditional weak convergence and quantile consistency imply $c_{1-\alpha}^{*,\mathrm{raw}} \to_{\mathbb{P}} c_{1-\alpha}$, the corresponding quantile of $\|\mathbb{G}\|_{\infty}$. Equation (177) is then the continuous-mapping consequence of (174). Clipping cannot remove a true probability from an interval that already contains it. □

Floor-studentised bands Let

$$s(\tau) := \sqrt{\mathrm{Var}(\mathbb{G}(\tau))}, \qquad \widehat{s}(\tau) \xrightarrow{\mathbb{P}} s(\tau) \quad \text{uniformly.} \tag{178}$$

Ordinary studentisation fails at derivative zeros. Define

$$\widehat{s}_{\kappa}(\tau) := \max\left\{\widehat{s}(\tau), \kappa_{N,n}\left\|\widehat{s}\right\|_{\infty}\right\}. \tag{179}$$

Assumption 9.10 (Floor rate)**.** *For a fast family with rate* $b_{N,n}$*,*

$$\kappa_{N,n} \downarrow 0, \qquad b_{N,n}\kappa_{N,n} \to \infty.$$

For a slow family with rate $a_{N,n}$*,*

$$\kappa_{N,n} \downarrow 0, \qquad a_{N,n}\kappa_{N,n} \to \infty, \qquad \frac{a_{N,n}}{b_{N,n}\kappa_{N,n}} \to 0.$$

The target map has a uniform quadratic Taylor remainder.

Let

$$c_{1-\alpha}^{*,\mathrm{st}} := q_{1-\alpha}^{*}\left(\sup_{\tau\in\mathcal{T}} \frac{\left|\mathbb{Z}_{N,n}^{*}(\tau)\right|}{\widehat{s}_{\kappa}(\tau)}\right), \tag{180}$$

and define

$$\mathcal{C}_{N,n}^{\mathrm{st}}(\tau) := \left[\widehat{T}(\tau) - \frac{c_{1-\alpha}^{*,\mathrm{st}}\widehat{s}_{\kappa}(\tau)}{r_{N,n}}, \widehat{T}(\tau) + \frac{c_{1-\alpha}^{*,\mathrm{st}}\widehat{s}_{\kappa}(\tau)}{r_{N,n}}\right]. \tag{181}$$

Theorem 9.26 (Derivative-zero robust simultaneous coverage; establishes Theorem 6.3)**.** *As stated in Theorem 6.3 of the main text.*

Proof. Let r be the family rate ($b_{N,n}$ fast, $a_{N,n}$ slow). By the uniform second-order expansion of Part S4, uniformly in $\tau \in \mathcal{T}$,

$$r\big(\widehat{T}(\tau) - T_0(\tau)\big) = d(\tau)^{\top} H_{N,n} + R_{N,n}(\tau), \tag{182}$$

where $H_{N,n} := A_{N,n}(\widehat{\vartheta} - \vartheta_0) = O_{\mathbb{P}}(1)$ is the *common* two-dimensional estimation error, $d(\tau)$ is the family-specific loading with $s(\tau)^2 = d(\tau)^{\top} J^{-1} d(\tau)$ (for a slow family, d has a zero fast coordinate and $R_{N,n}$ contains the omitted fast linear term), and

$$\sup_{\tau}\left|R_{N,n}(\tau)\right| = \begin{cases} O_{\mathbb{P}}(b_{N,n}^{-1}), & \text{fast family,} \\ O_{\mathbb{P}}(a_{N,n}^{-1}) + O_{\mathbb{P}}(a_{N,n}/b_{N,n}), & \text{slow family.} \end{cases} \tag{183}$$

Step 1: the studentised first-order field is globally bounded. By Cauchy–Schwarz in the J^{-1} inner product, $|d(\tau)^{\top} H| \le s(\tau)\left\|J^{1/2}H\right\|$ for every τ, including derivative zeros (where both sides vanish). Consequently, for any positive weight $w(\tau) \ge s(\tau)$,

$$\sup_{\tau} \frac{|d(\tau)^{\top} H_{N,n}|}{w(\tau)} \le \left\|J^{1/2} H_{N,n}\right\| = O_{\mathbb{P}}(1). \tag{184}$$

The same bound holds for the multiplier draw H^{*} conditionally.

Step 2: the floor absorbs the remainder. Since $\widehat{s}_{\kappa} \ge \kappa_{N,n}\left\|\widehat{s}\right\|_{\infty}$ and $\left\|\widehat{s}\right\|_{\infty} \xrightarrow{\mathbb{P}} \|s\|_{\infty} > 0$, (183) gives

$$\sup_{\tau}\frac{|R_{N,n}(\tau)|}{\widehat{s}_{\kappa}(\tau)} \le \frac{\sup_{\tau}|R_{N,n}|}{\kappa_{N,n}\left\|\widehat{s}\right\|_{\infty}} = \begin{cases} O_{\mathbb{P}}\big((b_{N,n}\kappa_{N,n})^{-1}\big) \to 0, & \text{fast, using } b_{N,n}\kappa_{N,n} \to \infty, \\ O_{\mathbb{P}}\big((a_{N,n}\kappa_{N,n})^{-1}\big) + O_{\mathbb{P}}\big(\frac{a_{N,n}}{b_{N,n}\kappa_{N,n}}\big) \to 0, & \text{slow, using both slow-family conditions.} \end{cases}$$

Step 3: uniform ratio consistency of the studentiser. On $\{s \geq \varepsilon\}$, $\widehat{s}/s \to 1$ uniformly in probability (from $\widehat{J} \to J$, Lemma 9.12, and uniform consistency of the plug-in derivative surfaces), and the floor is asymptotically inactive there because $\kappa_{N,n} \to 0$; hence $\sup_{\{s \geq \varepsilon\}} |\widehat{s}_\kappa / s_\kappa - 1| \to 0$ in probability, where $s_\kappa := \max\{s, \kappa_{N,n} \|s\|_\infty\}$. On $\{s < \varepsilon\}$, both $\widehat{s}_\kappa$ and s_κ lie in $[\frac{1}{2}\kappa_{N,n} \|s\|_\infty, \ \varepsilon + o_{\mathbb{P}}(1)]$, and the uniform ratio bound $\widehat{s}_\kappa / s_\kappa \in [1 - \eta_\varepsilon, 1 + \eta_\varepsilon]$ holds wp$\to 1$ with η_ε controlled by the same plug-in consistency.

Step 4: quantile transfer. Combining Steps 1–3, the coverage statistic and its bootstrap counterpart differ by $o_{\mathbb{P}}(1)$:

$$\sup_\tau \frac{r|\widehat{T}(\tau) - T_0(\tau)|}{\widehat{s}_\kappa(\tau)} = \sup_\tau \frac{|d(\tau)^\top H_{N,n}|}{s_\kappa(\tau)} + o_{\mathbb{P}}(1), \qquad M^*_{N,n} = \sup_\tau \frac{|d(\tau)^\top H^*_{N,n}|}{s_\kappa(\tau)} + o_{\mathbb{P}^*}(1),$$

where the first equality uses Steps 2–3 and the Lipschitz property of $x \mapsto \sup_\tau |x(\tau)|/s_\kappa(\tau)$ under multiplicative perturbation of the denominator together with the global bound (184) (with $w = s_\kappa \geq$ each of s, $\kappa \|s\|_\infty$).

Step 5: identification of the limiting functional. For $\kappa > 0$ define $V_\kappa, V : \mathbb{R}^2 \to [0, \infty)$ by

$$V_\kappa(H) := \sup_{\tau \in \mathcal{T}} \frac{|d(\tau)^\top H|}{s_\kappa(\tau)}, \qquad V(H) := \sup_{\tau:\, s(\tau) > 0} \frac{|d(\tau)^\top H|}{s(\tau)},$$

with the convention $V(H) := 0$ if $s \equiv 0$ (excluded by $\|s\|_\infty > 0$). Three elementary facts close the argument. *(a) Zeros carry no signal:* since J is positive definite and $s(\tau)^2 = d(\tau)^\top J^{-1} d(\tau)$, $s(\tau) = 0$ iff $d(\tau) = 0$; at such τ the ratio $|d^\top H|/s_\kappa = 0$ for every $\kappa > 0$. Hence V_κ is unaffected by the zero set. *(b) Monotone pointwise convergence:* $\kappa \mapsto s_\kappa$ is non-decreasing, so $\kappa \downarrow 0$ makes V_κ non-decreasing; for fixed τ with $s(\tau) > 0$, $|d^\top H|/s_\kappa(\tau) \uparrow |d^\top H|/s(\tau)$. Suprema of pointwise non-decreasing families interchange, so $V_\kappa(H) \uparrow V(H)$ for every H as $\kappa \downarrow 0$. *(c) Uniform Lipschitz:* by (184), $|V_\kappa(H) - V_\kappa(H')| \leq \left\|J^{1/2}(H - H')\right\|$ for every κ, and the same for V. By (b)–(c) and Dini-type upgrading (an equi-Lipschitz, pointwise-monotone family converging to a Lipschitz limit converges uniformly on compacts), $V_{\kappa_{N,n}} \to V$ uniformly on compact subsets of $\mathbb{R}^2$.

Since $H_{N,n} \Rightarrow H \sim \mathcal{N}(0, J^{-1})$ and, conditionally in probability, $H^*_{N,n} \Rightarrow \mathcal{N}(0, J^{-1})$ (Theorem 9.23), the extended continuous mapping theorem [5, Theorem 1.11.1] — its hypotheses hold: the maps $V_{\kappa_{N,n}}$ are deterministic, converge uniformly on compacts to the continuous map V, and the arguments converge weakly — gives

$$V_{\kappa_{N,n}}(H_{N,n}) \Rightarrow V(H), \qquad V_{\kappa_{N,n}}(H^*_{N,n}) \Rightarrow V(H) \text{ conditionally in probability.}$$

The law of $V(H)$ is continuous at its $(1-\alpha)$ quantile by assumption, so the conditional quantiles of the bootstrap statistic converge in probability to $q_{1-\alpha}(V(H))$, and with Step 4,

$$\mathbb{P}\big(T_0(\tau) \in \mathcal{C}^{\mathrm{st}}_{N,n}(\tau)\ \forall \tau\big) = \mathbb{P}\big(V_{\kappa_{N,n}}(H_{N,n}) \leq q^*_{1-\alpha} + o_{\mathbb{P}}(1)\big) \longrightarrow 1 - \alpha.$$

□

Joint mixed-rate bands Let $\mathfrak{F}$ be a finite family of target surfaces, with family-specific rates r_f and floor standard errors $\widehat{s}_{f,\kappa}$. Define

$$M^*_{N,n} := \max_{f \in \mathfrak{F}} \sup_{\tau \in \mathcal{T}_f} \frac{\left|\mathbb{Z}^*_{f,N,n}(\tau)\right|}{\widehat{s}_{f,\kappa}(\tau)}. \tag{185}$$

Corollary 9.10 (Joint fast–slow coverage)**.** *One conditional quantile of* (185)*, applied with the family-specific widths*

$$\frac{c^{*,\mathrm{joint}}_{1-\alpha} \widehat{s}_{f,\kappa}(\tau)}{r_f},$$

gives asymptotic simultaneous coverage $1 - \alpha$ *across all target families in* $\mathfrak{F}$*. The common draw* $H^*_{N,n}$ *preserves finite-sample correlation between the estimated drift and scale coordinates.*

9.6.5 Equality and integrated tests

Let $G_0 \in C(\mathcal{T}_0)$ be an effect surface estimated by $\widehat{G}$ at rate $r_{N,n}$, with bootstrap process $\mathbb{Z}^*_G$.

For

$$H_0^= : \quad G_0(\tau) = 0 \quad \forall \tau \in \mathcal{T}_0,$$

define

$$S^{\mathrm{KS}}_{N,n} := r_{N,n} \left\|\widehat{G}\right\|_\infty, \tag{186}$$

$$S^{\mathrm{CvM}}_{N,n} := r^2_{N,n} \int_{\mathcal{T}_0} \widehat{G}(\tau)^2 \, d\mu(\tau), \tag{187}$$

where μ is a finite measure. Their bootstrap analogues are

$$S_{N,n}^{*,\mathrm{KS}} := \|\mathbb{Z}_G^*\|_\infty, \tag{188}$$

$$S_{N,n}^{*,\mathrm{CvM}} := \int_{\mathcal{T}_0} \mathbb{Z}_G^*(\tau)^2 \, d\mu(\tau). \tag{189}$$

Theorem 9.27 (Equality-test validity)**.** *Under $H_0^=$ and the process-bootstrap assumptions, bootstrap critical values from the preceding statistics yield asymptotic level α whenever the corresponding limit distribution is continuous at the critical quantile. Under every fixed alternative with $G_0 \not\equiv 0$, both tests are consistent.*

Proof. Under the null, apply the continuous mapping theorem to the sup norm and the continuous quadratic functional. Under a fixed alternative, $r_{N,n}\|G_0\|_\infty \to \infty$, and the integrated statistic diverges whenever G_0 is non-zero on a set of positive μ measure. □

These tests apply to policy-to-policy CDF contrasts, factual-versus-counterfactual distributional effects, tail effects, and quantile effects.

9.6.6 One-sided hypotheses and estimated contact sets

Consider

$$H_0^{\leq}: \quad G_0(\tau) \leq 0 \quad \forall \tau \in \mathcal{T}_0. \tag{190}$$

Define the studentised statistic

$$S_{N,n}^{\leq} := \sup_{\tau \in \mathcal{T}_0} \left[\frac{r_{N,n}\widehat{G}(\tau)}{\widehat{s}_\kappa(\tau)} \right]_+. \tag{191}$$

The population contact set is

$$\mathcal{C}_0 := \{\tau \in \mathcal{T}_0 : G_0(\tau) = 0\}.$$

Let $\eta_{N,n} \to \infty$ and $\eta_{N,n}/r_{N,n} \to 0$. Define

$$\widehat{\mathcal{C}}_{N,n} := \left\{ \tau \in \mathcal{T}_0 : \frac{r_{N,n}\widehat{G}(\tau)}{\widehat{s}_\kappa(\tau)} \geq -\eta_{N,n} \right\}. \tag{192}$$

If $\widehat{\mathcal{C}}_{N,n} = \varnothing$, the procedure does not reject. Otherwise define

$$S_{N,n}^{*,\leq} := \sup_{\tau \in \widehat{\mathcal{C}}_{N,n}} \left[\frac{\mathbb{Z}_G^*(\tau)}{\widehat{s}_\kappa(\tau)} \right]_+. \tag{193}$$

Assumption 9.11 (Contact-set separation)**.** *For every open neighbourhood $\mathcal{U}$ of $\mathcal{C}_0$, there exists $c(\mathcal{U}) > 0$ such that*

$$\sup_{\tau \in \mathcal{T}_0 \setminus \mathcal{U}} G_0(\tau) \leq -c(\mathcal{U}).$$

At least one regular contact point has positive first-order standard deviation.

Theorem 9.28 (Feasible one-sided inference)**.** *Under (190), Assumption 9.11, and the preceding bootstrap conditions:*

(i) $\mathcal{C}_0 \subseteq \widehat{\mathcal{C}}_{N,n}$ with probability tending to one;

(ii) for every fixed neighbourhood $\mathcal{U}$ of $\mathcal{C}_0$, $\widehat{\mathcal{C}}_{N,n} \subseteq \mathcal{U}$ with probability tending to one;

(iii) the test rejecting when $S_{N,n}^{\leq}$ exceeds the conditional $(1-\alpha)$ quantile of (193) has asymptotic size at most α;

(iv) at regular boundary nulls, the asymptotic size is α;

(v) if $\sup_{\tau \in \mathcal{T}_0} G_0(\tau) > 0$, the test is consistent.

Proof. Write $\widehat{G}_s(\tau) := r_{N,n}\widehat{G}(\tau)/\widehat{s}_\kappa(\tau)$ for the studentised surface. Under the expansions in the proof of Theorem 9.26, $r_{N,n}(\widehat{G} - G_0) = d^\top H_{N,n} + R_{N,n}$ with $\sup_\tau |R_{N,n}|/\widehat{s}_\kappa = o_{\mathbb{P}}(1)$ and $\sup_\tau |d^\top H_{N,n}|/\widehat{s}_\kappa = O_{\mathbb{P}}(1)$.

(i) *Inclusion.* For $\tau \in \mathcal{C}_0$, $G_0(\tau) = 0$, so $\widehat{G}_s(\tau) = d(\tau)^\top H_{N,n}/\widehat{s}_\kappa(\tau) + o_{\mathbb{P}}(1)$, whose supremum over $\mathcal{C}_0$ is $O_{\mathbb{P}}(1)$ by the global Cauchy–Schwarz bound. Since $\eta_{N,n} \to \infty$, $\mathbb{P}\big(\inf_{\mathcal{C}_0} \widehat{G}_s \geq -\eta_{N,n}\big) \to 1$, i.e. $\mathcal{C}_0 \subseteq \widehat{\mathcal{C}}_{N,n}$ wp→1.

(ii) *No over-inclusion.* Let $\mathcal{U} \supseteq \mathcal{C}_0$ be open in $\mathcal{T}_0$; by the separation assumption there is $c(\mathcal{U}) > 0$ with $G_0 \leq -c(\mathcal{U})$ off $\mathcal{U}$. Since $\|\widehat{s}_\kappa\|_\infty$ is bounded and bounded away from zero in probability,

$$\sup_{\tau \notin \mathcal{U}} \widehat{G}_s(\tau) \;\leq\; -\frac{r_{N,n}\, c(\mathcal{U})}{\|\widehat{s}_\kappa\|_\infty} + \sup_\tau \frac{|d(\tau)^\top H_{N,n}| + |R_{N,n}(\tau)|}{\widehat{s}_\kappa(\tau)} \;\leq\; -c'\, r_{N,n}\big(1 + o_{\mathbb{P}}(1)\big) + O_{\mathbb{P}}(1)$$

for some $c' > 0$, using the global Cauchy–Schwarz bound and Step 2 of the proof of Theorem 9.26. Since $\eta_{N,n}/r_{N,n} \to 0$, these indices satisfy $\widehat{G}_s < -\eta_{N,n}$ wp→1, so $\widehat{\mathcal{C}}_{N,n} \subseteq \mathcal{U}$ wp→1.

(iii)–(iv) *Size.* On the null, decompose the statistic over $\widehat{\mathcal{C}}_{N,n}$: by (i)–(ii), for every open $\mathcal{U} \supseteq \mathcal{C}_0$,

$$\sup_{\mathcal{C}_0} \left[\widehat{G}_s\right]_+ \le S^{\le}_{N,n} \le \sup_{\mathcal{U}\cap\{\text{indices kept}\}} \left[\widehat{G}_s\right]_+ + o_{\mathbb{P}}(1) \le \sup_{\mathcal{U}} \left[d^\top H_{N,n}/\widehat{s}_\kappa\right]_+ + o_{\mathbb{P}}(1),$$

using $G_0 \le 0$ throughout the null. The bootstrap statistic (193) is the same functional of H^* over the estimated set. By (i)–(ii) the estimated set is sandwiched between $\mathcal{C}_0$ and any fixed $\mathcal{U}$, wp→1; conditional convergence of H^* and the shrinking argument of the Part S4 contact-set closure (letting $\mathcal{U} \downarrow \mathcal{C}_0$ along the a.s.-continuity of the limiting Gaussian functional) give that the conditional critical value converges to the $(1-\alpha)$ quantile of $\sup_{\mathcal{C}_0}[d^\top H/s_\kappa]_+$, majorising the limit of the statistic; hence asymptotic size $\le \alpha$, with equality at regular boundary nulls where the limiting distribution is continuous and strictly increasing at its quantile.

(v) *Consistency.* If $\sup_{\mathcal{T}_0} G_0 =: c_1 > 0$, pick τ_1 with $G_0(\tau_1) \ge c_1/2$; then $S^{\le}_{N,n} \ge \widehat{G}_s(\tau_1) \ge c r_{N,n}(1 + o_{\mathbb{P}}(1)) \to \infty$, while the critical value is $O_{\mathbb{P}}(1)$ (a positive-part supremum of a studentised Gaussian over a set of indices, bounded by the global Cauchy–Schwarz bound). Rejection probability tends to one. □

9.6.7 Numerical approximation and index discretisation

Suppose the implemented target uses spatial truncation K and numerical time step δ, and let

$$\left\|\widehat{T}^{K,\delta} - \widehat{T}\right\|_\infty \le \varepsilon_{K,\delta}. \tag{194}$$

Theorem 9.29 (Undersmoothed numerical transfer)**.** *The bands and tests above retain their limits if*

$$r_{N,n}\varepsilon_{K,\delta} \to 0.$$

Thus the sufficient requirements are

$$b_{N,n}\varepsilon_{K,\delta} \to 0 \quad \textit{for fast targets,} \qquad a_{N,n}\varepsilon_{K,\delta} \to 0 \quad \textit{for slow targets.}$$

Uniform consistency of the bootstrap process additionally requires the corresponding numerical first-derivative errors to vanish uniformly.

Theorem 9.30 (Bias-enlarged bands)**.** *Suppose a deterministic or high-probability certified bound $B_{K,\delta}$ satisfies*

$$\left\|T_0^{K,\delta} - T_0\right\|_\infty \le B_{K,\delta}.$$

Enlarge either raw or studentised half-width by $B_{K,\delta}$. Then the resulting band has asymptotic simultaneous coverage at least $1-\alpha$ without requiring $r_{N,n}B_{K,\delta} \to 0$.

Proof. If the statistical band contains the approximate target $T_0^{K,\delta}$, the triangle inequality shows that the enlarged band contains T_0. The statistical event has probability tending to $1-\alpha$. □

Let $\mathcal{G}_m$ be a finite h_m-net of the implemented index set.

Proposition 9.17 (Grid approximation)**.** *For every continuous function f,*

$$\left|\sup_{\tau\in\mathcal{T}} f(\tau) - \max_{\tau\in\mathcal{G}_m} f(\tau)\right| \le \omega_f(h_m),$$

where ω_f is its modulus of continuity. Hence:

- *(i) bootstrap approximation of a continuous-process supremum requires $h_m \to 0$;*
- *(ii) continuum coverage obtained only from grid endpoints requires either*

$$r_{N,n}\omega_{T_0}(h_m) \to 0$$

  or an explicit interpolation allowance $\omega_{T_0}(h_m)$;
- *(iii) a finite scientific dictionary of weights is an exact special case;*
- *(iv) basis truncation of the compact weight ellipsoid must have target error negligible at the relevant fast or slow rate, unless a certified tail allowance is added.*

9.6.8 Implementation algorithm

For one data set:

1. Compute the exact conditional MLE $(\widehat{\theta}_{N,n}, \widehat{\sigma}_{N,n})$.
2. Compute the modal score blocks $S_{k,N,n}(\widehat{\vartheta}_{N,n})$.
3. Scale them by $D_{N,n}$ and form $\widehat{J}_{B,N,n}$.

4. Draw Gaussian multipliers and obtain $H^*_{N,n}$ from (164), or draw directly from $\mathcal{N}(0, \widehat{J}^{-1}_{B,N,n})$.
5. Evaluate every target derivative surface at the estimated parameter.
6. Form fast and slow bootstrap target processes using the same draw $H^*_{N,n}$.
7. Compute raw, floor-studentised, or joint critical values.
8. For one-sided tests, estimate the contact set using (192) before computing critical values.
9. Add a numerical-bias allowance when the Galerkin/time approximation is not undersmoothed.
10. Refine intervention, threshold, and functional grids until critical values and band widths stabilise.

9.7 Part S6: Operator sensitivity and response derivatives

9.7.1 Scope and the two perturbation regimes

Let H and U be real separable Hilbert spaces. The positive core uses a linear stochastic evolution equation

$$dU_t = AU_t\,dt + \sigma G\,dW_t, \qquad U_0 = u_0, \tag{195}$$

where W is a cylindrical Wiener process on U and $G \in \mathcal{L}(U, H)$.

Part S6 studies the local effect of changing the post-intervention generator. Two regimes must be distinguished.

(i) A bounded direction,

$$A_{\varepsilon,B} = A + \varepsilon B, \qquad B \in \mathcal{L}(H), \tag{196}$$

which may be non-diagonal and non-commuting.

(ii) A scalar rescaling,

$$A_a = aA_0, \tag{197}$$

which is the main diffusivity intervention in the paper. Its formal direction is A_0, generally unbounded.

9.7.2 Standing assumptions and target topology

Assumption 9.12 (Exponentially stable semigroup)**.** *The operator $A : D(A) \subset H \to H$ generates a strongly continuous semigroup $S = (S(t))_{t\geq 0}$ satisfying*

$$\|S(t)\|_{\mathcal{L}(H)} \leq Me^{-\omega t}, \qquad t \geq 0, \tag{198}$$

for some $M \geq 1$ and $\omega > 0$.

Assumption 9.13 (Admissible additive noise)**.** *The operator $G \in \mathcal{L}(U, H)$ satisfies*

$$\int_0^T \|S(t)G\|^2_{\mathcal{S}_2(U,H)}\,dt < \infty. \tag{199}$$

The mild solution of (195) is

$$U_t = S(t)u_0 + \sigma \int_0^t S(t-r)G\,dW_r. \tag{200}$$

Set

$$\mathcal{H}_T := L^2(0, T; H), \qquad \mathbb{X}_T := L^2(\Omega; \mathcal{H}_T). \tag{201}$$

For $w \in \mathcal{H}_T$, define

$$\Psi_w(X) := \int_0^T \langle w_t, X_t\rangle_H\,dt. \tag{202}$$

Then

$$\mathbb{E}\,|\Psi_w(X)|^2 \leq \|w\|^2_{\mathcal{H}_T}\,\|X\|^2_{\mathbb{X}_T}. \tag{203}$$

9.7.3 Bounded operator perturbations

Fix $R < \infty$ and let

$$\mathbb{B}_R := \{B \in \mathcal{L}(H) : \|B\| \leq R\}. \tag{204}$$

Choose $\varepsilon_0 > 0$ such that

$$M\varepsilon_0 R < \omega, \qquad \omega_* := \omega - M\varepsilon_0 R > 0. \tag{205}$$

Let $S_{\varepsilon,B}$ be the semigroup generated by $A + \varepsilon B$.

Lemma 9.13 (Uniform bounded-perturbation estimate)**.** *Under Assumption 9.12,*

$$\|S_{\varepsilon,B}(t)\| \leq Me^{-\omega_* t} \tag{206}$$

for $B \in \mathbb{B}_R$, $|\varepsilon| \leq \varepsilon_0$, and $t \geq 0$.

Proof. The bounded perturbation theorem [16, Theorem III.1.3] and the Dyson–Phillips series estimate [28, Chapter 3] give $\|S_{\varepsilon,B}(t)\| \le M \exp\{(-\omega + M|\varepsilon|\,\|B\|)t\}$. Use (205). (The hypotheses of the perturbation theorem hold: A generates a C_0-semigroup with $\|S(t)\| \le Me^{-\omega t}$ by Assumption 9.12, and εB is a bounded operator with $\|\varepsilon B\| \le \varepsilon_0 R$.) □

Define the Duhamel derivative

$$\mathcal{D}_B S(t) := \int_0^t S(t-s)BS(s)\,ds. \tag{207}$$

Theorem 9.31 (Semigroup Fréchet derivative in operator norm)**.** *Under Assumption 9.12, uniformly over $B \in \mathbb{B}_R$,*

$$S_{\varepsilon,B}(t) = S(t) + \varepsilon \mathcal{D}_B S(t) + \mathcal{R}_{\varepsilon,B}(t), \tag{208}$$

where

$$\sup_{0\le t\le T} \|\mathcal{R}_{\varepsilon,B}(t)\| \le C_T \varepsilon^2 \|B\|^2. \tag{209}$$

Consequently,

$$B \longmapsto \mathcal{D}_B S(\cdot) \tag{210}$$

is a bounded linear map from $\mathcal{L}(H)$ into $C([0,T];\mathcal{L}(H))$.

Proof. The variation-of-constants identity in the orientation needed for the noise estimate is

$$S_{\varepsilon,B}(t) - S(t) = \varepsilon \int_0^t S_{\varepsilon,B}(t-s)BS(s)\,ds. \tag{211}$$

Subtract $\varepsilon\mathcal{D}_B S(t)$ and apply (211) once more to the first semigroup factor. This gives

$$\mathcal{R}_{\varepsilon,B}(t) = \varepsilon^2 \int_0^t \int_0^{t-s} S_{\varepsilon,B}(t-s-r)BS(r)BS(s)\,dr\,ds. \tag{212}$$

The bounds (198) and (206) imply $\|\mathcal{R}_{\varepsilon,B}(t)\| \le C\varepsilon^2 \|B\|^2 t^2$ on $[0,T]$. Linearity is immediate from (207), and boundedness follows from

$$\|\mathcal{D}_B S(t)\| \le M^2 t e^{-\omega t} \|B\|.$$

□

Proposition 9.18 (Noise-transfer derivative)**.** *Under Assumptions 9.12–9.13,*

$$\int_0^T \|\mathcal{D}_B S(t)G\|_{\mathcal{S}_2}^2\,dt \le C_T \|B\|^2, \tag{213}$$

$$\int_0^T \|\mathcal{R}_{\varepsilon,B}(t)G\|_{\mathcal{S}_2}^2\,dt \le C_T \varepsilon^4 \|B\|^4. \tag{214}$$

Proof. Let $q(t) = \|S(t)G\|_{\mathcal{S}_2}$, which belongs to $L^2(0,T)$ by (199). Formula (207) yields

$$\|\mathcal{D}_B S(t)G\|_{\mathcal{S}_2} \le M\|B\| \int_0^t e^{-\omega(t-s)} q(s)\,ds.$$

Young's convolution inequality proves (213). In (212), keep $S(s)G$ as the rightmost factor. The remaining inner integration contributes a bounded kernel proportional to $t-s$ on $[0,T]$. A second application of Young's inequality proves (214). □

9.7.4 Derivative of the stochastic mild solution

For $B \in \mathbb{B}_R$, let

$$U_t^{\varepsilon,B} := S_{\varepsilon,B}(t)u_0 + \sigma \int_0^t S_{\varepsilon,B}(t-r)G\,dW_r. \tag{215}$$

Define

$$\dot{U}_t[B] := \mathcal{D}_B S(t)u_0 + \sigma \int_0^t \mathcal{D}_B S(t-r)G\,dW_r. \tag{216}$$

Theorem 9.32 (Fréchet derivative of the counterfactual field; establishes Theorem 7.1)**.** *Under Assumptions 9.12–9.13, the perturbation map is Fréchet differentiable at zero from $\mathcal{L}(H)$ into $\mathbb{X}_T$. More precisely,*

$$\left\|U^{\varepsilon,B} - U - \varepsilon\dot{U}[B]\right\|_{\mathbb{X}_T} \le C_T \varepsilon^2 \|B\|^2. \tag{217}$$

The derivative has the equivalent mild response form

$$\dot{U}_t[B] = \int_0^t S(t-s)BU_s\,ds \tag{218}$$

in $\mathbb{X}_T$*. Thus it solves*

$$d\dot{U}_t = A\dot{U}_t\,dt + BU_t\,dt, \qquad \dot{U}_0 = 0, \tag{219}$$

in the mild sense.

Proof. For the deterministic component, Theorem 9.31 gives

$$\int_0^T \left\|\mathcal{R}_{\varepsilon,B}(t)u_0\right\|^2 dt \le C_T\varepsilon^4 \left\|B\right\|^4 \left\|u_0\right\|^2.$$

For the stochastic component, Itô isometry and Fubini give

$$\begin{aligned}\mathbb{E}\int_0^T \left\|\sigma\int_0^t \mathcal{R}_{\varepsilon,B}(t-r)G\,dW_r\right\|^2 dt \\ = \sigma^2\int_0^T (T-s)\left\|\mathcal{R}_{\varepsilon,B}(s)G\right\|_{\mathcal{S}_2}^2 ds \le C_T\varepsilon^4\left\|B\right\|^4\end{aligned}$$

by Proposition 9.18. This proves (217).

To obtain (218), insert (200) into the right-hand side. The deterministic term is $\mathcal{D}_B S(t)u_0$. For the stochastic term, stochastic Fubini is justified by Assumption 9.13, the semigroup bounds, and boundedness of B. It yields

$$\begin{aligned}\sigma\int_0^t S(t-s)B\left(\int_0^s S(s-r)G\,dW_r\right)ds \\ = \sigma\int_0^t\left[\int_r^t S(t-s)BS(s-r)ds\right]G\,dW_r = \sigma\int_0^t \mathcal{D}_B S(t-r)G\,dW_r.\end{aligned}$$

□

Corollary 9.11 (Derivative of the field functional)**.** *For every* $w \in \mathcal{H}_T$*,*

$$\frac{\Psi_w(U^{\varepsilon,B}) - \Psi_w(U)}{\varepsilon} \longrightarrow \Psi_w(\dot{U}[B]) \tag{220}$$

in $L^2(\Omega)$*, uniformly for* B *in bounded subsets of* $\mathcal{L}(H)$ *after division by* $\|B\|$*.*

Proof. Apply (203) to (217). □

9.7.5 Mean and covariance response

For $w \in \mathcal{H}_T$, define

$$m_A(w) := \int_0^T \langle w_t, S(t)u_0\rangle_H\,dt. \tag{221}$$

Define the adjoint transfer

$$g_r^{A,w} := \int_r^T S^*(t-r)w_t dt, \qquad 0 \le r \le T, \tag{222}$$

and the variance

$$v_A(w) := \sigma^2\int_0^T \left\|G^* g_r^{A,w}\right\|_U^2 dr. \tag{223}$$

Stochastic Fubini gives

$$\Psi_w(U) \sim \mathcal{N}(m_A(w), v_A(w)). \tag{224}$$

Theorem 9.33 (Mean derivative and remainder)**.** *For* $B \in \mathcal{L}(H)$*,*

$$D_A m_A(w)[B] = \int_0^T \left\langle w_t, \int_0^t S(t-s)BS(s)u_0 ds\right\rangle_H dt. \tag{225}$$

For every $R_w < \infty$*,*

$$\sup_{\|w\|_{\mathcal{H}_T}\le R_w} |D_A m_A(w)[B]| \le C_T R_w \left\|u_0\right\|\left\|B\right\|, \tag{226}$$

$$\sup_{\|w\|_{\mathcal{H}_T}\le R_w} |m_{A+\varepsilon B}(w) - m_A(w) - \varepsilon D_A m_A(w)[B]| \le C_T\varepsilon^2 R_w \left\|u_0\right\|\left\|B\right\|^2. \tag{227}$$

Proof. Insert (208) into the mean and use Cauchy–Schwarz in time. The derivative is (225); bounds (226)–(227) follow from Theorem 9.31. □

Lemma 9.14 (Adjoint transfer derivative)**.** *The map $A \mapsto g^{A,w}$ is differentiable in $\mathcal{H}_T$, with*

$$D_A g_r^{A,w}[B] = \int_r^T \left[\int_0^{t-r} S^*(s) B^* S^*(t-r-s) ds \right] w_t dt. \tag{228}$$

Uniformly for bounded w-classes,

$$\left\| D_A g^{A,w}[B] \right\|_{\mathcal{H}_T} \le C_T \|w\|_{\mathcal{H}_T} \|B\|, \tag{229}$$

$$\left\| g^{A+\varepsilon B,w} - g^{A,w} - \varepsilon D_A g^{A,w}[B] \right\|_{\mathcal{H}_T} \le C_T \varepsilon^2 \|w\|_{\mathcal{H}_T} \|B\|^2. \tag{230}$$

Proof. Take adjoints in (207):

$$\mathcal{D}_B S(t)^* = \int_0^t S^*(s) B^* S^*(t-s) ds.$$

Insert this into (222). The stable semigroup bound and Young's inequality for the backward Volterra operator give (229). Apply the same argument to the adjoint of the remainder in (209) to obtain (230). □

Theorem 9.34 (Variance derivative and remainder)**.** *For $B \in \mathcal{L}(H)$,*

$$D_A v_A(w)[B] = 2\sigma^2 \int_0^T \left\langle G^* g_r^{A,w}, G^* D_A g_r^{A,w}[B] \right\rangle_U dr. \tag{231}$$

For every $R_w < \infty$,

$$\sup_{\|w\|_{\mathcal{H}_T} \le R_w} |D_A v_A(w)[B]| \le C_T R_w^2 \|B\|, \tag{232}$$

$$\sup_{\|w\|_{\mathcal{H}_T} \le R_w} |v_{A+\varepsilon B}(w) - v_A(w) - \varepsilon D_A v_A(w)[B]| \le C_T \varepsilon^2 R_w^2 \|B\|^2. \tag{233}$$

Proof. The map $g \mapsto \sigma^2 \|G^* g\|^2_{L^2(0,T;U)}$ is a continuous quadratic form on $\mathcal{H}_T$. Apply the product rule together with Lemma 9.14. The quadratic remainder is the sum of a term involving (230) and the square of the first-order increment. Both are bounded by the right-hand side of (233). □

9.7.6 Derivative of Gaussian counterfactual laws

For $v_A(w) > 0$, set

$$a_{A,w,z} := \frac{z - m_A(w)}{\sqrt{v_A(w)}}, \qquad F_{A,w}(z) := \Phi(a_{A,w,z}). \tag{234}$$

Let $\mathcal{W} \subset \mathcal{H}_T$ be bounded, let $\mathcal{Z} \subset \mathbb{R}$ be compact, and assume

$$\inf_{w \in \mathcal{W}} v_A(w) \ge v_- > 0. \tag{235}$$

Theorem 9.35 (Uniform CDF derivative; establishes Theorem 7.1)**.** *Under (235),*

$$D_A F_{A,w}(z)[B] = \phi(a_{A,w,z}) \left[-\frac{D_A m_A(w)[B]}{\sqrt{v_A(w)}} - \frac{z - m_A(w)}{2 v_A(w)^{3/2}} D_A v_A(w)[B] \right]. \tag{236}$$

Moreover, for sufficiently small $|\varepsilon|$,

$$\sup_{(w,z) \in \mathcal{W} \times \mathcal{Z}} |F_{A+\varepsilon B,w}(z) - F_{A,w}(z) - \varepsilon D_A F_{A,w}(z)[B]| \le C \varepsilon^2 \|B\|^2. \tag{237}$$

Proof. Theorems 9.33 and 9.34 give uniform first-order expansions with quadratic remainders. The lower variance bound and continuity imply that the perturbed variance remains above $v_-/2$ for small ε. The map

$$(m, v) \mapsto \Phi((z-m)/\sqrt{v})$$

has uniformly bounded first and second derivatives on the resulting compact set. The Banach-space chain rule gives (236); Taylor's theorem gives (237). □

Corollary 9.12 (Tail and quantile derivatives)**.** *The tail probability $R_{A,w}(z) = 1 - F_{A,w}(z)$ satisfies*

$$D_A R_{A,w}(z)[B] = -D_A F_{A,w}(z)[B].$$

For $p \in [p_-, p_+] \Subset (0,1)$, the Gaussian quantile

$$Q_{A,w}(p) = m_A(w) + \sqrt{v_A(w)} \Phi^{-1}(p) \tag{238}$$

has derivative

$$D_A Q_{A,w}(p)[B] = D_A m_A(w)[B] + \frac{\Phi^{-1}(p)}{2\sqrt{v_A(w)}} D_A v_A(w)[B]. \tag{239}$$

Both derivatives and their quadratic remainders are uniform over the declared compact target classes.

9.7.7 Wasserstein stability and tangent laws

Theorem 9.36 (Local W_2 stability; establishes Theorem 7.1)**.** *Under Assumptions 9.12–9.13,*

$$W_2\left(\mathcal{L}(\Psi_w(U^{\varepsilon,B})), \mathcal{L}(\Psi_w(U))\right) \le C_T\,|\varepsilon|\,\|B\|\,\|w\|_{\mathcal{H}_T} \tag{240}$$

uniformly over $B \in \mathbb{B}_R$. In addition,

$$W_2\left(\mathcal{L}\left(\frac{\Psi_w(U^{\varepsilon,B}) - \Psi_w(U)}{\varepsilon}\right), \mathcal{L}(\Psi_w(\dot{U}[B]))\right) \longrightarrow 0. \tag{241}$$

Proof. Use the coupling in which the perturbed and unperturbed fields share the same Wiener process. By the definition of W_2, (203), and the first-order semigroup bound,

$$\begin{aligned} W_2^2 &\le \mathbb{E}\left|\Psi_w(U^{\varepsilon,B} - U)\right|^2 \\ &\le \|w\|_{\mathcal{H}_T}^2 \left\|U^{\varepsilon,B} - U\right\|_{\mathbb{X}_T}^2 \le C_T \varepsilon^2 \|B\|^2 \|w\|_{\mathcal{H}_T}^2 . \end{aligned}$$

This proves (240). For (241), use the same coupling and Corollary 9.11. □

9.7.8 The unbounded scalar-rescaling intervention

Let A_0 generate an exponentially stable analytic semigroup. For

$$a \in [a_-, a_+] \Subset (0, \infty), \tag{242}$$

write

$$S_a(t) := e^{taA_0}. \tag{243}$$

Assumption 9.14 (Uniform analytic bounds)**.** *For $j = 1, 2$,*

$$\sup_{a\in[a_-,a_+]} \sup_{t>0} \left\| t^j A_0^j S_a(t) \right\| < \infty. \tag{244}$$

This holds for the negative self-adjoint generator in the paper's diagonal core and, more generally, for bounded analytic semigroups uniformly over compact positive rescalings.

Theorem 9.37 (Sensitivity of scalar generator rescaling)**.** *Under Assumption 9.14,*

$$\partial_a S_a(t) = tA_0 S_a(t), \qquad \partial_a^2 S_a(t) = t^2 A_0^2 S_a(t). \tag{245}$$

Uniformly for $a \in [a_-, a_+]$ and $t \in [0, T]$,

$$\|S_{a+h}(t) - S_a(t) - h\,tA_0 S_a(t)\| \le Ch^2. \tag{246}$$

If the corresponding noise-transfer derivatives are square-integrable in Hilbert-Schmidt norm, all conclusions of Theorems 9.32–9.36 hold with the scalar direction

$$\dot{S}_a(t) = tA_0 S_a(t). \tag{247}$$

Proof. For $t > 0$, analytic semigroup calculus gives (245). Taylor's formula in the scalar parameter yields

$$S_{a+h}(t) - S_a(t) - h\,tA_0 S_a(t) = h^2 \int_0^1 (1-r) t^2 A_0^2 S_{a+rh}(t) dr.$$

The uniform bound (244) proves (246); at $t = 0$ both sides vanish. The remaining claims follow by repeating the deterministic and stochastic arguments above with (247) and the stated noise-transfer integrability. □

Corollary 9.13 (Diagonal formulas used by Part S4)**.** *Suppose $A_0 e_k = -\lambda_k e_k$ and $Ge_k = q_k e_k$. Then*

$$\dot{m}_a(w) = \int_0^T \langle w_t, tA_0 S_a(t) u_0 \rangle\, dt, \tag{248}$$

$$(\dot{K}_a w)_k(r) = -q_k \int_r^T \lambda_k (t-r) e^{-a\lambda_k(t-r)} w_k(t) dt, \tag{249}$$

$$\dot{V}_a(w) = 2\left\langle K_a w, \dot{K}_a w \right\rangle_{L^2(0,T;H)}. \tag{250}$$

The resulting CDF derivative is exactly the intervention derivative used in the Part S4 process theorem.

9.8 Part S7: Total statistical–numerical error

9.8.1 Standing assumptions and target class

The paper's positive theorem chain is now

$$\text{semantics} \Rightarrow \text{identification} \Rightarrow \text{LAN and efficiency} \Rightarrow \text{uniform target processes}$$
$$\Rightarrow \text{feasible bands} \Rightarrow \text{total numerical–statistical error.}$$

The present part proves the final implication.

Let $H = L^2(D)$ and set $\Lambda = -A_0$. Assume

$$\Lambda e_k = \lambda_k e_k, \qquad 0 < \lambda_1 \le \lambda_2 \le \cdots, \qquad \lambda_k \asymp k^{2/d}. \tag{251}$$

For an absolute intervention $a \in \widetilde{\Theta} = [a_-, a_+] \Subset (0, \infty)$,

$$dU_t^a = -a\Lambda U_t^a\, dt + \sigma Q^{1/2} dW_t, \qquad U_0^a = u_0, \tag{252}$$

where $Q^{1/2} e_k = q_k e_k$ and $\sigma \in [\sigma_-, \sigma_+] \Subset (0, \infty)$.

For $\mathcal{H} = L^2(0, T; H)$ and $w \in \mathcal{H}$, define

$$X_{a,w} := \Psi_w(U^a) = \int_0^T \langle w_t, U_t^a \rangle\, dt. \tag{253}$$

It is Gaussian:

$$X_{a,w} \sim \mathcal{N}\big(m_a(w), \sigma^2 V_a(w)\big), \tag{254}$$

with

$$m_a(w) = \sum_{k \ge 1} u_{0,k} \int_0^T w_k(t) e^{-a\lambda_k t}\, dt, \tag{255}$$

$$V_a(w) = \sum_{k \ge 1} q_k^2 \int_0^T \left[\int_s^T w_k(t) e^{-a\lambda_k (t-s)}\, dt \right]^2 ds. \tag{256}$$

Assumption 9.15 (Target and regularity class)**.** *There exist $r, s > 0$, $R < \infty$, $r_0 \ge 0$, and $\beta \in \mathbb{R}$ such that*

$$\mathcal{W}_R^{r,s} := \left\{ w = \sum_{k,\ell} w_{k\ell} \varphi_\ell e_k : \sum_{k,\ell} (1 + \lambda_k)^{2s} (1 + \nu_\ell)^{2r} |w_{k\ell}|^2 \le R^2 \right\}, \tag{257}$$

$$u_0 \in D(\Lambda^{r_0/2}), \tag{258}$$

$$q_k^2 \le C_q \lambda_k^{-\beta}, \qquad \beta + 1 > \frac{d}{2}, \tag{259}$$

$$\sum_{k \ge 1} \frac{q_k^2}{\lambda_k} < \infty. \tag{260}$$

The operative class $\mathcal{W} \subset \mathcal{W}_R^{r,s}$ satisfies

$$0 < v_- \le V_a(w) \le v_+ < \infty \qquad \text{for all } (a, w) \in \widetilde{\Theta} \times \mathcal{W}. \tag{261}$$

The temporal weight in (257) is needed for compactness and temporal approximation. The lower variance bound is inherited from Part S4 and is the mechanism that upgrades the stochastic variance tail into a sharper Gaussian-law rate.

9.8.2 Exact Galerkin primitives

Let P_K be the spectral projection onto $\operatorname{span}\{e_1, \dots, e_K\}$ and define

$$X_{a,w}^K := \Psi_{P_K w}(P_K U^a). \tag{262}$$

Its mean and primitive variance are

$$m_{a,K}(w) = \sum_{k \le K} u_{0,k} \int_0^T w_k(t) e^{-a\lambda_k t}\, dt, \tag{263}$$

$$V_{a,K}(w) = \sum_{k \le K} q_k^2 \int_0^T \left[\int_s^T w_k(t) e^{-a\lambda_k (t-s)}\, dt \right]^2 ds. \tag{264}$$

In particular, $V_a(w) - V_{a,K}(w) \ge 0$.

Theorem 9.38 (Uniform deterministic mean tail)**.** *Under Assumption 9.15, with*

$$\mu_m := \frac{r_0 + 1 + 2s}{2}, \tag{265}$$

there exists $C < \infty$ such that

$$\sup_{a \in \widetilde{\Theta},\, w \in \mathcal{W}} |m_a(w) - m_{a,K}(w)| \le C \lambda_{K+1}^{-\mu_m}. \tag{266}$$

Proof. Since

$$\left\| e^{-a\lambda_k \cdot} \right\|_{L^2(0,T)} \le (2a_- \lambda_k)^{-1/2},$$

Cauchy–Schwarz gives

$$\begin{aligned} |m_a - m_{a,K}| &\le C \sum_{k>K} |u_{0,k}| \lambda_k^{-1/2} or m w_k L^2 \\ &\le C \left(\sum_{k>K} \lambda_k^{r_0} |u_{0,k}|^2 \right)^{1/2} \left(\sum_{k>K} \lambda_k^{-r_0-1} \|w_k\|_{L^2}^2 \right)^{1/2}. \end{aligned}$$

Insert $\lambda_k^{2s} \lambda_k^{-2s}$ in the second factor and bound the tail by $\lambda_{K+1}^{-(r_0+1+2s)/2}$. Assumption 9.15 supplies the uniform constants. □

Theorem 9.39 (Uniform stochastic variance tail)**.** *Set*

$$\mu_V := \beta + 2 + 2s. \tag{267}$$

Then

$$0 \le \sup_{a,w} \{V_a(w) - V_{a,K}(w)\} \le C \lambda_{K+1}^{-\mu_V}. \tag{268}$$

Proof. For the backward Volterra kernel, Young's inequality yields

$$\left\| \int_\cdot^T w_k(t) e^{-a\lambda_k(t-\cdot)} dt \right\|_{L^2(0,T)} \le \frac{1}{a_- \lambda_k} \|w_k\|_{L^2(0,T)}.$$

Therefore

$$\begin{aligned} V_a - V_{a,K} &\le C \sum_{k>K} q_k^2 \lambda_k^{-2} \|w_k\|_{L^2}^2 \\ &\le C \lambda_{K+1}^{-(\beta+2+2s)} \sum_{k>K} \lambda_k^{2s} \|w_k\|_{L^2}^2. \end{aligned}$$

The ellipsoid bound completes the proof. □

9.8.3 Strong functional error and Gaussian law error

Because the omitted stochastic modes are centred and orthogonal,

$$\mathbb{E}|X_{a,w} - X_{a,w}^K|^2 = |m_a - m_{a,K}|^2 + \sigma^2 (V_a - V_{a,K}). \tag{269}$$

Corollary 9.14 (Strong coupled functional rate)**.** *Let*

$$\mu_{\mathrm{str}} := \min\left\{ \mu_m, \frac{\mu_V}{2} \right\}. \tag{270}$$

Then

$$\sup_{a,w,\sigma} \left\| X_{a,w} - X_{a,w}^K \right\|_{L^2(\Omega)} \le C \lambda_{K+1}^{-\mu_{\mathrm{str}}}. \tag{271}$$

Under Weyl growth,

$$\rho_{\mathrm{str}} = \min\left\{ \frac{r_0 + 1 + 2s}{d}, \frac{\beta + 2 + 2s}{d} \right\}, \qquad \left\| X - X^K \right\|_{L^2} = O(K^{-\rho_{\mathrm{str}}}). \tag{272}$$

The natural shared-noise coupling is not generally optimal at the law level.

Lemma 9.15 (Normal-law comparison)**.** *Let $G_i = \mathcal{N}(m_i, s_i^2)$ with $s_i \ge s_- > 0$. Then*

$$\mathsf{W}_2^2(G_0, G_1) = |m_1 - m_0|^2 + |s_1 - s_0|^2, \tag{273}$$

$$\sup_z |G_1(-\infty, z] - G_0(-\infty, z]| \le \frac{|m_1 - m_0|}{s_- \sqrt{2\pi}} + \frac{|s_1 - s_0|}{s_- \sqrt{2\pi e}}. \tag{274}$$

For $p \in [p_-, p_+] \Subset (0,1)$,

$$|Q_1(p) - Q_0(p)| \le |m_1 - m_0| + C_{p_-,p_+} |s_1 - s_0|. \tag{275}$$

Proof. Equation (273) follows from the monotone quantile coupling. For the CDF, interpolate first in the mean and then in $\log s$. The derivatives are bounded by $1/(s_-\sqrt{2\pi})$ and $\sup_x |x\phi(x)| = (2\pi e)^{-1/2}$, respectively. Equation (275) follows from $Q_i(p) = m_i + s_i\Phi^{-1}(p)$. □

Theorem 9.40 (Improved Gaussian law rate)**.** *Let*

$$\mu_{\mathrm{law}} := \min\{\mu_m, \mu_V\}. \tag{276}$$

For all sufficiently large K,

$$\sup_{a,w,\sigma} \mathsf{W}_2\big(\mathcal{L}(X_{a,w}), \mathcal{L}(X^K_{a,w})\big) \le C\lambda_{K+1}^{-\mu_{\mathrm{law}}}, \tag{277}$$

$$\sup_{a,w,z,\sigma} |F_{a,w,K}(z) - F_{a,w}(z)| \le C\lambda_{K+1}^{-\mu_{\mathrm{law}}}, \tag{278}$$

$$\sup_{a,w,p,\sigma} |Q_{a,w,K}(p) - Q_{a,w}(p)| \le C\lambda_{K+1}^{-\mu_{\mathrm{law}}}. \tag{279}$$

Under Weyl growth, the common law exponent is

$$\rho_F = \min\left\{\frac{r_0 + 1 + 2s}{d}, \frac{2(\beta + 2 + 2s)}{d}\right\}. \tag{280}$$

Proof. Assumption 9.15 and Theorem 9.39 imply $V_{a,K}(w) \ge v_-/2$ for large K. Therefore

$$|\sqrt{V_a} - \sqrt{V_{a,K}}| = \frac{V_a - V_{a,K}}{\sqrt{V_a} + \sqrt{V_{a,K}}} \le C(V_a - V_{a,K}).$$

Apply Lemma 9.15 together with Theorems 9.38–9.39. □

9.8.4 Modal-decay corollaries and a refined time-smooth rate

Corollary 9.15 (Pointwise modal-decay rates)**.** *Suppose*

$$|u_{0,k}| \le C\lambda_k^{-r_0/2}, \qquad \|w_k\|_{L^2} \le C\lambda_k^{-\gamma/2}, \qquad q_k^2 \le C\lambda_k^{-\beta}.$$

If the displayed tails are summable, then

$$|m - m_K| = O(K^{-\rho_m^{\mathrm{pt}}}), \qquad \rho_m^{\mathrm{pt}} = \frac{r_0 + \gamma + 1 - d}{d}, \tag{281}$$

$$V - V_K = O(K^{-\rho_V^{\mathrm{pt}}}), \qquad \rho_V^{\mathrm{pt}} = \frac{2(\beta + \gamma + 2) - d}{d}. \tag{282}$$

Consequently,

$$\rho_{\mathrm{str}}^{\mathrm{pt}} = \min\left\{\rho_m^{\mathrm{pt}}, \frac{\rho_V^{\mathrm{pt}}}{2}\right\}, \qquad \rho_F^{\mathrm{pt}} = \min\left\{\rho_m^{\mathrm{pt}}, \rho_V^{\mathrm{pt}}\right\}. \tag{283}$$

Proof. Use the Weyl-tail estimate

$$\sum_{k>K} \lambda_k^{-p} \lesssim K^{1-2p/d}, \qquad p > d/2,$$

with $p = (r_0 + \gamma + 1)/2$ for the mean and $p = \beta + \gamma + 2$ for the variance. □

9.8.5 Temporal approximation

The target functional, rather than the whole state path, is the primary numerical object. This allows an exact-modal construction that avoids an unnecessary state-discretisation error.

Projected-weight exact-modal scheme Let Π_δ be a piecewise-polynomial projection on a partition of mesh δ. Define

$$X^{K,\delta}_{a,w} := \Psi_{\Pi_\delta w}(P_K U^a). \tag{284}$$

In the diagonal additive model, its Gaussian mean and covariance can be evaluated mode by mode; equivalently, integrated OU cell increments can be sampled exactly.

Assumption 9.16 (Temporal projection rate)**.** *For some $\eta > 0$,*

$$\sup_{w\in\mathcal{W}} \|w - \Pi_\delta w\|_{\mathcal{H}} \le C\delta^\eta. \tag{285}$$

Theorem 9.41 (Temporal functional and law rate)**.** *Under Assumptions 9.15 and 9.16, uniformly in* K, a, w, σ*,*

$$\left\|X^K_{a,w} - X^{K,\delta}_{a,w}\right\|_{L^2(\Omega)} \le C\delta^\eta, \tag{286}$$

$$\mathsf{W}_2\big(\mathcal{L}(X^K_{a,w}), \mathcal{L}(X^{K,\delta}_{a,w})\big) \le C\delta^\eta. \tag{287}$$

For sufficiently small δ*, the corresponding CDF and quantile errors are bounded by* $C\delta^\eta$ *uniformly over the Part S4 index sets.*

Proof. By Cauchy–Schwarz,

$$|\Psi_{w-\Pi_\delta w}(P_K U)|^2 \le \|w - \Pi_\delta w\|^2_{\mathcal{H}} \int_0^T \|P_K U_t\|^2 \, dt.$$

The integrated moment is uniformly bounded by the baseline Part S1 estimate. The coupling gives (287), and Lemma 9.15 gives the law bounds after preserving the lower variance bound. □

Left-point quadrature and approximate states Consider

$$\bar{X}^{K,\delta}_{a,w} = \delta \sum_{j=0}^{J-1} \left\langle w(t_j), U^{a,K}_{t_j} \right\rangle. \tag{288}$$

Assumption 9.17 (Time regularity)**.** *For exponents* $\eta_w, \chi > 0$*,*

$$\sup_{w\in\mathcal{W}} \|w_t - w_s\| \le C|t-s|^{\eta_w}, \qquad \sup_w \|w\|_{L^\infty(0,T;H)} < \infty, \tag{289}$$

$$\sup_{a,K} \mathbb{E}\left\|U^{a,K}_t - U^{a,K}_s\right\|^2 \le C|t-s|^{2\chi}. \tag{290}$$

A sufficient field condition for (290), with $0 < \chi \le 1/2$*, is*

$$u_0 \in D(\Lambda^\chi), \qquad \sum_k q_k^2 \lambda_k^{2\chi-1} < \infty. \tag{291}$$

Proposition 9.19 (Robust rectangle-rule rate)**.** *Under Assumption 9.17,*

$$\sup_{a,K,w} \left\|X^K_{a,w} - \bar{X}^{K,\delta}_{a,w}\right\|_{L^2} \le C\delta^{\min(\eta_w,\chi)}. \tag{292}$$

If an approximate state $\widehat{U}^{K,\delta}$ *satisfies*

$$\sup_{a,K} \mathbb{E}\int_0^T \left\|U^{a,K}_t - \widehat{U}^{a,K,\delta}_t\right\|^2 dt \le C\delta^{2\kappa_U}, \tag{293}$$

then the target-level rate is δ^κ*, where*

$$\kappa = \min\{\eta_w, \chi, \kappa_U\}. \tag{294}$$

For the projected-weight scheme, replace η_w *and* χ *by the projection exponent* η *and use* $\kappa = \min\{\eta, \kappa_U\}$*.*

Proof. Let $\pi_\delta(t)$ be the left grid point. Then

$$\begin{aligned}
&\mathbb{E}\left|\langle w_t, U_t\rangle - \left\langle w_{\pi_\delta(t)}, U_{\pi_\delta(t)}\right\rangle\right|^2 \\
&\quad \le 2\left\|w_t - w_{\pi_\delta(t)}\right\|^2 \mathbb{E}\left\|U_t\right\|^2 + 2\left\|w_{\pi_\delta(t)}\right\|^2 \mathbb{E}\left\|U_t - U_{\pi_\delta(t)}\right\|^2.
\end{aligned}$$

Integrate in time and use $|\int_0^T f|^2 \le T\int_0^T |f|^2$. The state-scheme term follows from Cauchy–Schwarz and (293). □

9.8.6 Bounded operator approximation

Let the numerical generator be

$$A_{a,M} = aA_0 + E_{a,M}, \qquad \sup_{a\in\widetilde{\Theta}} orm E_{a,M}{}_{\mathcal{L}(H)} \le \eta_M. \tag{295}$$

This term covers bounded operator perturbations only. Spectral projection and domain-changing differential approximations remain governed by the Galerkin analysis and must not be hidden in $E_{a,M}$.

Theorem 9.42 (Operator transfer and response correction)**.** *Let* T_A *denote any non-degenerate Gaussian CDF, tail, or quantile target considered in Part S6. Then*

$$\sup_\tau |T_{A_{a,M}}(\tau) - T_{aA_0}(\tau)| \le C\eta_M. \tag{296}$$

If $E_{a,M}$ is known and

$$T^{\mathrm{corr}}_{A_{a,M}} := T_{A_{a,M}} - D_A T_{aA_0}[E_{a,M}], \tag{297}$$

then

$$\sup_\tau |T^{\mathrm{corr}}_{A_{a,M}}(\tau) - T_{aA_0}(\tau)| \le C\eta_M^2. \tag{298}$$

For a scalar intervention grid a_M, take $\eta_M = |a_M - a|$ and use the Part S6 analytic derivative $\partial_a T$.

Proof. Apply the Part S6 uniform first-order operator Lipschitz theorem and its quadratic Fréchet remainder. The constants are uniform on the compact intervention and target classes. □

9.8.7 Uniform primitive and loading approximation

Let T be any Part S4 target family: fixed-intervention CDF, factual CDF, distributional effect, tail, or quantile surface. Set

$$B_{K,\delta,M} := C\left(K^{-\rho_F} + \delta^\kappa + \eta_M\right). \tag{299}$$

For a first-order operator-corrected implementation, replace η_M by η_M^2.

Theorem 9.43 (Target and derivative approximation)**.** *Uniformly over the compact parameter rectangle,*

$$\left\|T^{K,\delta,M}(\vartheta) - T(\vartheta)\right\|_\infty \le B_{K,\delta,M}, \tag{300}$$

$$\left\|DT^{K,\delta,M}(\vartheta) - DT(\vartheta)\right\|_\infty \le C\left(K^{-\rho_F} + \delta^\kappa + \eta_M\right). \tag{301}$$

The same spatial exponent applies to the scalar diffusivity derivatives through order two.

Proof. Target bound (300). By the triangle inequality,

$$\left\|T^{K,\delta,M} - T\right\|_\infty \le \left\|T^{K,\delta,M} - T^{K,\delta}\right\|_\infty + \left\|T^{K,\delta} - T^K\right\|_\infty + \left\|T^K - T\right\|_\infty,$$

and the three terms are bounded by $C\eta_M$ (Theorem 9.42, uniform on the compact classes; $C\eta_M^2$ after the first-order correction (297)), $C\delta^\kappa$ (Theorem 9.41 or Proposition 9.19), and $CK^{-\rho_F}$ (Theorem 9.40), respectively; all constants are uniform over the parameter rectangle because the underlying mean and variance tails are.

Derivative bound (301). It suffices to prove that the intervention derivatives of the *primitives* obey tail bounds with the same spectral exponents; the chain rule through the smooth Gaussian formulas then transfers them to DT exactly as in the zeroth-order proofs. For the mean: $m_a(w) = \sum_k u_{0,k} \int_0^T w_k(t) e^{-a\lambda_k t}\,dt$, so

$$\partial_a^j m_a(w) = \sum_k (-1)^j u_{0,k} \lambda_k^j \int_0^T t^j w_k(t) e^{-a\lambda_k t}\,dt, \qquad j = 1, 2,$$

and by Cauchy–Schwarz with $\left\|t^j e^{-a\lambda_k t}\right\|^2_{L^2(0,T)} = \int_0^T t^{2j} e^{-2a\lambda_k t} dt \le \Gamma(2j+1)(2\underline{a}\lambda_k)^{-(2j+1)}$,

$$\lambda_k^j \left\|t^j e^{-a\lambda_k t}\right\|_{L^2} \le C_j\, \lambda_k^j \lambda_k^{-(2j+1)/2} = C_j\, \lambda_k^{-1/2},$$

the same spectral order as the underlying kernel $\left\|e^{-a\lambda_k t}\right\|_{L^2} \le C\lambda_k^{-1/2}$. Hence the tail sums defining the mean error repeat verbatim with unchanged exponent $(r_0 + 1 + 2s)/2$, through second order in a. For the variance: with the transfer kernel $g_{a,k,w}(s) = \int_s^T w_k(t) e^{-a\lambda_k(t-s)} dt = (\mathcal{K}_{a,k} * w_k)(s)$ a Volterra convolution against $\mathcal{K}(u) = e^{-a\lambda_k u}\mathbf{1}_{u\ge 0}$,

$$\partial_a^j g_{a,k,w} = (-\lambda_k)^j \left(u^j \mathcal{K}\right) * w_k, \qquad \left\|\lambda_k^j u^j \mathcal{K}\right\|_{L^1(0,\infty)} = \lambda_k^j \frac{\Gamma(j+1)}{(a\lambda_k)^{j+1}} \le C_j \lambda_k^{-1},$$

which is the same L^1 order as $\|\mathcal{K}\|_{L^1} = 1/(a\lambda_k)$. Young's inequality ($L^1 * L^2 \to L^2$) then reproduces the variance-tail exponent $\beta + 2 + 2s$ for $\partial_a^j V_a(w)$, $j = 1, 2$. Temporal and bounded-operator errors for the derivative surfaces follow by applying the same projection and perturbation arguments to these differentiated kernels, which lie in the same kernel classes; and the derivative of the Gaussian CDF composition contributes bounded factors by the uniform non-degeneracy $\inf_{a,w} V_a(w) > 0$. Combining gives (301). □

9.8.8 Uniform total-error theorem

The statistical normalisations are

$$R^{\mathrm{fast}}_{N,n} = b_{N,n} := \sqrt{Nn}, \qquad R^{\mathrm{slow}}_{N,n} = a_{N,n} := \sqrt{\Lambda_N}, \qquad \Lambda_N := \sum_{k\le N} \lambda_k. \tag{302}$$

The fast rate applies to fixed absolute-intervention and policy-to-policy targets. The slow rate applies to factual and factual-versus-counterfactual targets.

Theorem 9.44 (Statistical–numerical decomposition; establishes Theorem 7.2)**.** *Let f be one of the Part S4 target families and let $R_{N,n}^{(f)}$ be its rate in (302). Then*

$$\left\| \widehat{T}_{N,n}^{K,\delta,M} - T_0 \right\|_\infty = \mathrm{O}_{\mathbb{P}}\big((R_{N,n}^{(f)})^{-1}\big) + O\left(K^{-\rho_F} + \delta^{\kappa} + \eta_M\right). \tag{303}$$

Let

$$b_{K,\delta,M} := T_0^{K,\delta,M} - T_0. \tag{304}$$

If (301) tends to zero, then in $C(\mathcal{T}_f)$,

$$R_{N,n}^{(f)}\big(\widehat{T}_{N,n}^{K,\delta,M} - T_0\big) = \mathbb{G}_f + R_{N,n}^{(f)} b_{K,\delta,M} + \mathrm{o}_{\mathbb{P}}(1), \tag{305}$$

where $\mathbb{G}_f$ is the corresponding Part S4 Gaussian process.

Proof. Decompose

$$\widehat{T}^{K,\delta,M} - T_0 = \{T^{K,\delta,M}(\widehat{\vartheta}) - T^{K,\delta,M}(\vartheta_0)\} + b_{K,\delta,M}.$$

The first term has the uniform Part S4 Taylor expansion with the numerical derivative. Equation (301) allows replacement of that derivative by the exact loading, and Part S3 supplies the parameter limit. The second term is deterministic numerical bias. This proves (305); the norm bound follows immediately. □

Corollary 9.16 (Numerical-bias trichotomy)**.** *For a given family f:*

(i) If $R_{N,n}^{(f)} \|b_{K,\delta,M}\|_\infty \to 0$, the centred Part S4–Part S5 limit is unchanged.

(ii) If $R_{N,n}^{(f)} b_{K,\delta,M} \to b_f$ in $C(\mathcal{T}_f)$, the limit is $\mathbb{G}_f + b_f$.

(iii) If $R_{N,n}^{(f)} \|b_{K,\delta,M}\|_\infty \to \infty$, centred root-information inference is invalid.

9.8.9 Optimal numerical allocation

Suppose

$$\mathrm{Bias}(K,\delta,M) \lesssim C_s K^{-\rho_F} + C_t \delta^{\kappa} + C_o M^{-\rho_A}, \tag{306}$$

so that $\eta_M \asymp M^{-\rho_A}$.

Proposition 9.20 (Balanced and undersmoothed choices)**.** *For statistical normalisation R:*

(i) Balancing each numerical term with R^{-1} gives

$$K_{\mathrm{bal}} \asymp R^{1/\rho_F}, \qquad \delta_{\mathrm{bal}} \asymp R^{-1/\kappa}, \qquad M_{\mathrm{bal}} \asymp R^{1/\rho_A}. \tag{307}$$

This is appropriate for first-order total-error control but generally produces local numerical bias at the inferential scale.

(ii) For any slowly diverging L_R,

$$K \asymp (RL_R)^{1/\rho_F}, \qquad \delta \asymp (RL_R)^{-1/\kappa}, \qquad M \asymp (RL_R)^{1/\rho_A} \tag{308}$$

yields $R\,\mathrm{Bias} = O(L_R^{-1})$.

Proposition 9.21 (Cost-optimal bias shares)**.** *Suppose*

$$\mathrm{Cost}(K,\delta,M) \asymp K^{\zeta}\delta^{-\xi}M^{\upsilon}. \tag{309}$$

Allocate a total numerical-bias budget ε through $x_s + x_t + x_o = \varepsilon$. Then

$$\mathrm{Cost} \asymp x_s^{-\zeta/\rho_F} x_t^{-\xi/\kappa} x_o^{-\upsilon/\rho_A}.$$

The minimiser is

$$x_i = \frac{a_i}{a_s + a_t + a_o}\varepsilon, \qquad a_s = \frac{\zeta}{\rho_F}, \quad a_t = \frac{\xi}{\kappa}, \quad a_o = \frac{\upsilon}{\rho_A}. \tag{310}$$

Proof. Minimise the logarithm of the cost under the linear budget constraint. The first-order conditions give a_i/x_i equal to a common Lagrange multiplier, which yields (310). □

9.9 Part S8: Non-identification boundary

9.9.1 Mechanisms and equivalence relations

Let H be a separable Hilbert space and let

$$\mathfrak{M} = (A, G, \Sigma, u_0, \mathcal{C}, \mathcal{E})$$

denote a mechanism. Here A generates a strongly continuous semigroup, $G : [0,T] \times H \to H$ is measurable and globally Lipschitz in its state argument, Σ is a diffusion operator, u_0 is the initial condition, $\mathcal{C}$ is an

observation mechanism, and $\mathcal{E}$ contains measurement noise and any other declared exogenous modules. The state solves

$$dU_t = \big(AU_t + G(t, U_t)\big)\, dt + \Sigma\, dW_t. \tag{311}$$

Definition 9.4 (Modular operator intervention)**.** *For a declared post-intervention generator $\widetilde{A}$, define*

$$I_{\widetilde{A}}\mathfrak{M} := (\widetilde{A}, G, \Sigma, u_0, \mathcal{C}, \mathcal{E}).$$

The operator module alone is replaced.

Definition 9.5 (Observational and target equivalence)**.** *For an experiment Y,*

$$\mathfrak{M}_1 \sim_{\text{obs}} \mathfrak{M}_2 \quad \Longleftrightarrow \quad \mathcal{L}_{\mathfrak{M}_1}(Y) = \mathcal{L}_{\mathfrak{M}_2}(Y).$$

For a counterfactual target $\tau = (\widetilde{A}, w, z)$,

$$\mathfrak{M}_1 \sim_\tau \mathfrak{M}_2 \quad \Longleftrightarrow \quad F_\tau(\mathfrak{M}_1) = F_\tau(\mathfrak{M}_2).$$

The target is identified precisely when every observational-equivalence class is contained in a target-equivalence class. Part S8 constructs explicit violations.

9.9.2 The operator–forcing decomposition gauge

Let A_1 generate a C_0-semigroup and let $L \in \mathcal{L}(H)$. Define

$$A_2 := A_1 - L, \qquad G_2(t, u) := G_1(t, u) + Lu. \tag{312}$$

The bounded perturbation theorem ensures that A_2 is a generator with domain $D(A_2) = D(A_1)$.

Definition 9.6 (Drift-decomposition gauge)**.** *The transformation*

$$\mathcal{T}_L(A, G) := (A - L, G + L\,\cdot)$$

is called a drift-decomposition gauge transformation.

Theorem 9.45 (Exact factual gauge equivalence; establishes Theorem 8.1)**.** *As stated in Theorem 8.1 of the main text.*

Proof. For $u \in D(A_1)$,

$$(A_1 - L)u + G_1(t, u) + Lu = A_1 u + G_1(t, u).$$

Thus the two mechanisms define the same stochastic evolution equation, with the same initial condition and exogenous Wiener path. Well-posedness and pathwise uniqueness imply indistinguishability. □

Interpretation. Factual data can identify the total drift in (311). They cannot identify how that drift was partitioned into a module labelled "operator" and a module labelled "forcing". This is a structural equivalence, not a low-power asymptotic phenomenon.

Theorem 9.46 (The intervention is not gauge invariant; establishes Theorem 8.1)**.** *As stated in Theorem 8.1 of the main text.*

Proof. The displayed identities follow from modular replacement. If the intervention descended to the quotient, gauge-equivalent factual mechanisms would remain target equivalent after intervention. The rank-one construction below gives a counterexample whenever L acts on a target-relevant direction. □

Assumption carrying the burden. The positive model fixes the forcing module. This selects one representative of the gauge class. Without that restriction, an "operator intervention" is representative dependent.

9.9.3 Explicit adapted-feedback separation

Scalar OU model Let $a_0, \gamma, \sigma > 0$. Consider

$$\mathfrak{M}_1: \quad dX_t = -a_0 X_t\, dt + \sigma\, dW_t, \tag{313}$$

$$\mathfrak{M}_2: \quad dX_t = -(a_0 + \gamma) X_t\, dt + \gamma X_t\, dt + \sigma\, dW_t. \tag{314}$$

The factual equations are identical.

Intervene by replacing the operator rate by $\widetilde{a} > \gamma$. The post-intervention rates are

$$r_1 = \widetilde{a}, \qquad r_2 = \widetilde{a} - \gamma.$$

Lemma 9.16 (Integrated OU variance)**.** *For*

$$dX_t = -rX_t\,dt + \sigma\,dW_t, \qquad X_0 = 0,$$

the functional $\Psi(X) = \int_0^T X_t\,dt$ *is centred Gaussian with variance*

$$J_T(r) = \frac{\sigma^2}{r^2}\left[T - \frac{2(1-e^{-rT})}{r} + \frac{1-e^{-2rT}}{2r}\right]. \tag{315}$$

The map $r \mapsto J_T(r)$ *is strictly decreasing on* $(0,\infty)$*.*

Proof. Stochastic Fubini gives

$$\Psi(X) = \sigma\int_0^T \frac{1-e^{-r(T-s)}}{r}\,dW_s.$$

Itô isometry yields (315). For $x > 0$,

$$\frac{1-e^{-rx}}{r} = \int_0^x e^{-ru}\,du$$

is strictly decreasing in r. Squaring and integrating over $x = T - s$ proves strict monotonicity. □

Corollary 9.17 (Scalar counterfactual separation)**.** *The mechanisms* (313)–(314) *have identical factual path laws but*

$$\mathcal{L}\left(\int_0^T X_t^{(1),\mathrm{do}}\,dt\right) \neq \mathcal{L}\left(\int_0^T X_t^{(2),\mathrm{do}}\,dt\right),$$

because $J_T(\widetilde{a}) \neq J_T(\widetilde{a}-\gamma)$*.*

Rank-one SPDE construction Let $A_0 e_k = -\lambda_k e_k$ on H, and let $P_\ell = e_\ell \otimes e_\ell$. Set

$$A_1 = a_0 A_0, \qquad L = \gamma P_\ell, \qquad A_2 = A_1 - L, \qquad G_1 = 0, \qquad G_2(u) = Lu.$$

The factual SPDEs coincide by Theorem 9.45.

Theorem 9.47 (Field-level adapted-forcing non-identification)**.** *Let the common intervention be* $\widetilde{A} = \widetilde{a}A_0$*, assume*

$$0 < \gamma < \widetilde{a}\lambda_\ell, \qquad q_\ell > 0,$$

and take $w_t = e_\ell$*. Then the two factual field paths are indistinguishable, while the post-intervention laws of*

$$\Psi_w(U) = \int_0^T \langle e_\ell, U_t\rangle\,dt$$

are distinct centred Gaussian laws.

Proof. The factual total generator in both mechanisms is $a_0 A_0$. After intervention, mode ℓ has rate

$$r_1 = \widetilde{a}\lambda_\ell$$

under the first mechanism and

$$r_2 = \widetilde{a}\lambda_\ell - \gamma$$

under the second. The target uses only that mode and has variances

$$q_\ell^2 J_T(r_1) \quad \text{and} \quad q_\ell^2 J_T(r_2),$$

with the common scale absorbed in J_T. Lemma 9.16 gives strict inequality. □

Sharpness. The offset is harmful only when it reaches the target. If w annihilates the invariant subspace on which L acts and the perturbation cannot propagate into the target through the intervened dynamics, the two target laws can coincide.

9.9.4 Open-loop exogenous forcing

Complete factual path equivalence

Theorem 9.48 (Full-path open-loop confounding)**.** *Let* $a_1 \neq a_2$*, let* $h \in C^1([0,T])$ *satisfy* $h(0) = 0$*, and let* Z *be non-degenerate. Define*

$$F_i(t) := Z\big(h'(t) + a_i h(t)\big).$$

For

$$dX_t^{(i)} = -a_i X_t^{(i)}\,dt + F_i(t)\,dt, \qquad X_0^{(i)} = 0,$$

one has

$$X_t^{(i)} = Zh(t) \quad \textit{for all } t$$

under both mechanisms. Hence their complete factual path laws coincide.

After the common intervention $a_i \mapsto \widetilde{a}$, holding F_i fixed,

$$X_t^{(i),\mathrm{do}} = Z\left[h(t) + (a_i - \widetilde{a}) \int_0^t e^{-\widetilde{a}(t-s)} h(s)\, ds\right]. \tag{316}$$

Any field functional that does not annihilate the second term has a different post-intervention law for $i = 1$ and $i = 2$.

Proof. Substitution gives

$$\frac{d}{dt}[Zh(t)] = -a_i Zh(t) + Z[h'(t) + a_i h(t)].$$

Uniqueness proves factual equality. Variation of constants under rate $\widetilde{a}$ yields

$$X_t^{(i),\mathrm{do}} = Z \int_0^t e^{-\widetilde{a}(t-s)} [h'(s) + a_i h(s)]\, ds.$$

Integration by parts gives (316). The coefficient of $a_i - \widetilde{a}$ is non-zero for a target that sees the response direction. □

Consequence. State feedback is sufficient for failure but not necessary. Unknown open-loop forcing laws already destroy operator identification.

Positive-noise finite-grid confounding Consider

$$dX_t = -aX_t\, dt + F_t\, dt + \sigma\, dW_t, \qquad X_0 = 0. \tag{317}$$

Fix $0 = t_0 < t_1 < \cdots < t_m$ and let

$$F_t = Z_j, \qquad t \in (t_{j-1}, t_j], \qquad Z \sim N(0, K), \qquad Z \perp W.$$

For $Y = (X_{t_1}, \ldots, X_{t_m})^\top$,

$$Y = L_a Z + \varepsilon_a,$$

where

$$(L_a)_{j\ell} = \mathbf{1}\{\ell \le j\} e^{-a(t_j - t_\ell)} \frac{1 - e^{-a(t_\ell - t_{\ell-1})}}{a}, \tag{318}$$

and $\varepsilon_a \sim N(0, C_a^W)$. The lower-triangular matrix L_a is invertible.

Theorem 9.49 (Every finite Gaussian grid can be confounded)**.** *Fix $a_1 \neq a_2$. Choose*

$$C = MI_m, \qquad M > \max_{i=1,2} \lambda_{\max}(C_{a_i}^W),$$

and define

$$K_i = L_{a_i}^{-1}(C - C_{a_i}^W)L_{a_i}^{-\top}. \tag{319}$$

Then K_i is positive definite and both factual experiments satisfy

$$Y^{(i)} \sim N(0, C).$$

Under a common intervention $a_i \mapsto \widetilde{a}$, the covariance is

$$C_i^{\mathrm{do}} = C_{\widetilde{a}}^W + L_{\widetilde{a}} K_i L_{\widetilde{a}}^\top.$$

For all but at most one admissible value of M,

$$C_1^{\mathrm{do}} \neq C_2^{\mathrm{do}}.$$

Proof. Positive definiteness of $C - C_{a_i}^W$ and invertibility of L_{a_i} imply $K_i \succ 0$. Equation (319) gives

$$L_{a_i} K_i L_{a_i}^\top + C_{a_i}^W = C.$$

The counterfactual covariance difference is affine in M, with coefficient

$$L_{\widetilde{a}} \left[L_{a_1}^{-1} L_{a_1}^{-\top} - L_{a_2}^{-1} L_{a_2}^{-\top}\right] L_{\widetilde{a}}^\top.$$

This coefficient is non-zero. Indeed, the diagonal entries

$$c_a(\Delta_j) = \frac{1 - e^{-a\Delta_j}}{a}$$

of L_a are strictly decreasing in a, so $L_{a_1} L_{a_1}^\top \neq L_{a_2} L_{a_2}^\top$. An affine matrix function with non-zero slope can vanish for at most one scalar M. □

Interpretation. No finite increase in temporal resolution repairs an unrestricted-input model class. The positive experiment in Part S3 is regular because the forcing law is fixed, not because the grid alone rules out latent inputs.

Corollary 9.18 (One-time explicit construction)**.** *At time Δ, take $F_t = Z$ with $Z \sim N(0, \tau^2)$ independent of W. Put*

$$c_a(\Delta) = \frac{1 - e^{-a\Delta}}{a}, \qquad h_a(\Delta) = \frac{1 - e^{-2a\Delta}}{2a}.$$

Then

$$X_\Delta \sim N\left(0, \tau^2 c_a(\Delta)^2 + \sigma^2 h_a(\Delta)\right).$$

For any

$$V > \max_i \sigma^2 h_{a_i}(\Delta),$$

choose

$$\tau_i^2 = \frac{V - \sigma^2 h_{a_i}(\Delta)}{c_{a_i}(\Delta)^2}.$$

Both factual observations are $N(0, V)$, while after a common intervention to $\widetilde{a}$ their variances are

$$\tau_i^2 c_{\widetilde{a}}(\Delta)^2 + \sigma^2 h_{\widetilde{a}}(\Delta),$$

which are distinct whenever $\tau_1^2 \neq \tau_2^2$.

Proposition 9.22 (Unknown random initial condition)**.** *Let $dX_t = -aX_t\,dt$ and $X_0 \sim N(0, \nu_a^2)$. At a factual time t_*, choose*

$$\nu_a^2 = V e^{2at_*}.$$

Then $X_{t_*} \sim N(0, V)$ for every a. Under a common intervention to $\widetilde{a}$, holding the initial law fixed,

$$\mathrm{Var}(X_t^{\mathrm{do}}) = e^{-2\widetilde{a}t} \nu_a^2,$$

which depends on the factual representative. The deterministic-known-initial-state assumption removes this ambiguity.

9.9.5 Measurement failures

Rank-deficient observation Let

$$H = H_{\mathrm{obs}} \oplus H_{\mathrm{lat}}, \qquad \mathcal{C} = P_{\mathrm{obs}}.$$

Theorem 9.50 (Free latent target direction)**.** *Suppose the model class fixes the observed block but permits the generator, noise covariance, or initial law on H_{lat} to vary. Then any two such mechanisms have the same complete observed factual path law. If a counterfactual target has a non-zero latent loading and the two post-intervention latent laws differ, the target is not identified.*

Proof. The observation is a function only of the common observed block. Its law is therefore unchanged by latent-block modifications. A target with latent loading depends on the modified block and can separate the mechanisms. □

Unknown sensor calibration Consider

$$dX_t = -\theta X_t\,dt + \sigma\,dW_t, \qquad Y_t = cX_t.$$

Then

$$dY_t = -\theta Y_t\,dt + (c\sigma)\,dW_t.$$

Proposition 9.23 (Calibration confounds field counterfactuals)**.** *Let θ be common and choose*

$$c_1\sigma_1 = c_2\sigma_2, \qquad \sigma_1 \neq \sigma_2.$$

The complete factual sensor-path laws coincide. Under $\theta \mapsto \widetilde{\theta}$, the field law of X^{do} depends on σ_i and therefore differs. The sensor law of $Y^{\mathrm{do}} = c_i X^{\mathrm{do}}$ still depends only on $c_i\sigma_i$ and remains equal if calibration is invariant.

Target-specific lesson. Unknown calibration destroys the field-scale target but need not destroy the invariant sensor-scale target.

Intervention-dependent measurement

Proposition 9.24 (Measurement modularity failure)**.** *Suppose both factual mechanisms use the same sensor $\mathcal{C}_0$, while the post-intervention sensor is unspecified and can be either $\mathcal{C}_1^{\mathrm{do}}$ or $\mathcal{C}_2^{\mathrm{do}}$. Even if $\mathcal{L}(U^{\mathrm{do}})$ is identified,*

$$\mathcal{L}(\mathcal{C}_1^{\mathrm{do}} U^{\mathrm{do}}) \neq \mathcal{L}(\mathcal{C}_2^{\mathrm{do}} U^{\mathrm{do}})$$

in general. This is an under-specified joint intervention, not a failure of the field law.

State–measurement noise decomposition

Proposition 9.25 (Noise decomposition)**.** *Let*

$$Y = X + \varepsilon, \qquad X \sim N(0, q), \qquad \varepsilon \sim N(0, r), \qquad X \perp \varepsilon.$$

The factual law identifies only $q + r$. Suppose an intervention maps q to gq, with $g \neq 1$, and leaves r fixed. Two decompositions with the same $q+r$ but different q have the same factual law and distinct post-intervention variances $gq + r$.

Scope. Longer time series can identify process and measurement noise under sufficient dynamic restrictions. Proposition 9.25 shows that the decomposition is not identified automatically.

9.9.6 Marginal laws versus cross-world coupling

Proposition 9.26 (Counterfactual marginals do not identify unit-level effects)**.** *Suppose*

$$X^a \sim N(0, v_a), \qquad X^b \sim N(0, v_b)$$

are identified marginal counterfactual laws. For every $\rho \in [-1, 1]$, there exists a joint Gaussian coupling with

$$\mathrm{Corr}(X^a, X^b) = \rho.$$

All such couplings preserve the marginals, while

$$X^a - X^b \sim N\left(0, v_a + v_b - 2\rho\sqrt{v_a v_b}\right).$$

Thus a unit-level contrast is not identified without a cross-world coupling assumption.

Relation to Part S1. The shared-Wiener construction selects a structural coupling. Independent resampling selects another. The marginal CDF, quantile, and tail results of Part S4–Part S7 are unaffected; only joint cross-world claims require the additional convention.

9.9.7 Assumption-to-failure map

Positive assumption	Ambiguity removed	Failure when removed
Forcing module fixed	operator/forcing decomposition	$(A, G) \sim (A - L, G + L\,\cdot)$
No unrestricted feedback	exact state-dependent offset	Theorems 9.45–9.47
Open-loop forcing law restricted	operator/input covariance trade-off	Theorems 9.48–9.49
Known deterministic u_0	initial-law/operator trade-off	Proposition 9.22
Target-relevant observability or cross-mode restrictions	free latent mechanism	Theorem 9.50
Known calibration for field targets	$c\sigma$ confounding	Proposition 9.23
Measurement invariance	undefined observed intervention	Proposition 9.24
Identifiable process/measurement noise split	$q + r$ confounding	Proposition 9.25
Shared primitive noise for unit-level effects	cross-world correlation	Proposition 9.26

9.9.8 Completion and paper-facing statement

Theorem 9.51 (Sharp non-identification boundary; establishes Theorem 8.2)**.** *As stated in Theorem 8.2 of the main text.*

Proof. Items (i)–(vii) are Theorems 9.45, 9.48, 9.49, and 9.50, together with Propositions 9.22, 9.23, 9.24, and 9.25. The cross-world statement is Proposition 9.26. □

Table 2: Information-standardised estimation diagnostics.

N	n	$\mathrm{sd}(Z_\theta)$	$\mathrm{sd}(Z_\sigma)$	$\mathrm{bias}(\widehat{\theta})$
5	2000	1.0180	1.0062	3.41×10^{-3}
10	8000	1.0219	0.9990	4.5×10^{-4}
20	32000	0.9995	0.9955	-9.86×10^{-5}

What the theorem says. The positive assumptions are not merely convenient regularity restrictions. They remove concrete observational symmetries that otherwise make the declared operator intervention representative dependent.

What remains identifiable. Every negative result is target specific. Sensor-scale targets can survive unknown calibration; observed-subspace targets can survive latent dynamics; marginal targets can survive cross-world ambiguity. The correct conclusion is not that causal inference for SPDEs is impossible, but that its identifying modules must be stated.

Programme status. Part S8 completes the theorem system. the replication package must now consolidate all existing Part S3–Part S8 numerical assets into one command, add the planned high-replication heat-equation experiments, and verify every formal regime. Part S10 can then reconstruct the paper around the final dependency chain.

10 Numerical validation

The replication package separates theorem-aligned simulation, deterministic rate checks, and deliberate assumption-failure experiments. The one-dimensional heat equation uses

$$\lambda_k = \pi^2 k^2.$$

Remark 10.1 (Verification of the standing assumptions for the benchmark model). All abstract hypotheses are fulfilled for the one-dimensional Dirichlet heat equation with $q_k \equiv 1$, $u_0 = 0$, $\sigma_0 > 0$: (i) $A_0 = \partial_{xx}$ on $D = (0,1)$ with Dirichlet boundary is self-adjoint, negative, with compact resolvent, $\lambda_k = \pi^2 k^2$, $e_k = \sqrt{2}\sin(k\pi x)$, and Weyl growth holds with $d = 1$; (ii) trace condition: $\sum_k q_k^2/\lambda_k = \pi^{-2}\sum_k k^{-2} < \infty$ (Assumption 2.1); (iii) pathwise Galerkin condition: $\sum_k q_k^2 \lambda_k^{2(\alpha+\delta)-1} < \infty$ iff $\alpha + \delta < 1/4$, as recorded after Theorem 3.1; (iv) regime (Assumption 5.1): $\lambda_N \Delta_n = \pi^2 N^2 T/n \to 0$ iff $N^2/n \to 0$; the design path $n = 80N^2$ holds $\lambda_N \Delta_n = \pi^2 T/80 \approx 0.123$ *constant* (as recorded in the archived design table), so the experiments probe the joint regime at a fixed small resolution while N and n grow — strict theorem alignment would let n/N^2 diverge slowly, at higher cost and, per the reported diagnostics, no visible change; the initial-energy condition holds trivially with $u_0 = 0$; (v) spectral-growth condition of the consistency proof: $N\sqrt{\lambda_N} = \pi N^2 = o(\Lambda_N)$ since $\Lambda_N \sim \pi^2 N^3/3$; (vi) target class: $\widetilde{\Theta} \Subset (0,\infty)$, compact threshold and weight sets, and $\inf_{a,w} V_a(w) > 0$ on the chosen ellipsoid, checked numerically alongside the runs.

Because the LAN condition is $\lambda_N \Delta_n \to 0$, a naive Cartesian grid in (N, n) is not automatically theorem aligned. The main designs pair n with N^2 so that $\lambda_N \Delta_n$ remains small.

10.1 Mixed-rate likelihood

The publication validation uses the paired designs

$$(N, n) \in \{(5, 2000), (10, 8000), (20, 32000)\}$$

with 1000 replications per design. The information-standardised empirical standard deviations are within approximately 2.2% of one. Drift bias and likelihood-ratio remainders contract with N.

10.2 Counterfactual process and bootstrap

Fixed-intervention CDF sup-norm RMSE decreases from approximately 1.72×10^{-3} to 2.13×10^{-4}. Distributional-effect sup-norm RMSE decreases from 1.64×10^{-2} to 2.26×10^{-3}. Simultaneous coverage remains close to 0.95 for both rate families.

Beyond the headline coverage, the multiplier construction itself was stress-tested at $(N, n) = (10, 8000)$ with 300 replications per variant: the direct Gaussian draw, the modal-score multiplier with Gaussian weights, and the modal-score multiplier with Rademacher weights give statistically indistinguishable simultaneous coverage, and coverage is flat in the number of draws and in the threshold-grid resolution.

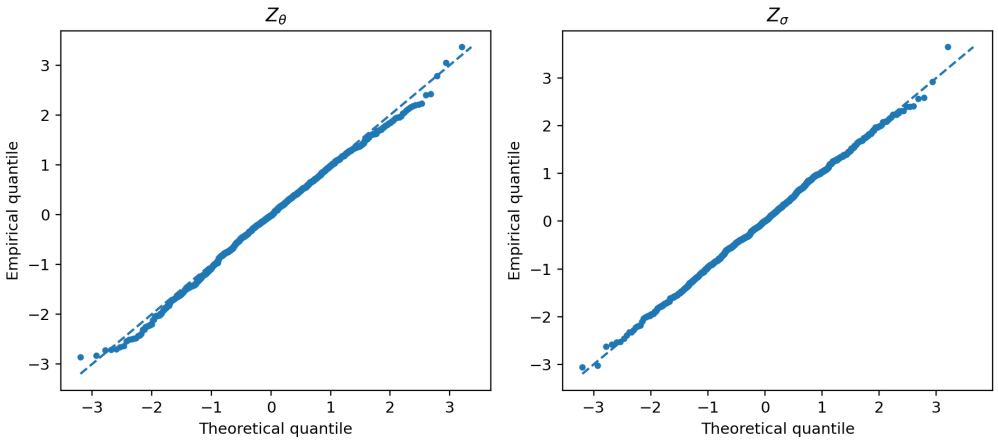


Figure 8: Normal quantile–quantile diagnostics for the standardised estimators. The visible right-skew of Z_θ at $N = 5$ (skewness 0.49, Kolmogorov–Smirnov $p = 0.021$) dissipates along the design path ($p = 0.42$ at $N = 10$, $p = 0.75$ at $N = 20$), consistent with the first-order asymptotics.

Table 3: Uniform target error and simultaneous coverage.

N	CDF sup RMSE	Effect sup RMSE	Fast coverage	Slow coverage
5	1.72×10^{-3}	1.64×10^{-2}	0.946	0.950
10	6.04×10^{-4}	6.29×10^{-3}	0.955	0.948
20	2.13×10^{-4}	2.26×10^{-3}	0.951	0.951

10.3 Approximation and sensitivity

For the baseline spatial design, fitted log–log slopes are -2.580 for the mean tail, -4.564 for the variance tail, -2.307 for the strong RMS error, and -2.580 for the Gaussian CDF error, close to the theoretical values $-2.6, -4.6, -2.3, -2.6$. Temporal slopes lie between 0.994 and 0.999.

For a non-commuting bounded perturbation, the central finite-difference derivative error decreases from 4.27×10^{-6} at $\varepsilon = 0.1$ to 1.71×10^{-9} at $\varepsilon = 0.002$. The CDF linearisation remainder is quadratic and the exact Gaussian W_2 distance remains below the derived Lipschitz envelope.

10.4 Impossibility constructions

The non-identification experiments are exact calculations rather than power studies. The adapted-feedback factual variance gap is zero while the post-intervention Kolmogorov distance is 0.0359. The one-time open-loop factual variance gap is numerical zero and its post-intervention distance is 0.0984. On a four-time grid, the factual covariance gap is 8.2×10^{-16} while the post-intervention covariance gap is 0.7003.

All tables and the headline diagnostics are regenerated by `make reproduce-publication`; the paper figures are regenerated end-to-end by the seeded validation pipeline (`make reproduce-paper`). The code records seeds, profiles, runtime, environment versions, and checksums.

11 Discussion

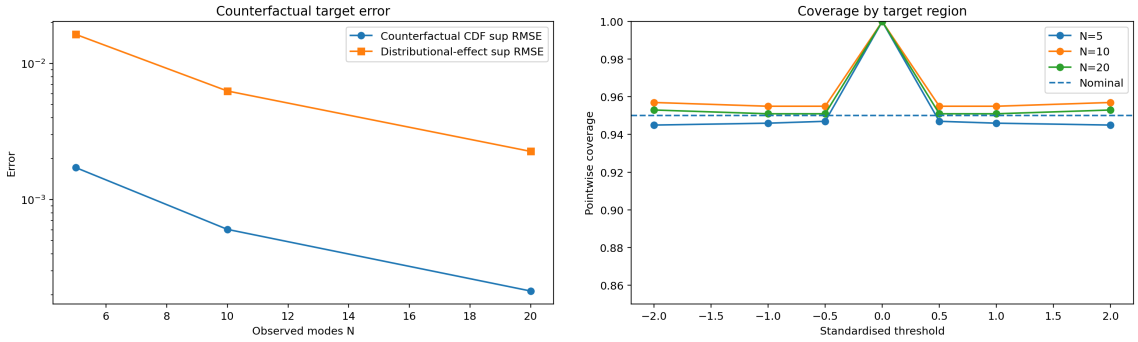
The paper places a narrow causal object on a complete mathematical and inferential footing. The intervention is an operator-module replacement inside a declared linear Gaussian SPDE class. The target is a counterfactual field-functional distribution. Identification is target specific. The inferential limit is mixed rate. Numerical approximation must be controlled at the target family's root-information scale.

Three distinctions are particularly important.

First, semantics and identification are separate. Section 3 provides a coherent shared-noise intervention even before one asks whether the resulting law is recoverable from factual data; Sections 4 and 8 then characterise the positive and negative identification results.

Second, strong field approximation and Gaussian law approximation need not have the same rate. Under non-degeneracy, the variance tail enters normal CDF, quantile, and optimal Wasserstein comparisons linearly. Using the common-noise strong rate for every law target can be unnecessarily conservative.

Third, the phrase "causal SPDE" should not be read as unrestricted causal discovery. The intervention and invariant modules must be supplied. The feedback-gauge and open-loop constructions show that passive



(a) Uniform target errors.

(b) Coverage across thresholds.

Figure 9: Counterfactual target accuracy and coverage.

Table 4: Bootstrap implementation sensitivity at $(N, n) = (10, 8000)$, 300 replications per row (Monte Carlo s.e. ≈ 0.013); nominal level 0.95.

axis	variant	draws	fast cov.	slow cov.
multiplier	direct Gaussian	499	0.933	0.930
multiplier	modal Gaussian	499	0.960	0.947
multiplier	modal Rademacher	499	0.957	0.947
draws	direct Gaussian	199	0.940	0.937
draws	direct Gaussian	999	0.940	0.927
grid (11 pts)	direct Gaussian	499	0.933	0.927
grid (61 pts)	direct Gaussian	499	0.937	0.920

equality of factual laws cannot decide which part of a drift is the module that should be replaced.

Several extensions remain. Non-diagonal unbounded perturbations, including spatially varying divergence-form coefficients, require domain-sensitive perturbation theory. Semilinear equations require differentiability of nonlinear solution maps. Non-Gaussian noise calls for characteristic-function, Wasserstein, or quantile-specific methods rather than Gaussian mean–variance reduction. Physical-space observations require a separate statistical experiment and may produce different rates.

The present results are nevertheless sufficient for a coherent paper-level conclusion:

$$\begin{aligned} &\text{specified structural intervention} + \text{identified factual experiment} \\ &\quad + \text{controlled numerical approximation} \\ &\qquad \Longrightarrow \text{uniformly inferable counterfactual laws.} \end{aligned}$$

The impossibility results define the boundary of that conclusion rather than an exception to it.

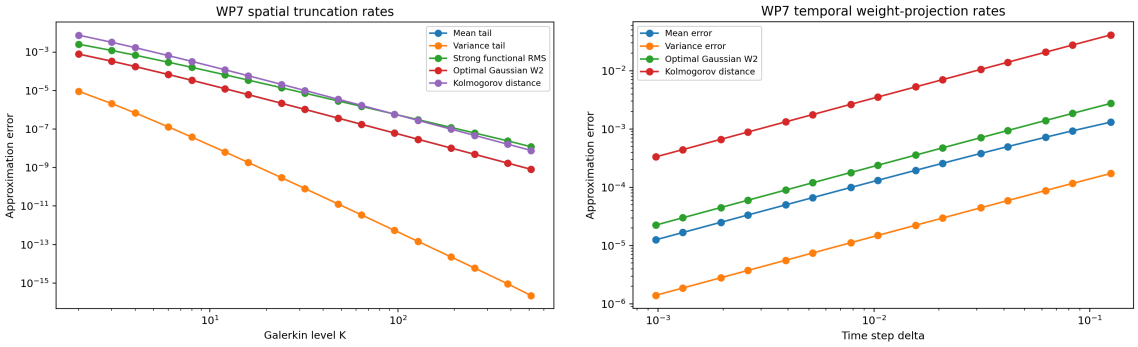
Declarations

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Competing interests. The author declares no competing interests.

Data and code availability. All numerical results are generated by the seeded replication package (Python reference implementation with a Julia parity implementation, deterministic profiles, SHA-256 manifests, and continuous-integration checks) archived with the manuscript sources; the figures and tables of Section 10 are reproduced end-to-end by the documented `make` targets. [Repository DOI to be added upon archival deposition.]

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(a) Galerkin rates.

(b) Temporal rates.

Figure 10: Deterministic approximation diagnostics.

References

[1] J. Cárcamo, A. Cuevas, and L.-A. Rodríguez. Directional differentiability for supremum-type functionals: Statistical applications. *Bernoulli*, 26(3):2143–2175, 2020. doi:10.3150/19-BEJ1188.

[2] G. Da Prato, S. Kwapień, and J. Zabczyk, "Regularity of solutions of linear stochastic equations in Hilbert spaces", *Stochastics*, 23 (1987), 1–23.

[3] G. Da Prato and J. Zabczyk, *Stochastic Equations in Infinite Dimensions*, 2nd ed., Cambridge University Press, 2014.

[4] C. Villani. *Optimal Transport: Old and New*. Springer, 2009.

[5] A. W. van der Vaart and J. A. Wellner. *Weak Convergence and Empirical Processes*. Springer, 1996.

[6] Randolf Altmeyer and Markus Reiß. Nonparametric estimation for linear SPDEs from local measurements. *The Annals of Applied Probability*, 31(1):1–38, 2021. doi: 10.1214/20-AAP1581.

[7] Donald W. K. Andrews and Gustavo Soares. Inference for parameters defined by moment inequalities using generalized moment selection. *Econometrica*, 78(1):119–157, 2010. doi: 10.3982/ECTA7502.

[8] Markus Bibinger and Mathias Trabs. Volatility estimation for stochastic PDEs using high-frequency observations. *Stochastic Processes and their Applications*, 130(5):3005–3052, 2020. doi: 10.1016/j.spa.2019.09.002.

[9] Stephan Bongers and Joris M. Mooij. From random differential equations to structural causal models: the stochastic case, 2018.

[10] Victor Chernozhukov, Denis Chetverikov, and Kengo Kato. Gaussian approximations and multiplier bootstrap for maxima of sums of high-dimensional random vectors. *The Annals of Statistics*, 41(6):2786–2819, 2013. doi: 10.1214/13-AOS1161.

[11] Victor Chernozhukov, Iván Fernández-Val, and Blaise Melly. Inference on counterfactual distributions. *Econometrica*, 81(6):2205–2268, 2013. doi: 10.3982/ECTA10582.

[12] Carsten Chong. High-frequency analysis of parabolic stochastic PDEs. *The Annals of Statistics*, 48(2):1143–1167, 2020. doi: 10.1214/19-AOS1841.

[13] Igor Cialenco. Statistical inference for SPDEs: an overview. *Statistical Inference for Stochastic Processes*, 21: 309–329, 2018. doi: 10.1007/s11203-018-9177-9.

[14] Igor Cialenco, Francisco Delgado-Vences, and Hyun-Jung Kim. Drift estimation for discretely sampled SPDEs. *Stochastics and Partial Differential Equations: Analysis and Computations*, 8:895–920, 2020. doi: 10.1007/s40072-019-00164-4.

[15] Giuseppe Da Prato, Stanisław Kwapień, and Jerzy Zabczyk. Regularity of solutions of linear stochastic equations in Hilbert spaces. *Stochastics*, 23(1):1–23, 1987.

[16] Klaus-Jochen Engel and Rainer Nagel. *One-Parameter Semigroups for Linear Evolution Equations*. Springer, 2000. doi: 10.1007/b97696.

[17] Zheng Fang and Andres Santos. Inference on directionally differentiable functions. *Review of Economic Studies*, 86(1):377–412, 2019. doi: 10.1093/restud/rdy012.

[18] Trygve Haavelmo. The probability approach in econometrics. *Econometrica*, 12:iii–115, 1944. Supplement.

[19] Peter Hall and Christopher C. Heyde. *Martingale Limit Theory and Its Application*. Academic Press, New York, 1980.

[20] Lars Peter Hansen and Thomas J. Sargent. The dimensionality of the aliasing problem in models with rational spectral densities. *Econometrica*, 51(2):377–387, 1983.

[21] Florian Hildebrandt and Mathias Trabs. Parameter estimation for SPDEs based on discrete observations in time and space. *Electronic Journal of Statistics*, 15(1):2716–2776, 2021. doi: 10.1214/21-EJS1848.

[22] Markus Hübner and Boris L. Rozovskii. On asymptotic properties of maximum likelihood estimators for parabolic stochastic PDEs. *Probability Theory and Related Fields*, 103:143–163, 1995. doi: 10.1007/BF01204212.

[23] Yusuke Kaino and Masayuki Uchida. Parametric estimation for a parabolic linear SPDE model based on discrete observations. *Journal of Statistical Planning and Inference*, 211:190–220, 2021. doi: 10.1016/j.jspi.2020.05.004.

[24] Oliver Linton, Kyungchul Song, and Yoon-Jae Whang. An improved bootstrap test of stochastic dominance. *Journal of Econometrics*, 154(2):186–202, 2010. doi: 10.1016/j.jeconom.2009.07.004.

[25] Sergey V. Lototsky. Statistical inference for stochastic parabolic equations: a spectral approach. *Publicacions Matemàtiques*, 53(1):3–45, 2009.

[26] Joris M. Mooij, Dominik Janzing, and Bernhard Schölkopf. From ordinary differential equations to structural causal models: the deterministic case. In *Proceedings of the 29th Conference on Uncertainty in Artificial Intelligence (UAI)*, pages 440–448, 2013.

[27] Gregor Pasemann and Wilhelm Stannat. Drift estimation for stochastic reaction-diffusion systems. *Electronic Journal of Statistics*, 14(1):547–579, 2020. doi: 10.1214/19-EJS1665.

[28] Amnon Pazy. *Semigroups of Linear Operators and Applications to Partial Differential Equations*. Springer, 1983. doi: 10.1007/978-1-4612-5561-1.

[29] Jonas Peters, Stefan Bauer, and Niklas Pfister. Causal models for dynamical systems. In *Probabilistic and Causal Inference: The Works of Judea Pearl*, pages 671–690. ACM, 2022.

[30] Peter C. B. Phillips. The problem of identification in finite parameter continuous time models. *Journal of Econometrics*, 1(4):351–362, 1973.

[31] Paul K. Rubenstein, Stephan Bongers, Bernhard Schölkopf, and Joris M. Mooij. From deterministic ODEs to dynamic structural causal models. In *Proceedings of the 34th Conference on Uncertainty in Artificial Intelligence (UAI)*, 2018.

[32] Mark Rudelson and Roman Vershynin. Hanson–wright inequality and sub-Gaussian concentration. *Electronic Communications in Probability*, 18(82):1–9, 2013.

[33] Alexander Sokol and Niels Richard Hansen. Causal interpretation of stochastic differential equations. *Electronic Journal of Probability*, 19(100):1–24, 2014. doi: 10.1214/EJP.v19-2891.

[34] Yozo Tonaki, Yusuke Kaino, and Masayuki Uchida. Parameter estimation for linear parabolic SPDEs in two space dimensions based on high frequency data. *Scandinavian Journal of Statistics*, 50(4):1568–1589, 2023. doi: 10.1111/sjos.12663.

[35] Yozo Tonaki, Yusuke Kaino, and Masayuki Uchida. Parametric estimation for linear parabolic SPDEs in two space dimensions based on temporal and spatial increments. *Metrika*, 2024. doi: 10.1007/s00184-024-00969-x. Published online; update volume/pages at submission.

[36] Aad W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, 1998. doi: 10.1017/CBO9780511802256.

[37] Aad W. van der Vaart and Jon A. Wellner. *Weak Convergence and Empirical Processes*. Springer, 1996. doi: 10.1007/978-1-4757-2545-2.